



Statistical Architecture Framework

8 jun 2026 21:45:16

Purpose

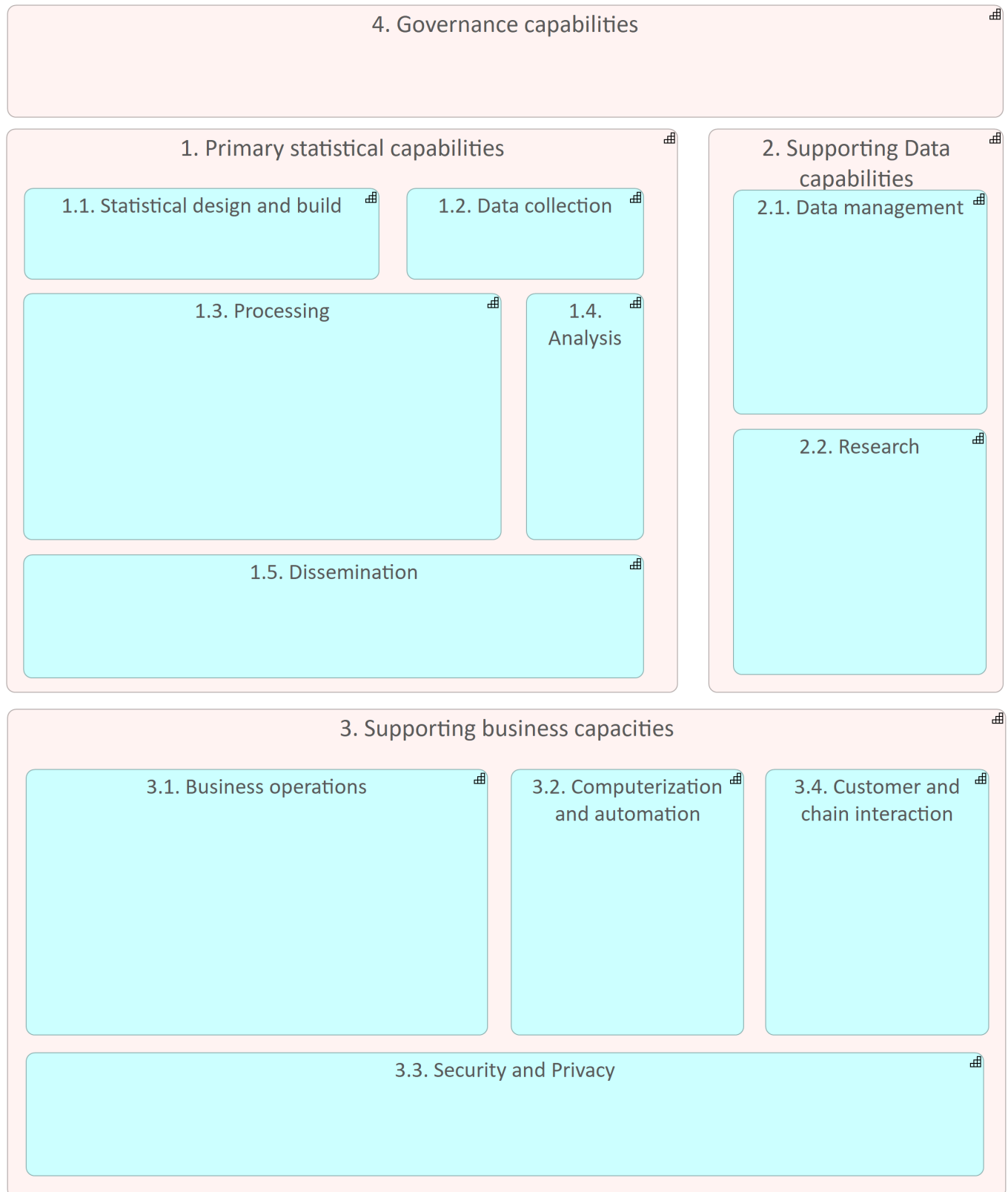
Properties

SSDefaultToolbox	ArchiMate3::Motivation
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Views

Capability Map Viewpoint - Level 1,2

No viewpoint



Documentation

A capability map provides a structured, visual overview of the organization's core abilities required to achieve its mission of producing and disseminating official statistics. This map displays capabilities as stable, business-centric building blocks, grouped to align with strategic goals and is useful to highlight areas for development and investment.

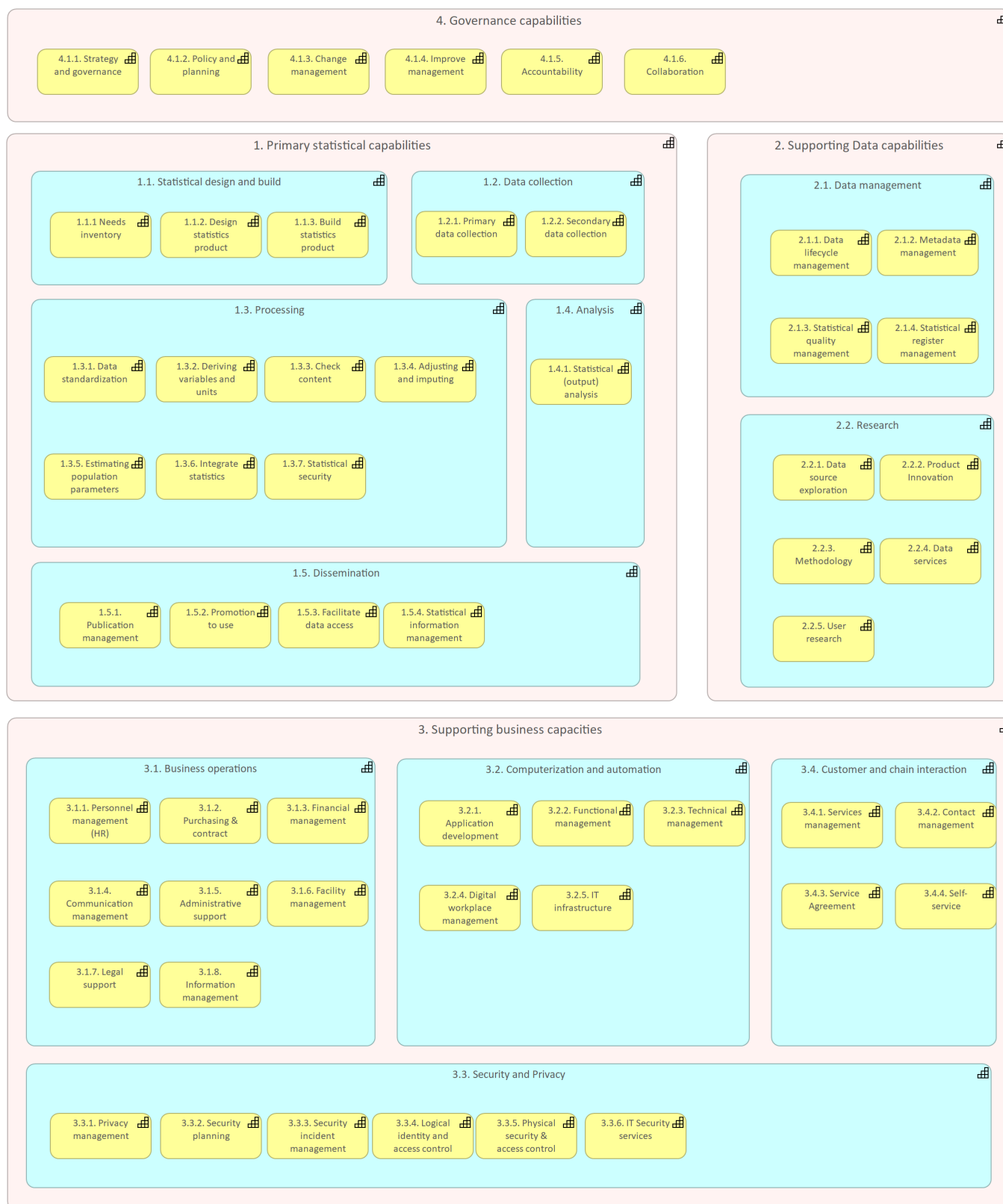
A capability map helps a statistical bureau assess and communicate its strengths, gaps, and areas needing improvement to support organizational objectives such as delivering high-quality, timely statistics, maintaining the trust of data providers, and promoting innovation in data processing.

Elements

Element	Type
1. Primary statistical capabilities	Capability
1.1. Statistical design and build	Capability
1.2. Data collection	Capability
1.3. Processing	Capability
1.4. Analysis	Capability
1.5. Dissemination	Capability
2. Supporting Data capabilities	Capability
2.1. Data management	Capability
2.2. Research	Capability
3. Supporting business capacities	Capability
3.1. Business operations	Capability
3.2. Computerization and automation	Capability
3.3. Security and Privacy	Capability
3.4. Customer and chain interaction	Capability
4. Governance capabilities	Capability

Capability Map Viewpoint - Level 1,2,3

No viewpoint



Documentation

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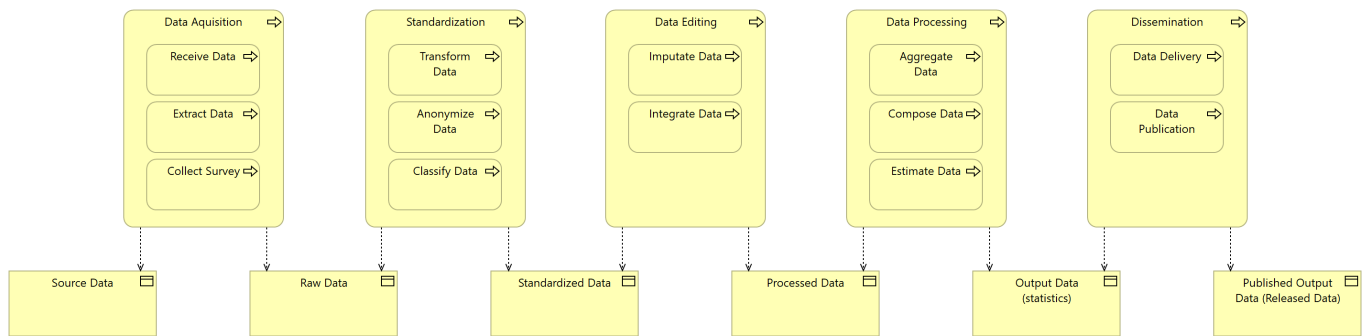
Elements

Element	Type
1. Primary statistical capabilities	Capability
1.1. Statistical design and build	Capability
1.1.1 Needs inventory	Capability
1.1.2. Design statistics product	Capability
1.1.3. Build statistics product	Capability
1.2. Data collection	Capability
1.2.1. Primary data collection	Capability
1.2.2. Secondary data collection	Capability
1.3. Processing	Capability
1.3.1. Data standardization	Capability
1.3.2. Deriving variables and units	Capability
1.3.3. Check content	Capability
1.3.4. Adjusting and imputing	Capability
1.3.5. Estimating population parameters	Capability
1.3.6. Integrate statistics	Capability
1.3.7. Statistical security	Capability
1.4. Analysis	Capability
1.4.1. Statistical (output) analysis	Capability
1.5. Dissemination	Capability
1.5.1. Publication management	Capability
1.5.2. Promotion to use	Capability
1.5.3. Facilitate data access	Capability
1.5.4. Statistical information management	Capability
2. Supporting Data capabilities	Capability
2.1. Data management	Capability
2.1.1. Data lifecycle management	Capability
2.1.2. Metadata management	Capability
2.1.3. Statistical quality management	Capability
2.1.4. Statistical register management	Capability
2.2. Research	Capability
2.2.1. Data source exploration	Capability

Element	Type
2.2.2. Product Innovation	Capability
2.2.3. Methodology	Capability
2.2.4. Data services	Capability
2.2.5. User research	Capability
3. Supporting business capacities	Capability
3.1. Business operations	Capability
3.1.1. Personnel management (HR)	Capability
3.1.2. Purchasing & contract management	Capability
3.1.3. Financial management	Capability
3.1.4. Communication management	Capability
3.1.5. Administrative support	Capability
3.1.6. Facility management	Capability
3.1.7. Legal support	Capability
3.1.8. Information management and archive management	Capability
3.2. Computerization and automation	Capability
3.2.1. Application development	Capability
3.2.2. Functional management	Capability
3.2.3. Technical management	Capability
3.2.4. Digital workplace management	Capability
3.2.5. IT infrastructure	Capability
3.3. Security and Privacy	Capability
3.3.1. Privacy management	Capability
3.3.2. Security planning	Capability
3.3.3. Security incident management	Capability
3.3.4. Logical identity and access control	Capability
3.3.5. Physical security & access control	Capability
3.3.6. IT Security services	Capability
3.4. Customer and chain interaction	Capability
3.4.1. Services management	Capability
3.4.2. Contact management	Capability
3.4.3. Service Agreement	Capability
3.4.4. Self-service	Capability
4. Governance capabilities	Capability
4.1.1. Strategy and governance	Capability
4.1.2. Policy and planning	Capability
4.1.3. Change management	Capability
4.1.4. Improve management	Capability
4.1.5. Accountability	Capability
4.1.6. Collaboration	Capability

SAF Business Architecture

No viewpoint



Documentation

This diagram describes a generic business architecture for statistical production that transforms data from external sources into published statistics through a series of well-defined functions and quality states. The architecture decomposes the statistical production process into five domain business functions (Data Acquisition, Standardization, Data Editing, Data Processing, and Dissemination) and four supporting functions (Process Design, Process Implementation, Process Execution, and Process Management), complemented by Metadata Management and Data Management capabilities. The key principle of this architecture is its focus on clearly defined data states (from State #0 Source Data through State #6 Consumed Data) that act as interfaces between functions. Each state represents data at a specific quality level with explicit requirements, enabling different statistical domains to share and reuse data at standardized quality milestones. This state-based approach promotes interoperability by establishing common data formats and classifications, enhances transparency through comprehensive data lineage and quality documentation at each stage, supports reproducibility by clearly defining the transformations and processes applied to move between states, and enables innovation by allowing teams to independently develop and improve their processes while adhering to standardized input and output specifications. By separating concerns between domain functions (what is done) and supporting functions (how it is designed, implemented, executed, and managed), the architecture provides flexibility for statistical organizations to modernize specific capabilities without disrupting the entire production chain, fostering continuous improvement and adaptation to new methodologies and technologies.

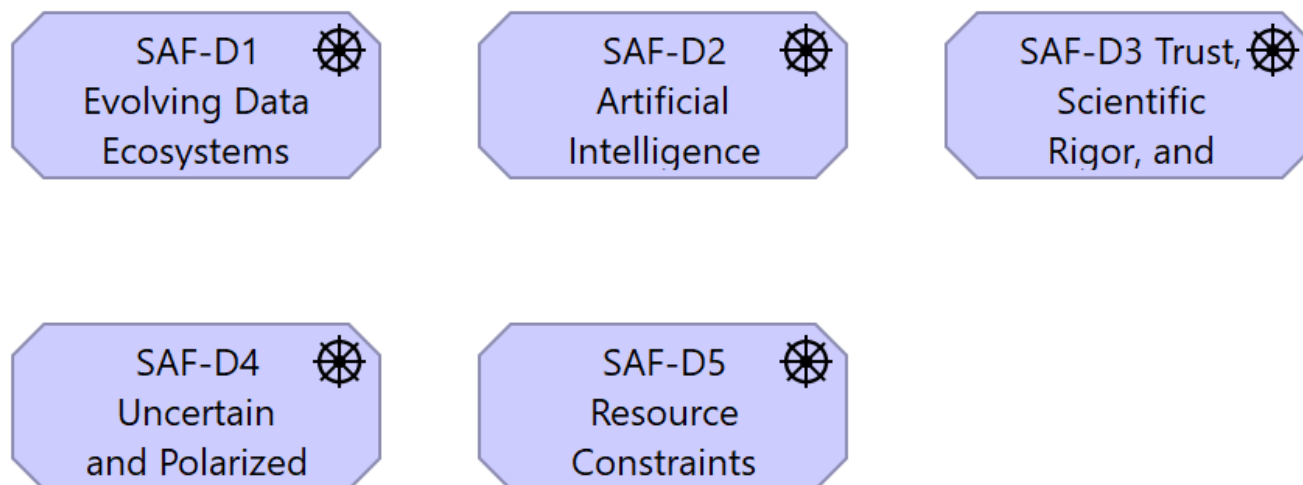
Elements

Element	Type
Aggregate Data	Business Process
Anonymize Data	Business Process
Classify Data	Business Process
Collect Survey	Business Process
Compose Data	Business Process
Data Aquisition	Business Process
Data Delivery	Business Process
Data Editing	Business Process
Data Processing	Business Process

Element	Type
Data Publication	Business Process
Dissemination	Business Process
Estimate Data	Business Process
Extract Data	Business Process
Imputate Data	Business Process
Integrate Data	Business Process
Output Data (statistics)	Business Object
Processed Data	Business Object
Published Output Data (Released Data)	Business Object
Raw Data	Business Object
Receive Data	Business Process
Source Data	Business Object
Standardization	Business Process
Standardized Data	Business Object
Transform Data	Business Process

SAF Drivers

Motivation viewpoint



Documentation

Architecture drivers are the forces that make change necessary. They capture the pressures, trends, and constraints that statistical offices face and which the SAF is designed to address. Drivers can be external, such as technological advancements or new regulations, or internal, such as the need to replace legacy systems or respond to stakeholder demands. By documenting these drivers, SAF ensures that its architecture is not an abstract ideal but a practical response to real-world challenges.

For statistical organizations, these drivers are particularly intense. Rapid technological change requires continuous adaptation, while political and societal pressures demand timely, relevant, and transparent data. At the same time, many organizations face limited budgets and must balance modernization with cost control. Recognizing these drivers helps statistical offices prioritize investments and avoid the pitfalls of reactive decision-making, where short-term fixes can lead to long-term inefficiencies.

Including architecture drivers in the framework also strengthens communication with stakeholders. By clearly articulating the external and internal factors shaping architecture, CIOs, CTOs, and enterprise architects can build a stronger case for change, justify strategic initiatives, and ensure that all parties understand the urgency of modernization. In this way, SAF creates a shared understanding of why action is necessary and what risks are avoided by adopting a structured framework.

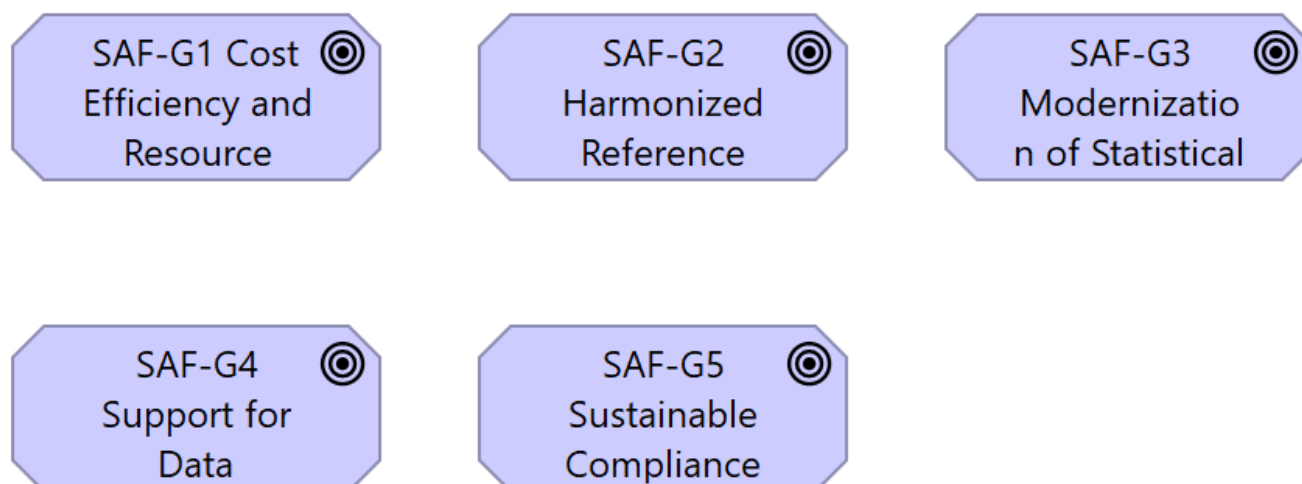
Elements

Element	Type
SAF-D1 Evolving Data Ecosystems and Partnerships	Driver
SAF-D2 Artificial Intelligence and Technological Acceleration	Driver
SAF-D3 Trust, Scientific Rigor, and Data Protection	Driver

Element	Type
SAF-D4 Uncertain and Polarized Global Environment	Driver
SAF-D5 Resource Constraints and Organizational Agility	Driver

SAF Goals

Motivation viewpoint



Documentation

Architecture goals represent the outcomes an organization aims to achieve by adopting and implementing the framework. They define what success looks like and serve as the link between strategy and action. In the case of the SAF, these goals are not just about modernizing IT systems, but about ensuring that national statistical institutes can continue to provide trustworthy, relevant, and high-quality statistics in a rapidly changing environment.

The goals of SAF address both operational and strategic needs. On one hand, they focus on efficiency, modernization, and compliance, which are essential for ensuring sustainability. On the other hand, they support innovation, adaptability, and collaboration, enabling statistical offices to embrace new opportunities such as integrating alternative data sources or experimenting with advanced analytics. Together, these goals ensure that modernization efforts do not compromise the fundamental values of official statistics — neutrality, accuracy, and reliability.

By defining goals explicitly, SAF provides a roadmap that can be adapted by each statistical organization according to its maturity and local context. This ensures that while each office may have unique challenges and resources, they are all working toward a shared vision that promotes harmonization, comparability, and global collaboration in official statistics.

Elements

Element	Type
SAF-G1 Cost Efficiency and Resource Sharing	Goal
SAF-G2 Harmonized Reference Framework	Goal
SAF-G3 Modernization of Statistical Systems	Goal
SAF-G4 Support for Data Innovation	Goal
SAF-G5 Sustainable Compliance	Goal



SAF Meaning

Motivation viewpoint

SAF-M1 Bridge 
Between
Strategy and

SAF-M2 
Catalyst for
Collaboration

SAF-M3 
Foundation for
Digital

SAF-M4 
Unifying
Language and

Documentation

Architecture meaning provides the “why” behind the framework. While goals and outcomes describe what SAF achieves, meaning explains why it matters. For statistical organizations, architecture is not just a technical exercise but a way of ensuring their continued relevance in a digital society. SAF’s meaning lies in its role as the foundation for modernization, trust, and international collaboration.

One dimension of this meaning is its role as a bridge between strategy and technology. Without such a framework, investments in IT risk being disconnected from the mission of providing high-quality statistics. SAF ensures that every technological decision — from adopting a new platform to restructuring business processes — can be traced back to strategic objectives, creating a coherent organizational narrative.

Another dimension is collaboration. The meaning of SAF extends beyond individual organizations: it is a unifying language for the UNECE statistical community. By providing shared models, practices, and principles, SAF fosters international collaboration, reduces duplication, and amplifies collective impact. In this sense, SAF is not just meaningful to IT professionals but to the broader community of statisticians, policymakers, and citizens who depend on trustworthy statistics.

Elements

Element	Type
SAF-M1 Bridge Between Strategy and Technology	Meaning
SAF-M2 Catalyst for Collaboration	Meaning
SAF-M3 Foundation for Digital Transformation	Meaning
SAF-M4 Unifying Language and Notation	Meaning

SAF Outcomes

Motivation viewpoint

SAF-O1 
Accelerated
Modernizati

SAF-O2 
Enhanced
Data Quality

SAF-O3 
Improved
Interoperabi

SAF-O4 
Improved
Trust and

SAF-O5 
Increased
Efficiency

SAF-O6 
Shared
Knowledge

Documentation

Architecture outcomes describe the tangible benefits that organizations can expect when adopting SAF. They translate principles, goals, and drivers into measurable impacts such as improved interoperability, reduced costs, and enhanced data quality. Outcomes are critical for demonstrating the value of architecture to stakeholders, as they provide a clear “return on investment” for adopting the framework.

In the SAF, outcomes are both technical and societal. On the technical side, NSIs benefit from streamlined processes, better metadata management, and more efficient system development. On the societal side, SAF supports trust, transparency, and comparability of statistics, which are essential for informed policy-making and democratic accountability. This dual perspective ensures that SAF delivers benefits beyond IT, directly supporting the mission of statistical organizations.

By emphasizing outcomes, SAF also reinforces a culture of continuous improvement. Outcomes are not static — they evolve as organizations mature and as new challenges emerge. Making outcomes explicit allows NSIs to assess progress, measure success, and adjust their modernization strategies in line with both internal priorities and external demands.

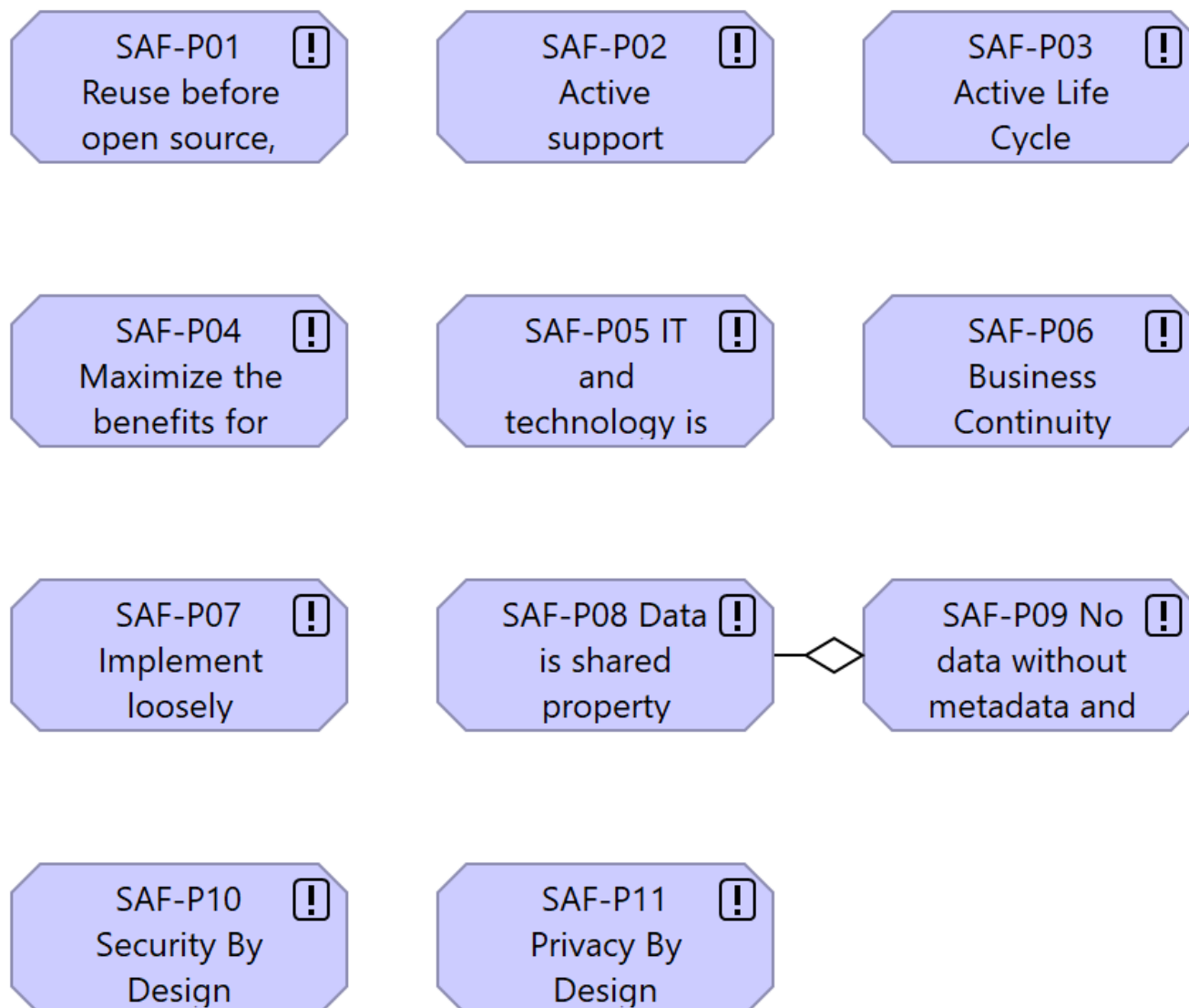
Elements

Element	Type
SAF-O1 Accelerated Modernization	Outcome
SAF-O2 Enhanced Data Quality	Outcome
SAF-O3 Improved Interoperability	Outcome
SAF-O4 Improved Trust and Transparency	Outcome
SAF-O5 Increased Efficiency	Outcome
SAF-O6 Shared Knowledge Base	Outcome



SAF Principles

Motivation viewpoint



Documentation

Architecture principles are the basis for any enterprise architecture framework. They provide the guiding rules and fundamental truths that define the way organizations design, implement, and manage their IT landscapes. In the context of official statistics, principles are particularly important because they ensure consistency across an interconnected environment.

For the SAF, architecture principles act as a compass for decision-making. They ensure that whenever there is a choice to be made between technologies, processes, or solutions, decisions are not taken in isolation but in accordance with the overarching vision of the statistical community. They also prevent short-term convenience from undermining long-term sustainability, a challenge that statistical offices often face given their reliance on legacy systems and resource constraints.

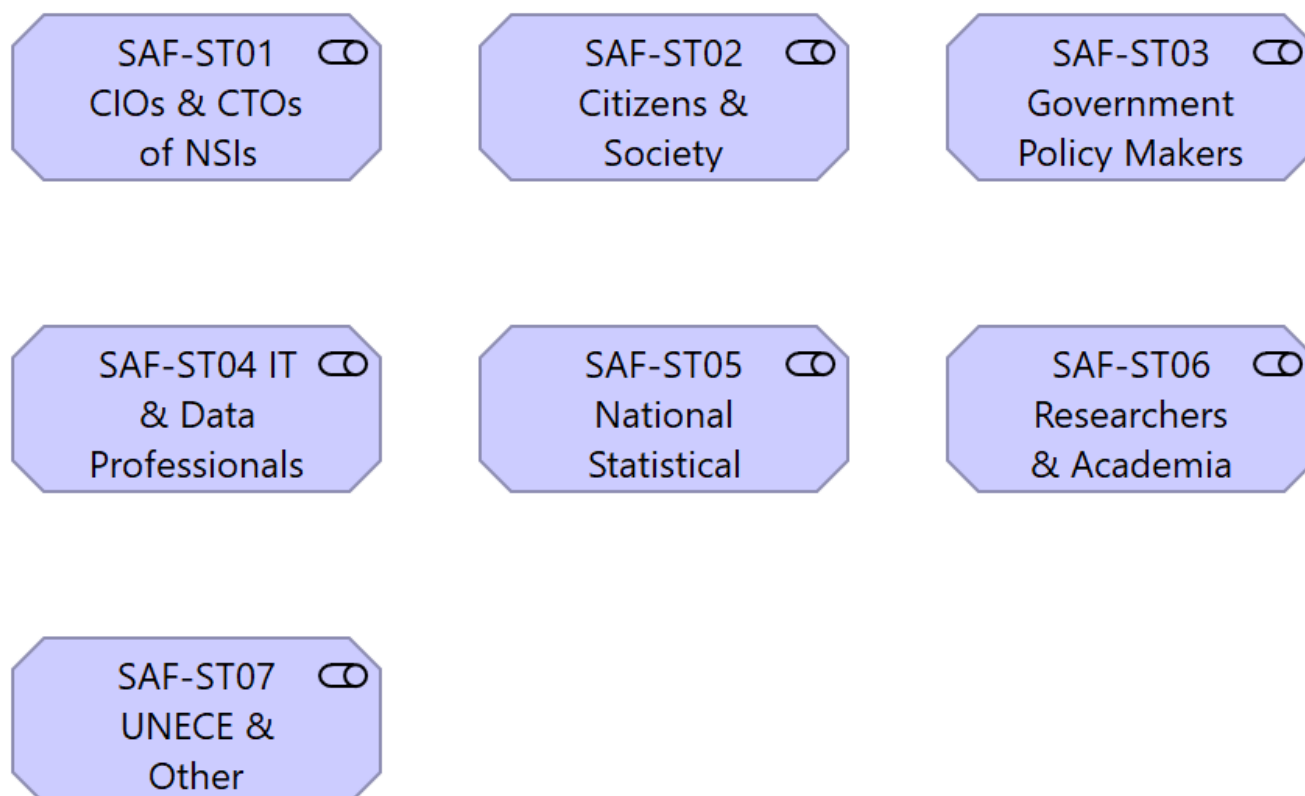
These principles also provide an element of accountability. When stakeholders understand the principles behind architectural decisions, it becomes easier to justify investments, explain trade-offs, and demonstrate that architecture is not a purely technical exercise but a strategic enabler of the organization's mission. In this way, principles make architecture transparent, predictable, and anchored in trust.

Elements

Element	Type
SAF-P01 Reuse before open source, open source before buying, buying before making it yourself	Principle
SAF-P02 Active support	Principle
SAF-P03 Active Life Cycle management	Principle
SAF-P04 Maximize the benefits for the organization	Principle
SAF-P05 IT and technology is everyone's business	Principle
SAF-P06 Business Continuity	Principle
SAF-P07 Implement loosely coupled processes and systems	Principle
SAF-P08 Data is shared property	Principle
SAF-P09 No data without metadata and classification	Principle
SAF-P10 Security By Design	Principle
SAF-P11 Privacy By Design	Principle

SAF Stakeholders

Motivation viewpoint



Documentation

Architecture is never developed in isolation. Stakeholders — the individuals, groups, or organizations that influence or are influenced by architecture — are central to its success. In the SAF, stakeholders range from technical experts and CIOs within NSIs to international organizations, policymakers, researchers, and ultimately, society at large. Each of these stakeholders has unique interests, expectations, and contributions, and the framework must balance these to be effective.

Identifying stakeholders clarifies who should be involved in governance, decision-making, and implementation. For example, IT professionals are responsible for translating SAF principles into concrete systems, while policymakers are concerned with the outcomes those systems enable. International organizations such as UNECE play a custodial role, ensuring that SAF remains harmonized and relevant across countries. By mapping these roles, SAF ensures that no perspective is overlooked and that architecture decisions are inclusive and robust.

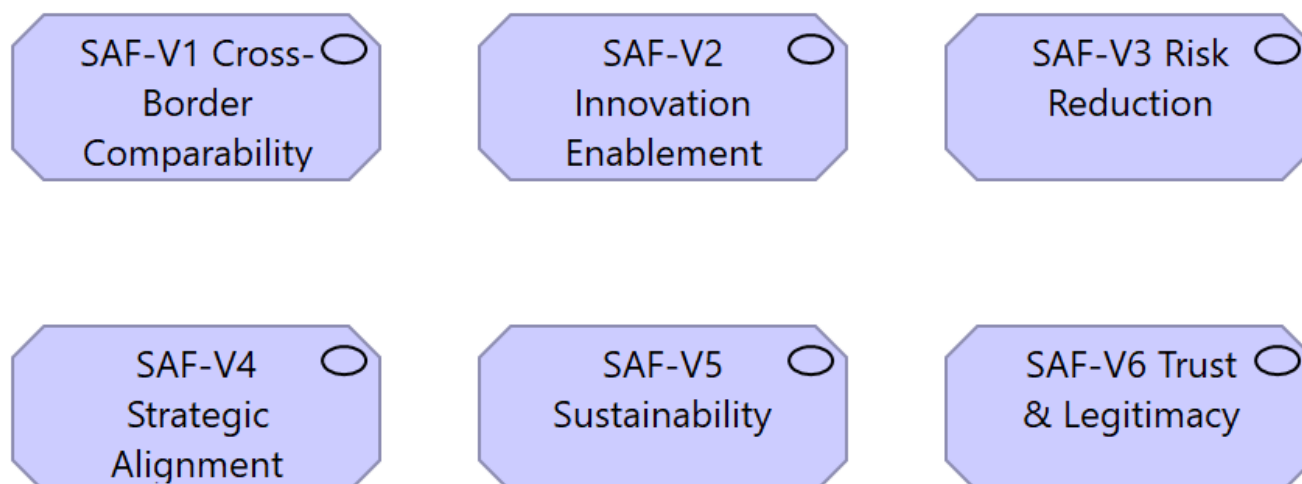
Engaging stakeholders also strengthens legitimacy. When those affected by architecture decisions are part of the conversation, they are more likely to support the outcomes. For statistical organizations, where public trust is essential, engaging stakeholders such as researchers, academia, and citizens indirectly ensures that the architecture remains aligned with societal expectations of transparency, security, and fairness.

Elements

Element	Type
SAF-ST01 CIOs & CTOs of NSIs	Stakeholder
SAF-ST02 Citizens & Society	Stakeholder
SAF-ST03 Government Policy Makers	Stakeholder
SAF-ST04 IT & Data Professionals	Stakeholder
SAF-ST05 National Statistical Institutes (NSIs)	Stakeholder
SAF-ST06 Researchers & Academia	Stakeholder
SAF-ST07 UNECE & Other International Bodies	Stakeholder

SAF Value

Motivation viewpoint



Documentation

The value of architecture lies in the benefits it creates for stakeholders. While outcomes describe specific impacts, value captures the broader, long-term importance of SAF to statistical organizations and society. It is about demonstrating that investing in architecture is not merely a technical necessity but a strategic decision that creates trust, efficiency, and sustainability.

For NSIs, the value of SAF includes reduced risks, greater agility, and improved alignment between IT and business objectives. For international organizations, it provides harmonization and comparability, enabling more effective collaboration across borders. For citizens and policymakers, the value lies in the trustworthiness of the statistics produced, ensuring that public policy and democratic processes are grounded in reliable data.

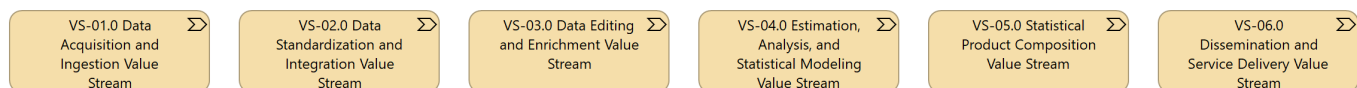
Value is also about sustainability. Without a framework, modernization efforts can be fragmented, costly, and short-lived. SAF provides a stable foundation that reduces technical debt, improves governance, and ensures that investments yield long-term benefits. In doing so, it secures the role of statistical organizations as trusted institutions in the digital age.

Elements

Element	Type
SAF-V1 Cross-Border Comparability	Value
SAF-V2 Innovation Enablement	Value
SAF-V3 Risk Reduction	Value
SAF-V4 Strategic Alignment	Value
SAF-V5 Sustainability	Value
SAF-V6 Trust & Legitimacy	Value

Value Stream to Process Realization View

No viewpoint



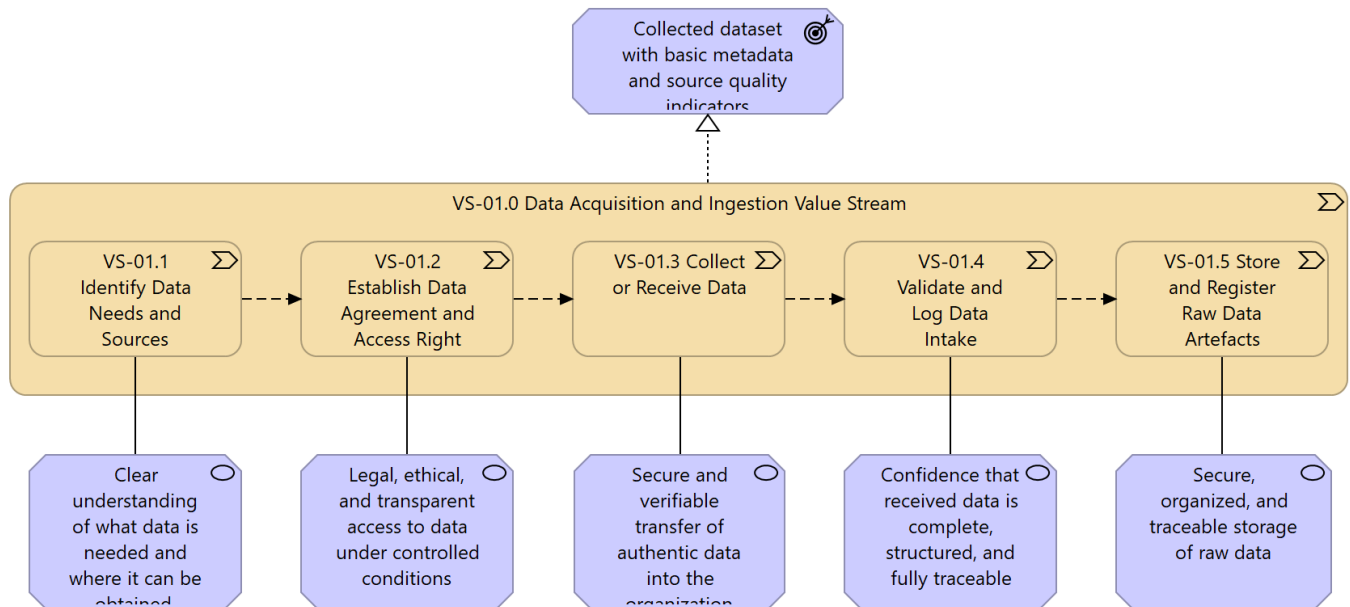
Elements

Element	Type
VS-01.0 Data Acquisition and Ingestion Value Stream	Value Stream
VS-02.0 Data Standardization and Integration Value Stream	Value Stream
VS-03.0 Data Editing and Enrichment Value Stream	Value Stream
VS-04.0 Estimation, Analysis, and Statistical Modeling Value Stream	Value Stream
VS-05.0 Statistical Product Composition Value Stream	Value Stream
VS-06.0 Dissemination and Service Delivery Value Stream	Value Stream

VS-01.0 Data Acquisition and Ingestion Value Stream

No viewpoint

Purpose: To obtain raw data from diverse sources and bring it securely into the statistical environment under controlled and compliant conditions.



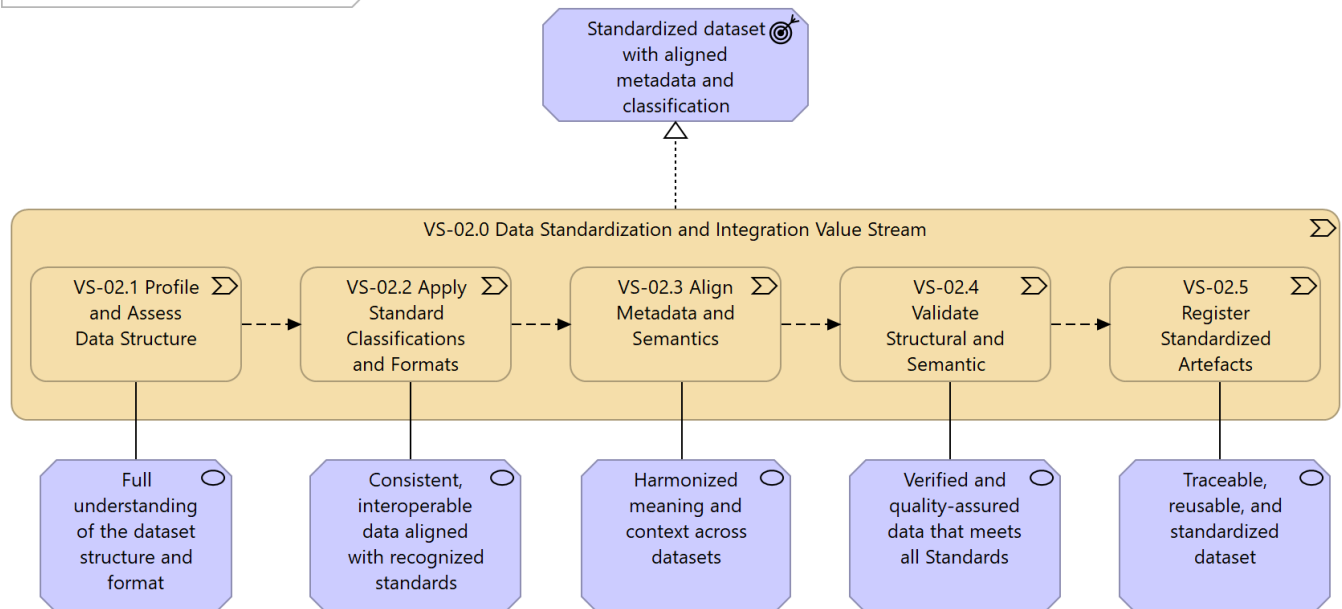
Elements

Element	Type
Clear understanding of what data is needed and where it can be obtained.	Value
Collected dataset with basic metadata and source quality indicators	Outcome
Confidence that received data is complete, structured, and fully traceable	Value
Legal, ethical, and transparent access to data under controlled conditions	Value
Secure and verifiable transfer of authentic data into the organization	Value
Secure, organized, and traceable storage of raw data	Value
VS-01.0 Data Acquisition and Ingestion Value Stream	Value Stream
VS-01.1 Identify Data Needs and Sources	Value Stream
VS-01.2 Establish Data Agreement and Access Right	Value Stream
VS-01.3 Collect or Receive Data	Value Stream
VS-01.4 Validate and Log Data Intake	Value Stream
VS-01.5 Store and Register Raw Data Artefacts	Value Stream

VS-02.0 Data Standardization and Integration Value Stream

No viewpoint

Purpose: To harmonize, structure, and standardize collected data for interoperability and comparability without altering its informational content.



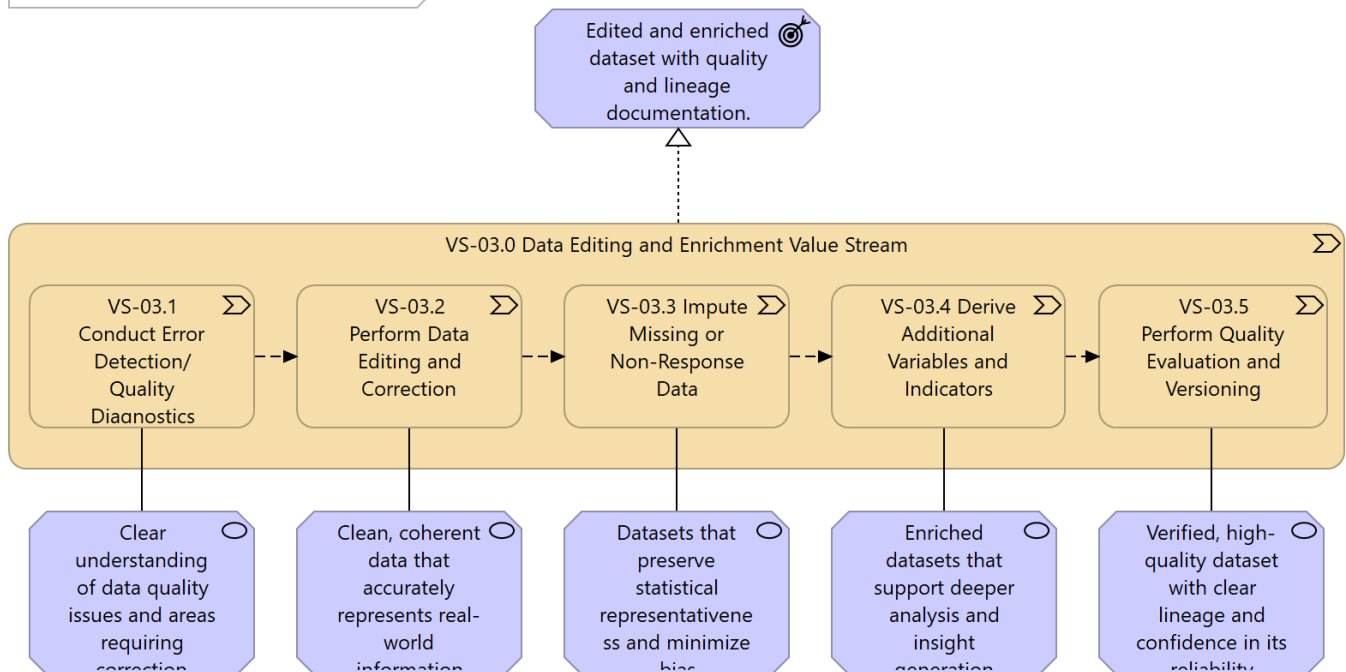
Elements

Element	Type
Consistent, interoperable data aligned with recognized standards	Value
Full understanding of the dataset structure and format	Value
Harmonized meaning and context across datasets	Value
Standardized dataset with aligned metadata and classification mappings.	Outcome
Traceable, reusable, and standardized dataset	Value
Verified and quality-assured data that meets all Standards	Value
VS-02.0 Data Standardization and Integration Value Stream	Value Stream
VS-02.1 Profile and Assess Data Structure	Value Stream
VS-02.2 Apply Standard Classifications and Formats	Value Stream
VS-02.3 Align Metadata and Semantics	Value Stream
VS-02.4 Validate Structural and Semantic Conformance	Value Stream
VS-02.5 Register Standardized Artefacts	Value Stream

VS-03.0 Data Editing and Enrichment Value Stream

No viewpoint

Purpose: To detect and correct errors, fill gaps, and derive new variables, improving the accuracy and completeness of data.



Elements

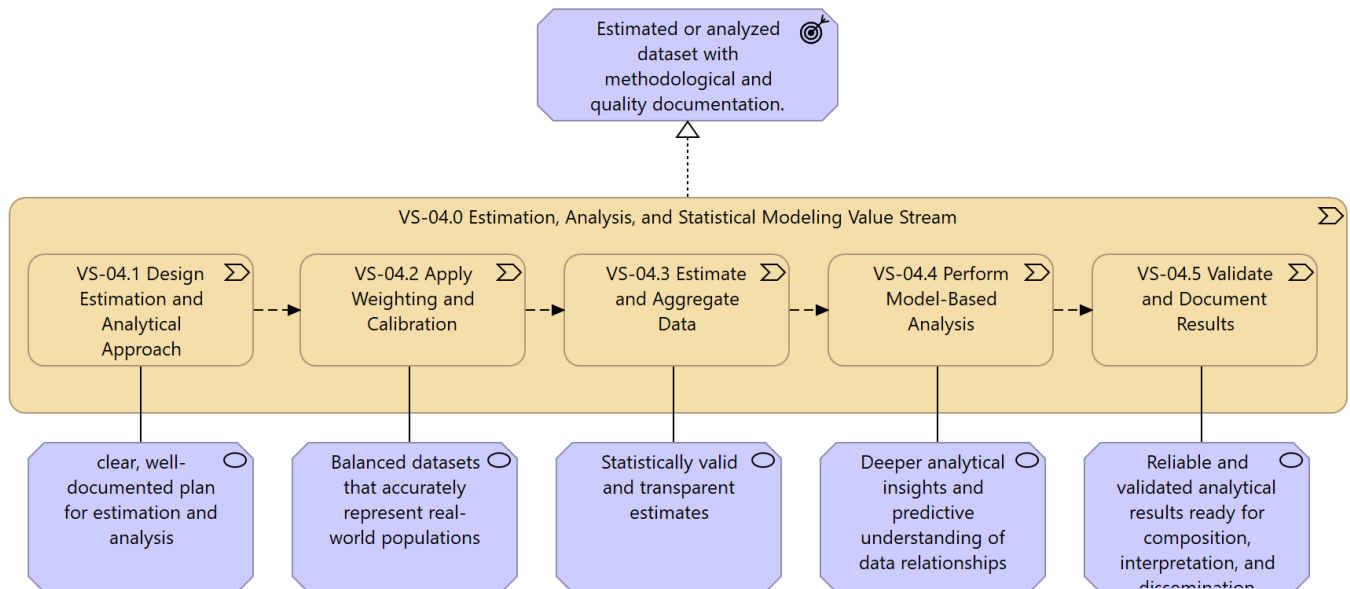
Element	Type
Clean, coherent data that accurately represents real-world information	Value
Clear understanding of data quality issues and areas requiring correction	Value
Datasets that preserve statistical representativeness and minimize bias	Value
Edited and enriched dataset with quality and lineage documentation.	Outcome
Enriched datasets that support deeper analysis and insight generation	Value
Verified, high-quality dataset with clear lineage and confidence in its reliability	Value
VS-03.0 Data Editing and Enrichment Value Stream	Value Stream
VS-03.1 Conduct Error Detection/ Quality Diagnostics	Value Stream
VS-03.2 Perform Data Editing and Correction	Value Stream
VS-03.3 Impute Missing or Non-Response Data	Value Stream
VS-03.4 Derive Additional Variables and Indicators	Value Stream
VS-03.5 Perform Quality Evaluation and Versioning	Value Stream



VS-04.0 Estimation, Analysis, and Statistical Modeling Value Stream

No viewpoint

Purpose: To transform prepared data into analytical insights, estimates, and statistical measures that represent real-world phenomena accurately and impartially.



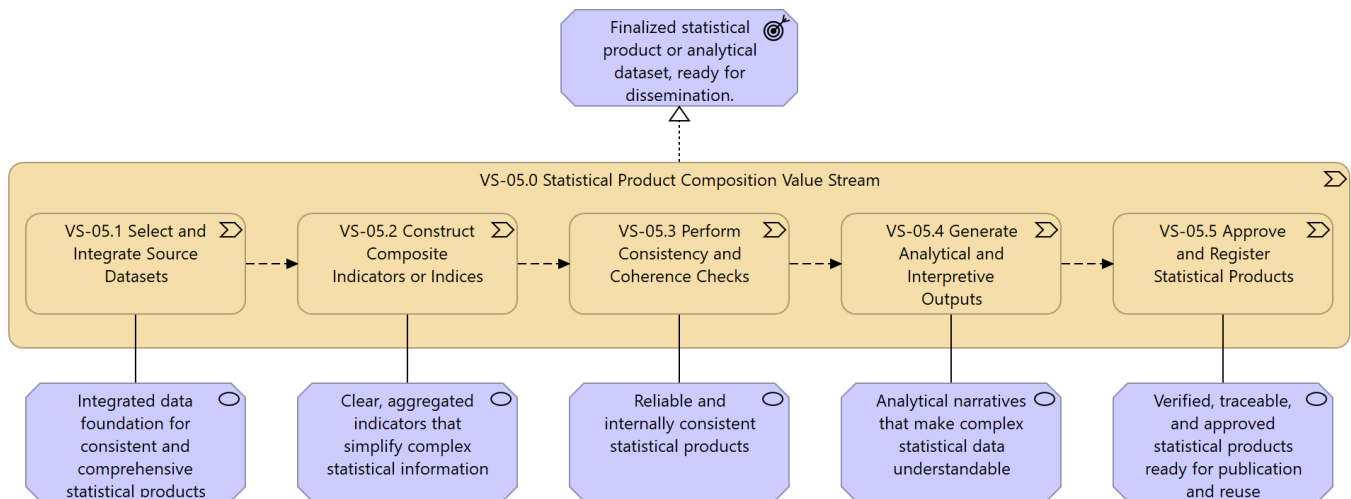
Elements

Element	Type
Balanced datasets that accurately represent real-world populations	Value
clear, well-documented plan for estimation and analysis	Value
Deeper analytical insights and predictive understanding of data relationships	Value
Estimated or analyzed dataset with methodological and quality documentation.	Outcome
Reliable and validated analytical results ready for composition, interpretation, and dissemination	Value
Statistically valid and transparent estimates	Value
VS-04.0 Estimation, Analysis, and Statistical Modeling Value Stream	Value Stream
VS-04.1 Design Estimation and Analytical Approach	Value Stream
VS-04.2 Apply Weighting and Calibration	Value Stream
VS-04.3 Estimate and Aggregate Data	Value Stream
VS-04.4 Perform Model-Based Analysis	Value Stream
VS-04.5 Validate and Document Results	Value Stream

VS-05.0 Statistical Product Composition Value Stream

No viewpoint

Purpose: To integrate multiple datasets or analytical results into finalized, multi-domain, or composite statistical products.



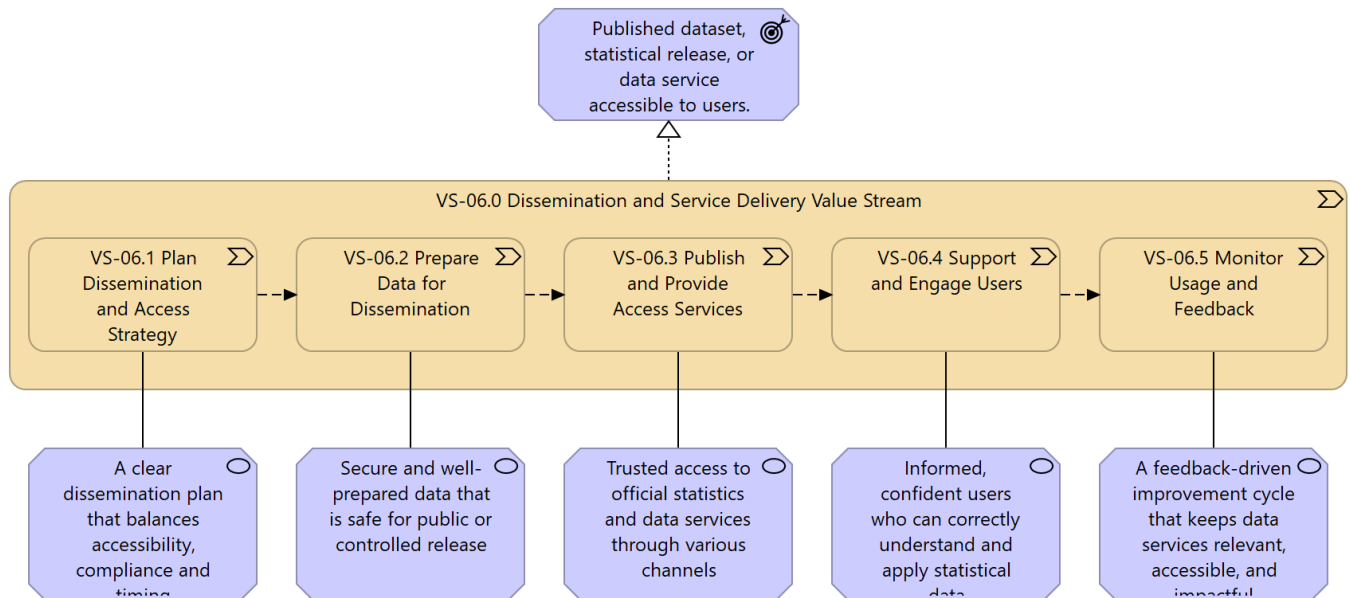
Elements

Element	Type
Analytical narratives that make complex statistical data understandable	Value
Clear, aggregated indicators that simplify complex statistical information	Value
Finalized statistical product or analytical dataset, ready for dissemination.	Outcome
Integrated data foundation for consistent and comprehensive statistical products	Value
Reliable and internally consistent statistical products	Value
Verified, traceable, and approved statistical products ready for publication and reuse	Value
VS-05.0 Statistical Product Composition Value Stream	Value Stream
VS-05.1 Select and Integrate Source Datasets	Value Stream
VS-05.2 Construct Composite Indicators or Indices	Value Stream
VS-05.3 Perform Consistency and Coherence Checks	Value Stream
VS-05.4 Generate Analytical and Interpretive Outputs	Value Stream
VS-05.5 Approve and Register Statistical Products	Value Stream

VS-06.0 Dissemination and Service Delivery Value Stream

No viewpoint

Purpose: To provide users with secure, accessible, and user-centric access to statistical outputs, ensuring transparency, usability, and impact.



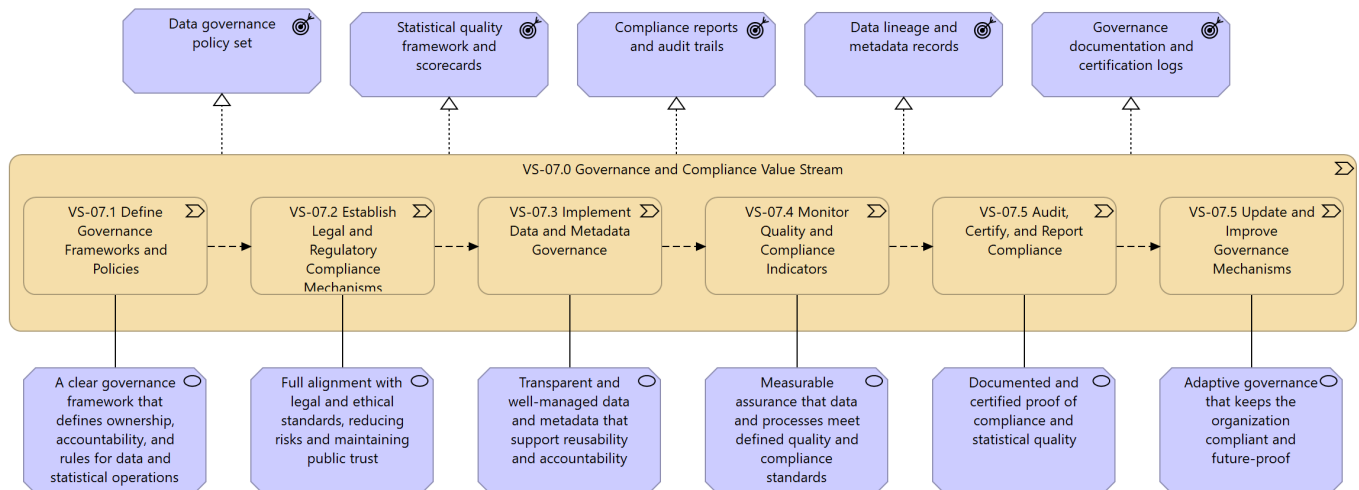
Elements

Element	Type
A clear dissemination plan that balances accessibility, compliance and timing	Value
A feedback-driven improvement cycle that keeps data services relevant, accessible, and impactful	Value
Informed, confident users who can correctly understand and apply statistical data	Value
Published dataset, statistical release, or data service accessible to users.	Outcome
Secure and well-prepared data that is safe for public or controlled release	Value
Trusted access to official statistics and data services through various channels	Value
VS-06.0 Dissemination and Service Delivery Value Stream	Value Stream
VS-06.1 Plan Dissemination and Access Strategy	Value Stream
VS-06.2 Prepare Data for Dissemination	Value Stream
VS-06.3 Publish and Provide Access Services	Value Stream
VS-06.4 Support and Engage Users	Value Stream
VS-06.5 Monitor Usage and Feedback	Value Stream

VS-07.0 Supporting Value Stream: Governance and Compliance

No viewpoint

Purpose: Ensure that all activities within the statistical organization are performed within a coherent framework of legal, ethical, methodological, and quality standards.



Elements

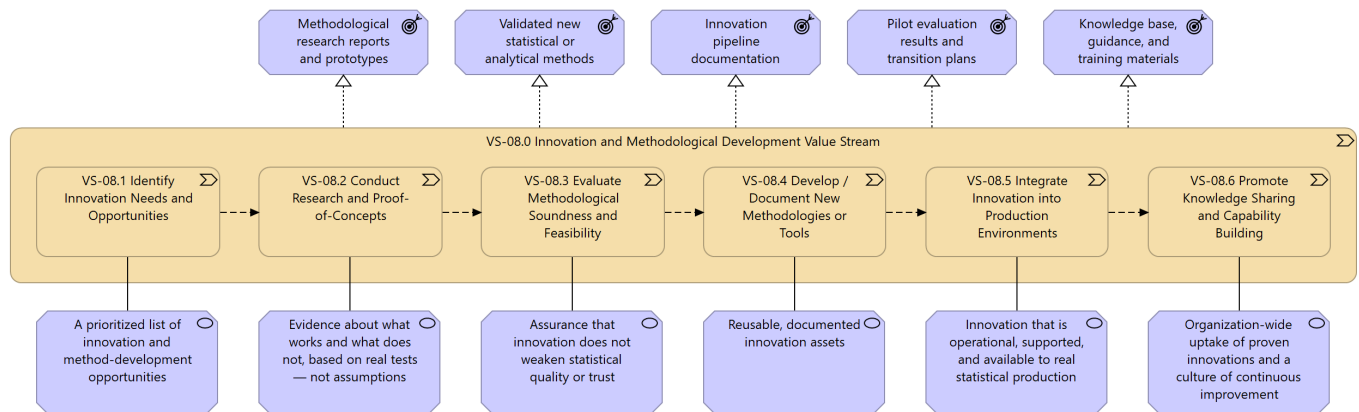
Element	Type
A clear governance framework that defines ownership, accountability, and rules for data and statistical operations	Value
Adaptive governance that keeps the organization compliant and future-proof	Value
Compliance reports and audit trails	Outcome
Data governance policy set	Outcome
Data lineage and metadata records	Outcome
Documented and certified proof of compliance and statistical quality	Value
Full alignment with legal and ethical standards, reducing risks and maintaining public trust	Value
Governance documentation and certification logs	Outcome
Measurable assurance that data and processes meet defined quality and compliance standards	Value
Statistical quality framework and scorecards	Outcome
Transparent and well-managed data and metadata that support reusability and accountability	Value
VS-07.0 Governance and Compliance Value Stream	Value Stream
VS-07.1 Define Governance Frameworks and Policies	Value Stream
VS-07.2 Establish Legal and Regulatory Compliance Mechanisms	Value Stream
VS-07.3 Implement Data and Metadata Governance	Value Stream
VS-07.4 Monitor Quality and Compliance	Value Stream

Element	Type
Indicators	
VS-07.5 Audit, Certify, and Report Compliance	Value Stream
VS-07.5 Update and Improve Governance Mechanisms	Value Stream

VS-08.0 Supporting Value Stream: Innovation and Methodological Development

No viewpoint

Purpose: drives continuous improvement and modernization across the statistical organization.



Elements

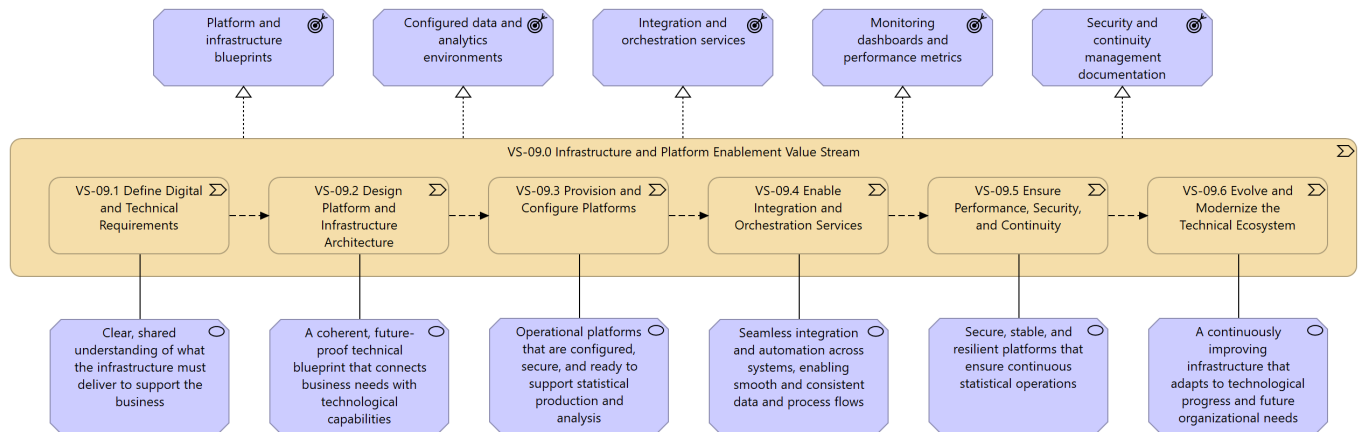
Element	Type
A prioritized list of innovation and method-development opportunities	Value
Assurance that innovation does not weaken statistical quality or trust	Value
Evidence about what works and what does not, based on real tests — not assumptions	Value
Innovation pipeline documentation	Outcome
Innovation that is operational, supported, and available to real statistical production	Value
Knowledge base, guidance, and training materials	Outcome
Methodological research reports and prototypes	Outcome
Organization-wide uptake of proven innovations and a culture of continuous improvement	Value
Pilot evaluation results and transition plans	Outcome
Reusable, documented innovation assets	Value
Validated new statistical or analytical methods	Outcome
VS-08.0 Innovation and Methodological Development Value Stream	Value Stream
VS-08.1 Identify Innovation Needs and Opportunities	Value Stream
VS-08.2 Conduct Research and Proof-of-Concepts	Value Stream
VS-08.3 Evaluate Methodological Soundness and Feasibility	Value Stream
VS-08.4 Develop / Document New Methodologies or Tools	Value Stream
VS-08.5 Integrate Innovation into Production Environments	Value Stream
VS-08.6 Promote Knowledge Sharing and Capability Building	Value Stream



VS-09.0 Supporting Value Stream: Infrastructure and Platform Enablement

No viewpoint

Purpose: Provides the digital and technical foundation that enables all other value streams to operate efficiently, securely, and at scale.



Elements

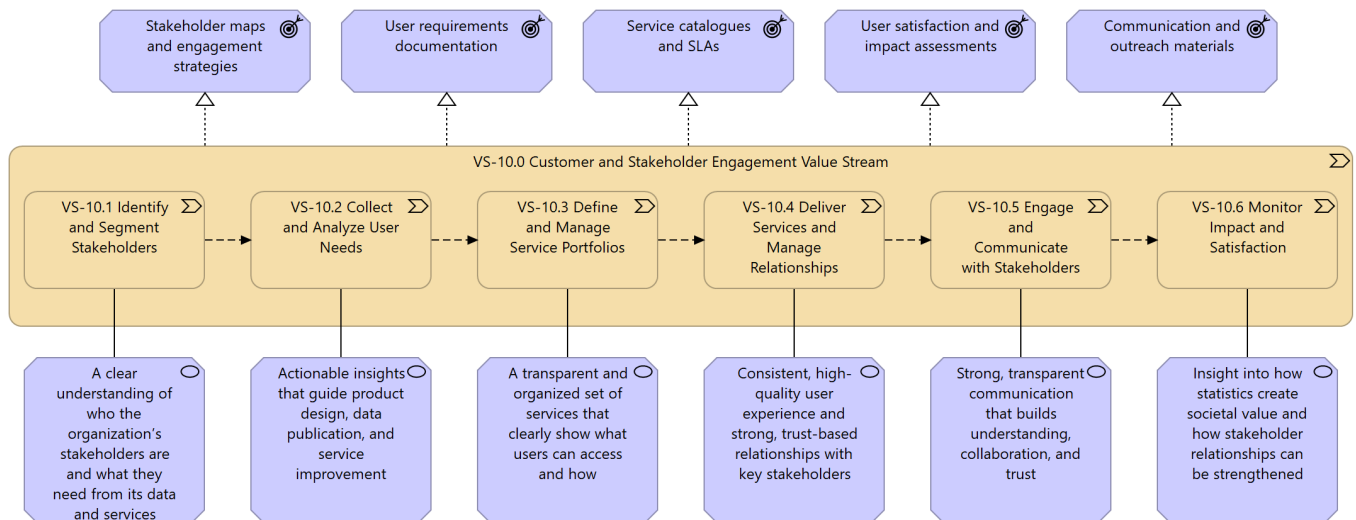
Element	Type
A coherent, future-proof technical blueprint that connects business needs with technological capabilities	Value
A continuously improving infrastructure that adapts to technological progress and future organizational needs	Value
Clear, shared understanding of what the infrastructure must deliver to support the business	Value
Configured data and analytics environments	Outcome
Integration and orchestration services	Outcome
Monitoring dashboards and performance metrics	Outcome
Operational platforms that are configured, secure, and ready to support statistical production and analysis	Value
Platform and infrastructure blueprints	Outcome
Seamless integration and automation across systems, enabling smooth and consistent data and process flows	Value
Secure, stable, and resilient platforms that ensure continuous statistical operations	Value
Security and continuity management documentation	Outcome
VS-09.0 Infrastructure and Platform Enablement Value Stream	Value Stream
VS-09.1 Define Digital and Technical Requirements	Value Stream
VS-09.2 Design Platform and Infrastructure Architecture	Value Stream

Element	Type
VS-09.3 Provision and Configure Platforms	Value Stream
VS-09.4 Enable Integration and Orchestration Services	Value Stream
VS-09.5 Ensure Performance, Security, and Continuity	Value Stream
VS-09.6 Evolve and Modernize the Technical Ecosystem	Value Stream

VS-10-0 Supporting Value Stream: Customer and Stakeholder Engagement

No viewpoint

Purpose: Ensures that the statistical organization remains connected to its users, partners, and data providers.



Elements

Element	Type
A clear understanding of who the organization's stakeholders are and what they need from its data and services	Value
A transparent and organized set of services that clearly show what users can access and how	Value
Actionable insights that guide product design, data publication, and service improvement	Value
Communication and outreach materials	Outcome
Consistent, high-quality user experience and strong, trust-based relationships with key stakeholders	Value
Insight into how statistics create societal value and how stakeholder relationships can be strengthened	Value
Service catalogues and SLAs	Outcome
Stakeholder maps and engagement strategies	Outcome
Strong, transparent communication that builds understanding, collaboration, and trust	Value
User requirements documentation	Outcome
User satisfaction and impact assessments	Outcome
VS-10.0 Customer and Stakeholder Engagement Value Stream	Value Stream
VS-10.1 Identify and Segment Stakeholders	Value Stream
VS-10.2 Collect and Analyze User Needs	Value Stream
VS-10.3 Define and Manage Service Portfolios	Value Stream

Element	Type
VS-10.4 Deliver Services and Manage Relationships	Value Stream
VS-10.5 Engage and Communicate with Stakeholders	Value Stream
VS-10.6 Monitor Impact and Satisfaction	Value Stream

Strategy Layer

1. Primary statistical capabilities

Type	Capability
archimate_element_identifier	EAID_F0239B39_14D6_4dd7_9F85_8F135A24F9E3
archimate_element_identifier	EAID_16E37B1B_7533_4827_B728_D926D688AC64
Capability level	0

Description: Contains all capacities required to perform the legally required tasks of a Statistical Institute

Explanation: -

Source: GAMSO 1.2, Production

1.1. Statistical design and build

Type	Capability
archimate_element_identifier	EAID_4E10FFB4_A285_4523_AA78_12AC1DA6BA74
archimate_element_identifier	EAID_6B81148A_F8CB_4c16_BC77_8D2B5C042999
Capability Level	1

Description: The ability to map information needs and set up statistical processes for this purpose.

Explanation: -

Source: GSBPM 5.2, Design

Link: <https://unece.github.io/GSBPM-5.2/#design-phase>

1.1.1 Needs inventory

Type	Capability
archimate_element_identifier	EAID_155083E4_6235_4ce3_9CE0_8E72B0988382
archimate_element_identifier	EAID_C219DA73_38F5_404c_9F8D_3D412C966DD5

Description: The ability to map out the information requirement (demand for information) and translate it into social tasks that we want to respond to (supply)

Explanation: We often do this through strategic relationships, see the social tasks mentioned in the MJP and the process preceding it.

Source: GSBPM 1.2

1.1.2. Design statistics product

Type	Capability
archimate_element_identifier	EAID_47DED069_E2BC_4720_B513_E1E6F2AE4D8B
archimate_element_identifier	EAID_E8D1F2F5_C38B_431f_9E21_52F2FFCC4586
Capability Level	2

Description: the ability to translate information needs into concrete designs

consisting of datasets, process and workflow models and rule sets/methodology.
This for the datasets that are released

Explanation: This concerns both the statistical product itself and the process that produces this product. Components of the process are various process steps, rule sets for the various process steps, methodology used, the process flow, the designs of the required datasets and the designs for the shared connection points for reuse. If the required datasets are not available, an observation assignment can follow. For primary observation, designs can then follow regarding sampling, questionnaire, etc.

Source: GSBPM

1.1.3. Build statistics product

Type	Capability
archimate_element_identifier	EAID_798B6B31_2C11_4998_932E_6132D2974311
archimate_element_identifier	EAID_3706D074_58D7_48b1_88B1_3A788C693380
Capability Level	2

Description: Building, testing, and managing a statistical product to the point where the product lifecycle ends

Explanation: The output from the "Design" phase is assembled and configured in this phase to create the complete operational environment to execute the process.

Source: GSBPM

1.2. Data collection

Type	Capability
archimate_element_identifier	EAID_382F11F1_7DB0_4581_A63E_F54470510477
archimate_element_identifier	EAID_52FDEFC3_BBA8_4d1e_92B4_920170CE895D
Capability Level	1

Description: The ability to collect data for official statistics.

Explanation: -

Source: GSBPM 5.2, Collect

Link : <https://unece.github.io/GSBPM-5.2/#collect-phase>

1.2.1. Primary data collection

Type	Capability
archimate_element_identifier	EAID_835D4A7F_193B_4cc1_9018_EBEB3B18E6E6
archimate_element_identifier	EAID_CA310974_379F_4911_8D3B_915C2C122259
Capability Level	2

Description: Primary observation describes the ability to set up and manage surveys, through to completion and transfer for processing and analysis. Primary observation collects both the data and the associated metadata.

Explanation: Primary observation is a domain within data collection and performs observations via one or more modes. The research population and the size of the sample are determined together with the client based on the needs. A distinction is

made between personal surveys and company surveys. Components of surveys include samples, questionnaires, approach strategies
Source: ESS capability model

1.2.2. Secondary data collection

Type	Capability
archimate_element_identifier	EAID_3A2A54E3_639C_4c60_8FF6_B881B7E9E28B
archimate_element_identifier	EAID_1033BE51_678B_4860_9AD9_2DF97852C5E1
Capability Level	2

Description: Secondary observation is the activity that describes the work required to collect data and associated metadata from third parties for the production of statistics.

Explanation: Within secondary observation, sources with data collected by third parties are used as much as possible.

Source: ESS capability model

1.3. Processing

Type	Capability
archimate_element_identifier	EAID_A5C4C347_6A82_483d_9409_D846184DAAEB
archimate_element_identifier	EAID_851565AE_148D_4f0d_A692_CA973F34C54D
Capability Level	1

Description: the ability to process raw data into statistical microdata and statistics.

Source : GSBPM5.2, Process

Link : <https://unece.github.io/GSBPM-5.2/#process-phase>

1.3.1. Data standardization

Type	Capability
archimate_element_identifier	EAID_6EDA9A76_47CB_4896_B17F_1618C7A950F0
archimate_element_identifier	EAID_3235770C_E7C7_49fc_B784_D681662A03FA
Capability Level	2

Description: the ability to standardize data without loss of content.

Explanation: This includes standardization of data format, codes for categories, (link) keys, possible units of measurement (for weight, distance, time). Data standardization without loss of content takes place immediately after receipt of the raw data because we want this standardization as early as possible in the processing process for efficiency reasons.

Source:

1.3.2. Deriving variables and units

Type	Capability
-------------	------------

archimate_element_identifier	EAID_C3F2E78A_DB03_4623_9775_57C934783FE9
archimate_element_identifier	EAID_F299FE7B_9611_4d48_9A8B_B8FB9C4B06C8
Capability Level	2

Description: the ability to create new units and values of new variables from existing units and values of existing variables

Explanation: We generally derive new variables with the aim of better describing reality; something the data source was not originally intended for.

Source:

1.3.3. Check content

Type	Capability
archimate_element_identifier	EAID_E8A4F1DA_0AF2_4e70_A4D9_C0269FB90BD8
archimate_element_identifier	EAID_2F069E1E_49DB_45fe_A9A6_AAF341D4E701
Capability Level	2

Description: the ability to check datasets for irregularities, such as missing data, possible errors and outliers.

Explanation: discovering possible errors can lead to adjustment, see also capacity "Adjust and impute", but can also lead to a quality indicator on the basis of which quality reports can be made. Discovering outliers leads to an 'outlier indicator' and is important for a good estimation method, see also capacity "Estimate population parameter".

Source: GSBPM

1.3.4. Adjusting and imputing

Type	Capability
archimate_element_identifier	EAID_38590372_4D2A_4428_BC90_3222D744B94C
archimate_element_identifier	EAID_17BB682F_D827_40a3_A2CA_FB187AB0290A
Capability Level	2

Description: the ability to adjust possible incorrect values and missing values.

Explanation: together with the "Content check" capacity, this capacity is often called editing. The check is then carried out using editing rules. The adjustment is then often based on imputation. The imputed values do not yet have to comply with the editing rules. To this end, a final 'adjustment' often takes place with the editing rules as a restriction.

Source: GSBPM, ESS capability model

1.3.5. Estimating population parameters

Type	Capability
archimate_element_identifier	EAID_C5A79B02_9C3B_4a58_AD55_23F08E70C05A
archimate_element_identifier	EAID_604050F2_23D1_419f_B66D_3C5FFB41A9D1
Capability Level	2

Description: the ability to estimate population parameters from microdata.
 Explanation: estimated population parameters often manifest themselves as observations (the cells) in a dimensional dataset (table). There are several estimation methods, including the traditional 'weight and aggregate'. There are several techniques to arrive at the weights. Sometimes weights are adjusted to give found outliers, see capacity "Content check" a smaller weight.
 Source: B&I architecture, CIO office

1.3.6. Integrate statistics

Type	Capability
archimate_element_identifier	EAID_047B0D64_6983_4ef1_8FE9_142DC84346C4
archimate_element_identifier	EAID_4A9E2EF1_2CFC_49c6_B255_9E05B4A2890C
Capability Level	2

Description: the ability to integrate separate statistics into one consistent whole.
 Explanation: We know integration systems for, for example, the national accounts, sector accounts and labour accounts
 Source:

1.3.7. Statistical security

Type	Capability
archimate_element_identifier	EAID_1E2470A6_7CAC_4fab_938F_AA717BC96C50
archimate_element_identifier	EAID_EF402C35_BD2D_4bce_94BC_9B99021C9437
Capability Level	2

Description: the ability to check datasets for privacy sensitivity and to adjust them in case of violation.
 Explanation: see the statistical security manual.
 Source: GSBPM 6.4

1.4. Analysis

Type	Capability
archimate_element_identifier	EAID_0B3327A9_E5F8_4350_91F6_DB7E882566AB
archimate_element_identifier	EAID_6DE22443_D575_4161_A460_DCD4803F47EC
Capability Level	1

Description: The ability to examine and understand data before it is disseminated.
 Explanation: Within statistical institutes, statistical analysis is also known as statistical research in which certain connections or conclusions can be drawn on the basis of relationships (such as correlation).
 Source: GSBPM 5.2, Analyse
 Link : <https://unece.github.io/GSBPM-5.2/#analyse-phase>

1.4.1. Statistical (output) analysis

Type	Capability
-------------	------------

archimate_element_identifier	EAID_10FDA9B4_CE49_43f2_AD72_3B7BD225C804
archimate_element_identifier	EAID_29A8770D_9F8C_4bd4_A439_69F1E355AF08
Capability Level	2

Description: The ability to validate output, interpret and explain statistical data and finalize output for dissemination using shared methodologies and processes.

Explanation: -

Source: ESS capability model

1.5. Dissemination

Type	Capability
archimate_element_identifier	EAID_30908CD3_2C09_4649_AA00_E68B52B1A6A6
archimate_element_identifier	EAID_566D08EA_E938_462e_A8E8_D85E098C2457
Capability Level	1

Description: The ability to manage the release of statistical products to users.

Explanation: This applies to both internal and external users.

Source: GSBPM 5.2, Disseminate

Link: <https://unece.github.io/GSBPM-5.2/#disseminate-phase>

1.5.1. Publication management

Type	Capability
archimate_element_identifier	EAID_C94D4D5B_878E_40eb_AEB2_9EF293F84DDD
archimate_element_identifier	EAID_49F98CB7_564F_4019_9094_001C777A6636
Capability Level	2

Description: The ability to manage the release of statistical output and associated content according to release schedules, so that users have predictable and equal access to data.

Explanation: -

Source: ESS capability model

1.5.2. Promotion to use

Type	Capability
archimate_element_identifier	EAID_FD089483_30D5_4a3d_AF6B_B81A18A70F6F
archimate_element_identifier	EAID_BD5725CB_7C2C_4e58_9747_B08D7C76D577
Capability Level	2

Description: The ability to promote the statistical output to potential users and to inform the press and other interested parties about the statistical output.

Explanation: -

Source: ESS capability model

1.5.3. Facilitate data access

Type	Capability
archimate_element_identifier	EAID_210DD12F_28D7_495a_868E_2454304959F5
archimate_element_identifier	EAID_FD5EA5E2_B444_4457_98E4_551580C66521
Capability Level	2

Description: The ability to make statistical data and metadata available to humans and machines through multiple channels.

Explanation: -

Source: ESS capability model

1.5.4. Statistical information management

Type	Capability
archimate_element_identifier	EAID_68CD439E_D29B_4d52_A820_7320994EFE4A
archimate_element_identifier	EAID_112A4A7E_756A_4a4e_BE79_A91028ADBDF1
Capability Level	2

Description: The ability to prepare and distribute publications related to the statistical output.

Explanation: This includes press releases, interpretations and reports.

Source: ESS capability model

2. Supporting Data capabilities

Type	Capability
archimate_element_identifier	EAID_4ED7C8CA_0E6D_4577_B617_6B5FFDE929B6
archimate_element_identifier	EAID_40B31561_493C_43d9_9E73_200B5DE17A82
Capability Level	0

Description: Includes additional statistical capabilities such as managing statistical data and conducting statistical research.

Explanation: -

Source: GAMSO 1.2, Capability Development

2.1. Data management

Type	Capability
archimate_element_identifier	EAID_9457C915_FE11_4d4b_A2B3_56965772E961
archimate_element_identifier	EAID_F7D5198E_E631_40f6_8D62_4451650882DD
Capability Level	1

Description: The ability to manage data and metadata.

Explanation: -

Source:

2.1.1. Data lifecycle management

Type	Capability
archimate_element_identifier	EAID_6C948EE4_3D9A_42f0_92D4_2E90056BECA9
archimate_element_identifier	EAID_5F4804DE_F95A_4bfe_92E9_97DE2F1894CF
Capability Level	2

Description: The ability to manage data throughout its entire lifecycle.

Explanation: -

Source: CIO-office

2.1.2. Metadata management

Type	Capability
archimate_element_identifier	EAID_F38D273E_6E79_4f18_AA91_EF53BC76CC67
archimate_element_identifier	EAID_43494633_0A45_482b_8A25_4D7D81CF62B6
Capability Level	2

Description: The ability to manage metadata throughout the lifecycle.

Explanation: -

Source: ESS Capabilities, CIO-office

2.1.3. Statistical quality management

Type	Capability
archimate_element_identifier	EAID_AA26E33C_B81E_4ac5_AD8B_E1E1A82858FD
archimate_element_identifier	EAID_D8C6E0F5_7DFB_49e5_9F36_584EBF412727
Capability Level	2

Description: The ability to perform quality assessments, implement control mechanisms and initiate quality improvement mechanisms for the statistical value chain.

Explanation: -

Source: CIO-office

2.1.4. Statistical register management

Type	Capability
archimate_element_identifier	EAID_687F1047_A2F2_4f8b_9144_819E513DE4B3
archimate_element_identifier	EAID_230E2475_604B_4356_8048_738E2DE56AFD
Capability Level	2

Description: The ability to set up, maintain and make available "registry" services for collecting and integrating data.

Explanation: Some examples are: EHO, SBB, ABR

Source: ESS capability model

2.2. Research

Type	Capability
archimate_element_identifier	EAID_568414F1_E1E5_4232_B164_920FEEEF29EB
archimate_element_identifier	EAID_8C47D894_4911_4899_B848_FDFA59A98FC0
Capability Level	1

Description: Includes acquiring and recording knowledge in relation to statistical methods, products, sources and techniques.

Explanation: -

Source: GAMSO 1.2, Develop Capability Improvements

2.2.1. Data source exploration

Type	Capability
archimate_element_identifier	EAID_95A5F99A_81FC_4352_A891_D0BB176C6886
archimate_element_identifier	EAID_253C89EA_CD07_4f3f_8C6A_2EC6110D7B48
Capability Level	2

Description: Exploring new data sources

Explanation: -

Source: ESS, CIO-office

2.2.2. Product Innovation

Type	Capability
archimate_element_identifier	EAID_68C985EA_36CB_4966_9997_75E97FF21447
archimate_element_identifier	EAID_135FF6C7_1E33_4eea_A162_ADE2511A513C
Capability Level	2

Description: The ability to innovate, i.e. create new statistical products based on existing data sources and explore new data sources that are useful and important to users.

Explanation: A design phase may be preceded by product innovation.

Source: ESS

2.2.3. Methodology

Type	Capability
archimate_element_identifier	EAID_49989740_FEF4_4140_A151_A85C81032A51
archimate_element_identifier	EAID_1C4E4906_54FD_4226_AC6D_2D1BF35E3336
Capability Level	2

Description: The ability to investigate new methods for the statistical process.

Explanation: -

Source: CIO-office

2.2.4. Data services

Type	Capability
archimate_element_identifier	EAID_2AB5B054_4AD4_433a_B7B5_68751F93C5B2
archimate_element_identifier	EAID_6A97AA57_8D61_4edf_A1E2_3D322E6A3DB5
Capability Level	2

Description: the ability to provide data for statistical or scientific research.

Explanation: This applies to both internal and external datasets. These services are realized with, among other things, Microdata services.

Source: CBS Law article 41

2.2.5. User research

Type	Capability
archimate_element_identifier	EAID_3426A9FA_BA04_4eb9_A239_4DA17FDF203D
archimate_element_identifier	EAID_2E214A33_5996_4876_8BA0_8EE42ED7338C
Capability Level	2

Description: The ability to collect, assess and translate user needs into statistical output

Explanation: -

Source: CIO-office

3. Supporting business capacities

Type	Capability
archimate_element_identifier	EAID_2F52C7EA_AC8B_4d00_BAFC_596F0A0D33ED
archimate_element_identifier	EAID_C7003D58_E0F7_4e03_B56D_D04B821510A0
Capability Level	0

Description: Includes the capabilities that ensure that the right people, information and resources are available to implement and support the statistical domains and the governing domain.

Explanation: -

Source: GAMS0 1.2, Corporate support

3.1. Business operations

Type	Capability
archimate_element_identifier	EAID_16162BB1_E272_4838_878C_8C29A145AA79
archimate_element_identifier	EAID_DF4E1E05_4B67_49e7_89CD_78C91A1C2D09
Capability Level	1

Description: Contains all activities required for managing personnel, purchasing & contracts, finances and housing and the associated systems and communication.

Explanation: -

Source: CEO

3.1.1. Personnel management (HR)

Type	Capability
archimate_element_identifier	EAID_D3D96F8A_F3D0_47d3_BB71_BD0B3B0A6E8E
archimate_element_identifier	EAID_41340930_A10D_4c5e_9A1C_CB950F3EF251
Capability Level	2

Description: Ensuring that sufficient qualified employees are available to carry out all business processes.

Explanation: -

Source:

3.1.2. Purchasing & contract management

Type	Capability
archimate_element_identifier	EAID_57A6610D_A9AE_4199_BF24_355895FBC6C3
archimate_element_identifier	EAID_0E90D6A9_66E6_4225_AD49_5634731ADF27
Capability Level	2

Description: Acquiring goods and services and monitoring the agreements made with suppliers.

Explanation: -

Source:

3.1.3. Financial management

Type	Capability
archimate_element_identifier	EAID_29F61E1F_9080_409b_BB89_D782993A5807
archimate_element_identifier	EAID_4DC78E14_9ADA_43f4_BFD2_E22C228AA969
Capability Level	2

Description: Includes the planning, organization, design, recording and control of the financial activities of the organization.

Explanation: -

Source: CIO-office

3.1.4. Communication management

Type	Capability
archimate_element_identifier	EAID_B67643B0_4405_420a_BC9E_D3FD10E69CF1
archimate_element_identifier	EAID_32261CB2_CBEB_4b68_8C31_FBE8AA4C1F05
Capability Level	2

Description: Ensuring that the organization communicates correctly to internal and external parties.

Explanation: -

Source:

3.1.5. Administrative support

Type	Capability
archimate_element_identifier	EAID_6206B8DC_1107_4380_8B6E_1322D678152E
archimate_element_identifier	EAID_24CDFDFD_376A_46d9_BF40_943E61366339
Capability Level	2

Description: Providing support with administrative tasks.

Explanation: -

Source:

3.1.6. Facility management

Type	Capability
archimate_element_identifier	EAID_8283AFE7_6205_4e8d_A72C_5EB1697DD69C
archimate_element_identifier	EAID_BA7FE5A8_6A8F_4358_A764_121A28A1CC80
Capability Level	2

Description: Organizational function that integrates people, place and process within the built environment.

Explanation: Description based on ISO 41011. The purpose of facility management is to improve the quality of life of people and the productivity of the core activity.

Source: ISO 41011

3.1.7. Legal support

Type	Capability
archimate_element_identifier	EAID_96E1E443_5FB6_4970_804B_922DC7FE0D6D
archimate_element_identifier	EAID_61484FAE_AE40_4a2e_A82F_65F46F4A0090
Capability Level	2

Description: Handling requests and applications, assessing agreements and advising on relevant legislation and regulations.

Explanation:

Source:

3.1.8. Information management and archive management

Type	Capability
archimate_element_identifier	EAID_B25CF2A8_E57A_41ad_A7B4_435702BE4A0C
archimate_element_identifier	EAID_18BDEE5F_09BC_410b_93BA_8D10CCC01A87
Capability Level	2

Description: Includes the activities to bring, maintain and preserve the

organization's information in a state of sustainable accessibility (FAIR).
 Explanation: This concerns information created in the supporting and management processes, not statistical information from the primary CBS process.
 Source:

3.2. Computerization and automation

Type	Capability
archimate_element_identifier	EAID_1353E4FD_A6E4_458c_A9A9_1D03BE703B4C
archimate_element_identifier	EAID_B1A8265C_39AC_40b7_A8E1_C552E46C6ADE
Capability Level	1

Description: Ensuring that the information needs of the organization are translated into desired functionalities of the information provision and ensuring that the information systems are available to support the business processes.

Explanation:

Source: GAMS0 1.2, Manage IT

3.2.1. Application development

Type	Capability
archimate_element_identifier	EAID_466167EA_1EAD_4a3b_8223_C7BEA5A83D3A
archimate_element_identifier	EAID_6355C941_B54F_450b_B7B3_87C2797E9A5F
Capability Level	1

Description: The ability to develop applications yourself.

Explanation: This skill is needed to fill gaps in functionality that are missing in the statistical process.

Source:

3.2.2. Functional management

Type	Capability
archimate_element_identifier	EAID_14B43D53_1BD9_443b_80D4_4FE292A28AD9
archimate_element_identifier	EAID_15D5DBBE_B822_4487_B49C_EC7B0066F93C
Capability Level	2

Description: Performing management tasks so that applications optimally support business processes.

Explanation: For example, by collecting existing user requirements and translating them into new application functionality.

Source:

3.2.3. Technical management

Type	Capability
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archimate_element_identifier	EAID_78FB96EF_460E_46a7_83FF_31E46F422D40
archimate_element_identifier	EAID_8DBC510D_3A0A_4db4_ACE6_FD55E693E184
Capability Level	2

Description: Performing management tasks regarding system software and applications.

Explanation: Installing and configuring hardware and software. Managing this hardware and software to ensure that it continues to function correctly, including all measures to help resolve disruptions in a timely manner.

Source:

3.2.4. Digital workplace management

Type	Capability
archimate_element_identifier	EAID_03DF985C_4551_44f8_8087_28FC5CB46820
archimate_element_identifier	EAID_DFF72ADC_9DA3_4cc9_A738_C0DF5BDC72FC
Capability Level	2

Description: Management of the resources and services that the employee uses for his digital workplace.

Explanation: Examples are: (Develop) Laptop / Desktops, Mobile devices, Virtual desktops, Video / audio conferencing, Printing, VPN, proxy services (for the benefit of the workplace).

Source:

3.2.5. IT infrastructure

Type	Capability
archimate_element_identifier	EAID_C70F5512_83E9_4d6e_B32C_1A73F5FE02ED
archimate_element_identifier	EAID_98AB0083_EDBC_4fa9_98F8_5965E68343CA
Capability Level	2

Description: Developing, maintaining and managing IT infrastructure

Explanation:

Source:

3.3. Security and Privacy

Type	Capability
archimate_element_identifier	EAID_E3CF79C3_C2F7_4fbe_B9DE_A154BB63B827
archimate_element_identifier	EAID_2B240688_886C_4a0f_AA3E_4A6C5D3EE7C2
Capability Level	1

Description: Contains all security and privacy related activities

Explanation: -

Source:

3.3.1. Privacy management

Type	Capability
archimate_element_identifier	EAID_DF55AD61_CC51_4e9e_9424_02B3F86256BA
archimate_element_identifier	EAID_5064C160_BC0C_4391_85A9_A6460E95D5D8
Capability Level	2

Description: Developing, managing, applying and testing the privacy policy of Statistics Netherlands in order to comply with, among other things, legislation and regulations in the field of privacy protection.

Explanation: -

Source:

3.3.2. Security planning

Type	Capability
archimate_element_identifier	EAID_5073D0AB_3BBC_4c65_B983_2F679BA21BA6
archimate_element_identifier	EAID_08DC37A5_EEDE_4cf5_9040_A33C9324A525
Capability Level	2

Description: Includes the activities of identifying, analyzing, and planning measures to ensure the security of a system, network, organization, and facility.

Explanation: -

Source:

3.3.3. Security incident management

Type	Capability
archimate_element_identifier	EAID_C490B350_7274_4914_9E26_62EFE8DD7450
archimate_element_identifier	EAID_BE041ADA_61B3_45c0_83D7_D28AA0FFCCAE
Capability Level	2

Description: Includes the activities of identifying, analyzing, responding to, and recovering from security incidents within an organization.

Explanation: -

Source:

3.3.4. Logical identity and access control

Type	Capability
archimate_element_identifier	EAID_B9A8B8C1_E4D7_440b_A354_E8EA3E98A3B7
archimate_element_identifier	EAID_7BCF6889_AF46_4db8_B159_5F15759DBEBC
Capability Level	2

Description: Includes all activities that ensure that a person, organization or IT facility can only use automated functions for which it has been granted access rights through an application process.

Explanation: -

Source:

3.3.5. Physical security & access control

Type	Capability
archimate_element_identifier	EAID_97EB9EFF_4B65_4327_A30C_B82D04399DF7
archimate_element_identifier	EAID_AB0C1929_2BF6_444c_9695_D5635179E9B6
Capability Level	2

Description: Includes all activities that ensure controlled and safe access to buildings and spaces and that people and goods in these buildings and spaces are safe.

Explanation: -

Source:

3.3.6. IT Security services

Type	Capability
archimate_element_identifier	EAID_B1B85816_9FB6_4bcb_8AE4_18F124988D88
archimate_element_identifier	EAID_E61329B5_7D3A_49fa_9E81_5E9C561FAE15
Capability Level	2

Description: IT services that serve to support information security

Explanation: Examples include: Endpoint protection, PKI, network security management,

Source:

3.4. Customer and chain interaction

Type	Capability
archimate_element_identifier	EAID_383F20F6_798B_4c56_B294_CA1F0B1DA837
archimate_element_identifier	EAID_6A7A0C22_A859_466d_A346_84DC4FFD3BA4
Capability Level	1

Description: The ability to interact with the customer and chain partners, ensuring that they have the correct data and information and receiving notifications, requests and data.

Explanation:

Source: GAMS0 1.2, Manage consumers

3.4.1. Services management

Type	Capability
archimate_element_identifier	EAID_B7714DB9_FBD0_4112_9574_48E7B5C2EC4F
archimate_element_identifier	EAID_DE8570F8_C506_4226_B69E_72931581651E
Capability Level	2

Description: Includes the design, development, implementation and management of services that the organization (parts) provides to customers and chain partners.

Explanation: -

Source:

3.4.2. Contact management

Type	Capability
archimate_element_identifier	EAID_0378F9EE_732E_4518_99D4_AFFAD4E1E1F8
archimate_element_identifier	EAID_452455B5_040E_4a58_A32B_C6834D56ED85
Capability Level	2

Description: Maintaining relationships with customers and chain partners.

Explanation: -

Source: GSBPM 1.2,

Link:

<https://statswiki.unece.org/spaces/GAMSO/pages/247302304/Corporate+Support#CorporateSupport-GAMSOv1.2ManageConsumersManageConsumers>

3.4.3. Service Agreement

Type	Capability
archimate_element_identifier	EAID_5EDBE693_625D_4cbd_BAB9_BB7050D7840C
archimate_element_identifier	EAID_0F1F1C8A_5593_41f1_BB16_B91E93D00299
Capability Level	2

Description: Includes drawing up, managing and enforcing service agreements with clients and chain partners.

Explanation: These contracts apply to clients to whom CBS provides services such as Remote Access for external researchers or for Additional Statistical Services, but also to parties that supply information for primary or secondary observation. If these parties fail to comply, CBS can take enforcement action.

Source:

3.4.4. Self-service

Type	Capability
archimate_element_identifier	EAID_F89DD359_1871_452a_92B6_00D4788B168E
archimate_element_identifier	EAID_E4C9BBC6_0877_4622_9A58_24BFEBE0014D
Capability Level	2

Description: Offers the opportunity to customers and chain partners to find solutions to questions about the services purchased themselves or to make justified adjustments to these services without the intervention of an employee.

Explanation: -

Source:

4. Governance capabilities

Type	Capability
archimate_element_identifier	EAID_D20E66A9_A277_4c5b_B734_5401D2433DB8
archimate_element_identifier	EAID_72269981_6969_40ab_B766_491D8249D76A
Capability Level	0

Description: Includes the capabilities that provide direction to the organization by determining the desired strategy and the framework within which changes must take place.

Explanation:

Source: GAMS0 1.2, Strategy and Leadership

4.1.1. Strategy and governance

Type	Capability
archimate_element_identifier	EAID_37509EFB_9BB2_4f6d_A20D_05205930DF38
archimate_element_identifier	EAID_57E04237_50BE_4cdd_9DED_BFB300D0018A
Capability Level	1

Description: Developing a vision and setting up and monitoring the organization and its management.

Explanation: -

Source: GAMS0 1.2, Govern and Lead

4.1.2. Policy and planning

Type	Capability
archimate_element_identifier	EAID_3791B6E5_4437_477a_891E_603165FE852E
archimate_element_identifier	EAID_1A2C1E9D_0514_4e5a_829E_02B284DB6776
Capability Level	1

Description: Translating the strategy into more concrete objectives, principles and plans.

Explanation: -

Source:

4.1.3. Change management

Type	Capability
archimate_element_identifier	EAID_A8B4ADA3_EEC9_45a0_BE98_07A00C9CF5BE
archimate_element_identifier	EAID_892FE782_755D_4dee_B22C_4D3258CE3D5D

Description: Determining and managing major changes so that maximum contribution is made to the objectives.

Explanation: -

Source:

4.1.4. Improve management

Type	Capability
archimate_element_identifier	EAID_4180D2A7_ECB4_4d30_8F51_7602302EC948
archimate_element_identifier	EAID_23EDD12F_27A3_4f49_BC7E_2934E4C63A84
Capability Level	1

Description: Managing the daily operation and working on identifying areas for improvement and implementing improvement measures based on daily operations.

Explanation: -

Source:

4.1.5. Accountability

Type	Capability
archimate_element_identifier	EAID_E8113F80_6C4D_4606_ABDD_77E8BC04E3B3
archimate_element_identifier	EAID_11A25114_8CAF_42f5_B0B5_85A73B8AEE3C
Capability Level	1

Description: Reporting to stakeholders inside and outside the organization on the extent to which obligations and agreements are met.

Explanation: -

Source:

4.1.6. Collaboration

Type	Capability
archimate_element_identifier	EAID_21169A06_D3C8_4592_BDB9_03BE818EAB5C
archimate_element_identifier	EAID_F06DB547_3E59_4e0f_99EC_A860777FCCF7
Capability Level	1

Description: The ability to maintain and consolidate strategic relationships with external stakeholders.

Explanation: These activities involve collaboration and coordination with other statistical organisations and other external stakeholders. They may involve coordination within a statistical system, which may be based on a geographical hierarchy of entities (local, regional, national, multinational), or a division of responsibilities between organisations based on activities. They include activities undertaken to identify new opportunities for data exchange or integration. They provide opportunities for the statistical community to exchange knowledge, improve statistical infrastructure and practices, and influence statistical standards

Source: GAMS0 1.2, Manage Strategic Collaboration and Cooperation

VS-01.0 Data Acquisition and Ingestion Value Stream

Type	Value Stream
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The Data Acquisition & Ingestion Value Stream is the first step in the statistical value chain. It focuses on bringing raw data into the statistical system in a secure, traceable, and compliant way.

The purpose of this value stream is to make sure that data — no matter where the data comes from — enters the organization in a state that is authentic, well-documented, and ready for use. This value stream sets the foundation for everything that follows. It makes sure that the right data is collected, that legal requirements and ethical guidelines are observed, and that the data is safely stored and traceable from its source.

This value chain identifies the following value steps:

- Identify Data Needs and Sources: Define the statistical or analytical requirement and identify potential data sources (surveys, registers, administrative systems, sensors, commercial or open data).
- Establish Data Agreements and Access Rights: Negotiate and formalize data-sharing agreements, define legal and ethical frameworks, and ensure privacy and compliance.
- Collect or Receive Data: Execute data collection through surveys, APIs, secure data transfers, or data space connectors. Capture metadata during ingestion.
- Validate and Log Data Intake: Verify completeness, structural integrity, and compliance with expected formats; log the ingestion process for traceability.
- Store and Register Raw Data Artefacts: Persist data in controlled repositories, assign identifiers, and register associated metadata, ownership, and provenance.

The value created here lies in trust and readiness — trust that the data has been acquired correctly, and readiness for the data to move forward into later stages of the statistical process.

Once data has been safely acquired and registered, it is ready to move into the next phase - the Data Standardization & Integration Value Stream. Here, the focus shifts from collecting data to structuring it, ensuring that all incoming data can be compared, combined, and analyzed.

VS-01.1 Identify Data Needs and Sources

Type	Value Stream
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Every data journey begins by understanding what data is needed and where it can be found. In this step, analysts, subject-matter experts, and data stewards work together to define the statistical or analytical purpose. They clarify what questions must be answered and what information is required to do so. Next, they identify potential data sources — these may include surveys, administrative registers, external databases, sensors, or even open data portals. Each potential source is evaluated for relevance, quality, and accessibility. This step ensures that only meaningful and reliable data is considered. It also helps align the organization's data needs with available sources, avoiding unnecessary duplication or data collection.

Value created: Clear understanding of what data is needed and where it can be obtained.

VS-01.2 Establish Data Agreement and Access Right

Type	Value Stream
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Once the data sources are known, the organization must secure the rights and permissions to use them. This involves both legal and ethical considerations. Teams

negotiate and formalize data-sharing agreements with the data providers. These agreements define what data will be shared, how it will be used, who owns it, and how privacy will be protected. They ensure compliance with data protection laws (e.g., GDPR) and internal governance frameworks. At the same time, access rules and security measures are defined — specifying who can access which datasets, under what conditions, and for what purpose. This step guarantees that all future use of the data is both authorized and transparent.

Value created: Legal, ethical, and transparent access to data under controlled conditions.

VS-01.3 Collect or Receive Data

Type	Value Stream
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After agreements are in place, the actual data collection or transfer begins. Depending on the data type and source, this may happen through surveys, APIs, secure data transfers, or data space connectors. During this step, the organization focuses on data integrity and security. Data is transferred through controlled channels that ensure confidentiality and prevent tampering.

At the same time, metadata is captured — information about where the data came from, how it was collected, and under what conditions. This metadata provides context and traceability, linking the data back to its original source and collection method.

Value created: Secure and verifiable transfer of authentic data into the organization.

VS-01.4 Validate and Log Data Intake

Type	Value Stream
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Once the data has been received, it must be checked before it can be used. In this step, data managers validate the completeness, structure, and format of the incoming data. They verify that all expected files and records are present, that field types and values match the expected structure, and that the data follows the agreed-upon standards. Any inconsistencies or missing information are recorded for follow-up. Next, all intake activities are logged — documenting who received the data, when it was received, from which source, and under which agreement. This log creates an audit trail, ensuring that every dataset entering the system can be traced and verified later if needed.

Value created: Confidence that received data is complete, structured, and fully traceable.

VS-01.5 Store and Register Raw Data Artefacts

Type	Value Stream
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Finally, the validated data is securely stored in the organization's controlled repositories. Each dataset is assigned a unique identifier and linked with its metadata, ownership, and provenance information.

This step ensures that the data is not only stored safely but also discoverable and managed according to data governance rules. It becomes part of the organization's data inventory — the official record of all raw data available for processing and

analysis. Access permissions are applied according to the rules established earlier, ensuring that only authorized users can view or handle the data.

Value created: Secure, organized, and traceable storage of raw data, ready for standardization and further processing.

VS-02.0 Data Standardization and Integration Value Stream

Type	Value Stream
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The Data Standardization & Integration Value Stream is the second step in the statistical value chain. After raw data has been collected and securely stored, it now needs to be standardized, harmonized, and prepared for further processing. In this value stream, the organization ensures that data from different sources — surveys, administrative registers, or external datasets — can work together seamlessly. This is achieved through consistent structures, common formats, and shared definitions. The main goal is interoperability: data that looks different at the source must be aligned so it can be understood, compared, and reused across systems, domains, and time periods.

This value chain identifies the following value steps:

- Profile and Assess Data Structure: Analyze data structure, identify inconsistencies, and align schema with target models.
- Apply Standard Classifications and Formats: Map variables, codes, and identifiers to standardized taxonomies and reference data (e.g., NACE, ISCO, SDMX, GSIM concepts).
- Align Metadata and Semantics: Harmonize descriptive metadata, ensuring coherence in definitions, classifications, and quality descriptors.
- Validate Structural and Semantic Conformance: Execute validation routines to ensure the data conforms to standards and classifications.
- Register Standardized Artefacts: Store the standardized dataset and update the metadata catalogue to reflect structural and semantic alignment.

The value created here is coherence and comparability — transforming raw data into a structured and standardized asset that can flow smoothly into editing, analysis, and modeling.

Once data is standardized and validated, it can safely move to the Data Editing & Enrichment Value Stream. In that stage, attention shifts from structure and standards to quality and completeness. The organization begins improving data accuracy, filling gaps, and enriching datasets with derived or corrected information.

VS-02.1 Profile and Assess Data Structure

Type	Value Stream
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Before any harmonization can begin, the data must be understood in detail. In this step, teams profile the dataset to identify its structure, content, and potential issues. They look at variables, data types, field names, coding systems, and metadata to see how they compare to existing standards. They assess whether the dataset uses consistent formats, whether codes and classifications are aligned, and whether any transformations will be needed. This step provides a clear picture of the “as-is” structure of the data, serving as the foundation for all subsequent standardization efforts.

Value created: Full understanding of the dataset's structure, format, and readiness for integration.

VS-02.2 Apply Standard Classifications and Formats

Type	Value Stream
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Next, the organization begins to align the data with established standards. This means mapping local variables and codes to standardized classifications, such as NACE for industries, ISCO for occupations, or other domain-specific taxonomies. Data formats are also standardized — dates, numbers, and identifiers are converted to consistent patterns that match the organization's reference frameworks. This ensures that the data can be combined with other datasets or reused in future analyses without additional reformatting. By using common standards, the organization makes sure that data can be shared and understood across internal and external systems.

Value created: Consistent, interoperable data aligned with recognized standards.

VS-02.3 Align Metadata and Semantics

Type	Value Stream
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Once structural and format alignment is complete, the next step focuses on semantic alignment — ensuring that data means the same thing across different contexts. Teams review and adjust metadata to confirm that definitions, measurement units, and concepts are consistent with organizational and international standards (e.g., GSIM, SDMX). They ensure that variable names, descriptions, and reference metadata reflect a shared understanding of what each data element represents. This step is critical for semantic interoperability, allowing data from different sources to be correctly interpreted and compared without ambiguity.

Value created: Harmonized meaning and context across datasets — a shared language for all data.

VS-02.4 Validate Structural and Semantic Conformance

Type	Value Stream
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After standardization and alignment, the data must be validated to ensure everything is correct and complete. Validation tools check that each dataset conforms to the defined standards, classifications, and metadata rules. Automated validation routines flag missing codes, invalid values, or inconsistent classifications. If discrepancies are found, they are reviewed and corrected before the dataset moves forward. This validation step ensures that every dataset entering the next stage of the statistical process is both technically and semantically sound.

Value created: Verified and quality-assured data that meets all structural and semantic standards.

VS-02.5 Register Standardized Artefacts

Type	Value Stream
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Finally, once data is validated, it is registered and stored as a standardized artefact within the organization's controlled repositories. Each dataset is given a version identifier, linked with its metadata, and catalogued for discovery and reuse. The registration process ensures traceability — it documents how the data was transformed, what standards were applied, and where it originated.

This makes it possible to track lineage and maintain consistency across time and versions. At this point, the data is ready to move into the next phase: Data Editing & Enrichment — where its quality and completeness will be further refined.

Value created: Traceable, reusable, and standardized dataset ready for editing and analysis.

VS-03.0 Data Editing and Enrichment Value Stream

Type	Value Stream
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The Data Editing & Enrichment Value Stream is the third step in the statistical value chain. After data has been standardized and aligned, the next task is to improve its quality, completeness, and usability.

At this stage, the organization focuses on cleaning, correcting, and enriching data so that it becomes trustworthy and analytically sound. Errors are detected, inconsistencies are corrected, missing values are filled, and new variables are derived to add more meaning and analytical value. This is where raw, standardized data begins to transform into high-quality statistical input — ready for estimation, modeling, and analysis.

This value chain identifies the following value steps:

- Conduct Error Detection and Quality Diagnostics: Identify logical errors, duplicates, missing values, or outliers through rule-based or machine-learning validation.
- Perform Data Editing and Correction: Apply predefined rules and algorithms to correct errors or replace invalid entries.
- Impute Missing or Non-Response Data: Estimate missing values using statistical imputation techniques or external data sources.
- Derive Additional Variables and Indicators: Enrich datasets by calculating derived measures or additional analytical variables.
- Perform Quality Evaluation and Versioning: Assess improvement in data quality, document changes, and create traceable versions of edited datasets.

The main value created here is data reliability and analytical readiness. By systematically improving data quality, the organization ensures that every subsequent result — every estimate, model, and publication — rests on a solid foundation.

VS-03.1 Conduct Error Detection/ Quality Diagnostics

Type	Value Stream
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The process begins with examining the data for errors and inconsistencies. Teams use automated checks, business rules, and sometimes machine learning models to identify suspicious or incorrect values. These may include impossible combinations (e.g., negative ages), missing responses, duplicate records, or outliers. The data is also tested against expected patterns, reference data, and statistical distributions

to detect anomalies. Each detected issue is flagged and categorized — for example, by type (missing value, outlier) or severity (critical, minor). This diagnostic step builds an accurate picture of the dataset's current quality state, showing where corrections or enrichments are needed.

Value created: Clear understanding of data quality issues and areas requiring correction.

VS-03.2 Perform Data Editing and Correction

Type	Value Stream
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Next, the organization focuses on correcting identified errors. This involves systematically updating incorrect or inconsistent values to restore internal data coherence. Corrections may be applied automatically using predefined rules, or manually when expert judgment is needed. For example, a misreported code may be replaced with a valid one, or inconsistent entries may be harmonized with reference data. Every correction is documented and versioned, ensuring that data lineage and accountability are maintained. This step ensures that datasets are internally consistent, eliminating structural or logical defects before further processing.

Value created: Clean, coherent data that accurately represents real-world information.

VS-03.3 Impute Missing or Non-Response Data

Type	Value Stream
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After correcting known errors, the next challenge is missing information. Incomplete data can reduce the reliability of statistical analysis, so this step aims to fill gaps responsibly. The organization uses statistical imputation techniques — such as mean substitution, regression, or model-based estimation — to replace missing or non-response values. When possible, external sources or related datasets are used to enrich or cross-check the imputed information. All imputed values are flagged to ensure transparency and to differentiate between observed and derived data. By carefully imputing missing data, the organization restores completeness and analytical balance to the dataset.

Value created: Complete datasets that preserve statistical representativeness and minimize bias.

VS-03.4 Derive Additional Variables and Indicators

Type	Value Stream
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Once errors are corrected and gaps filled, the data can be enriched with new variables. This involves deriving new attributes or indicators that enhance analytical potential — for example, calculating age groups from birthdates, income brackets from earnings, or regional indicators from addresses. These derived variables help make the dataset more meaningful and easier to use in analysis and modeling. They also allow for consistent comparisons across studies, time periods, or regions. This step transforms the dataset from a purely factual record into a structured analytical resource.

Value created: Enriched datasets that support deeper analysis and insight generation.

VS-03.5 Perform Quality Evaluation and Versioning

Type	Value Stream
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Finally, after all editing, correction, and enrichment, the dataset undergoes a quality evaluation. This step checks whether the actions taken have improved data accuracy, consistency, and completeness.

Teams compare pre- and post-editing quality metrics to confirm that known issues have been resolved. Documentation is updated, and a new version of the dataset is created — fully traceable and ready for use in estimation or modeling. This versioned dataset becomes the organization's official edited dataset, forming the trusted baseline for statistical computation.

Value created: Verified, high-quality dataset with clear lineage and confidence in its reliability.

VS-04.0 Estimation, Analysis, and Statistical Modeling Value Stream

Type	Value Stream
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The Estimation & Analysis Value Stream is the fourth stage of the statistical value chain. Here, the focus shifts from improving data quality to extracting insight and meaning. At this stage, the organization applies statistical methods, estimation techniques, and analytical models to transform edited and enriched data into representative results that describe real-world phenomena. This is where data becomes statistical information — summarized, interpreted, and validated to reflect populations, trends, and behaviors. Every decision taken in this stage must balance methodological detail with transparency and reproducibility.

This value chain identifies the following value steps:

- Design Estimation and Analytical Approach: Select methodology, model type, and parameters based on survey design and analytical objectives.
- Apply Weighting and Calibration: Adjust for sample bias, non-response, or coverage errors through statistical weighting or calibration methods.
- Estimate and Aggregate Data: Generate aggregated statistics, measures, and derived indicators.
- Perform Model-Based Analysis: Conduct modeling, regression, or forecasting analyses as applicable to the domain.
- Validate and Document Results: Review statistical soundness, check for consistency, document methodology and quality metrics.

The main value created here is analytical insight and statistical credibility. Through careful estimation and analysis, data is turned into knowledge that policymakers, researchers, and society can trust.

After estimation and analysis, the organization moves to the Statistical Product Composition Value Stream. Here, the focus shifts from individual analyses to building integrated and composite products.

In that next stage, data from multiple sources and analyses are combined into coherent statistical publications, dashboards, and reports — turning analytical results into accessible products that provide a complete picture.

VS-04.1 Design Estimation and Analytical Approach

Type	Value Stream
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The process begins with defining how the analysis will be done. In this step, statisticians and methodologists work together to design the estimation strategy that fits the data and the analytical goals. They choose appropriate methods — such as regression analysis, calibration, model-based estimation, or weighting — and define parameters, sampling adjustments, and quality criteria. They also establish the assumptions, constraints, and validation metrics that will guide the analytical process. By doing this design work upfront, the organization ensures that every analytical result is methodologically sound and reproducible.

Value created: A clear, well-documented plan for estimation and analysis that ensures methodological integrity.

VS-04.2 Apply Weighting and Calibration

Type	Value Stream
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Next, the organization ensures that the dataset properly represents the population or phenomenon being studied. This is achieved through weighting and calibration. Weighting adjusts the influence of each observation in the dataset so that sample results reflect the true characteristics of the population. Calibration fine-tunes these weights using known reference totals or auxiliary variables — for example, demographic or economic distributions from trusted sources. These adjustments correct for sampling bias, non-response, or undercoverage. The result is data that is statistically representative and ready for estimation.

Value created: Balanced datasets that accurately represent real-world populations or domains.

VS-04.3 Estimate and Aggregate Data

Type	Value Stream
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After calibration, the organization performs the core estimation work — calculating measures, aggregates, and indicators from the prepared data. This may include averages, totals, ratios, growth rates, or more complex derived metrics. These calculations can be performed at different levels of detail — by region, sector, or demographic group — depending on the analytical objectives. At this point, data transitions from detailed records to summarized statistical outputs. All methods, parameters, and results are documented to ensure reproducibility and auditability.

Value created: Statistically valid and transparent estimates derived from high-quality data.

VS-04.4 Perform Model-Based Analysis

Type	Value Stream
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Once basic estimates are available, analysts may take a further step — performing model-based analysis. This involves applying statistical or econometric models to reveal deeper insights, identify relationships, or make predictions. Examples include

regression models, time-series forecasting, small-area estimation, or machine-learning models for classification and clustering. Model-based analysis adds explanatory and predictive value to the data, helping to understand underlying patterns rather than just describe observed values. All model assumptions, limitations, and performance metrics are reviewed and documented to maintain statistical credibility.

Value created: Deeper analytical insights and predictive understanding of complex data relationships.

VS-04.5 Validate and Document Results

Type	Value Stream
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Finally, all results must be validated and documented before being used or shared. This includes checking whether estimates and models meet quality standards, reviewing for plausibility, and comparing results against external benchmarks or historical data. Quality indicators — such as standard errors, confidence intervals, or R^2 values — are calculated and attached to each result. If anomalies are found, the results are revised or annotated to explain them. Once validated, the outputs are documented with detailed metadata that describes the methodology, assumptions, and quality measures. This documentation ensures transparency, reproducibility, and trust in the published results.

Value created: Reliable and validated analytical results ready for composition, interpretation, and dissemination.

VS-05.0 Statistical Product Composition Value Stream

Type	Value Stream
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The Statistical Product Composition Value Stream is the fifth stage in the statistical value chain. After estimation and analysis, the focus now shifts to bringing multiple analytical results together to create coherent, meaningful, and high-value statistical products. At this stage, data from different sources, domains, and analyses are combined, refined, and formatted into composite datasets, time series, and indicators that can be used for reporting, policymaking, and research. This value stream transforms intermediate analytical outputs into final statistical products — ready for validation, interpretation, and dissemination.

This value chain identifies the following value steps:

- Select and Integrate Source Datasets: Combine outputs from multiple domains, surveys, or time periods.
- Construct Composite Indicators or Indices: Create derived measures such as indices, ratios, or time-series aggregates.
- Perform Consistency and Coherence Checks: Ensure integrated datasets are internally consistent and methodologically sound.
- Generate Analytical and Interpretive Outputs: Produce thematic reports, visualizations, or analytical interpretations.
- Approve and Register Statistical Products: Validate products against dissemination and confidentiality standards and register them for publication.

The key value created here is coherence and interpretability. Through careful composition, integration, and quality control, statistical outputs become consistent,

accessible, and ready to serve their intended purpose.

After products are approved and registered, they move into the Dissemination & Service Delivery Value Stream. Here, the focus shifts from producing statistics to sharing them — making them available through publications, APIs, dashboards, and data services. That next stage ensures that the work done throughout the entire value chain ultimately reaches users in a secure, accessible, and user-friendly way.

VS-05.1 Select and Integrate Source Datasets

Type	Value Stream
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The process starts with selecting the datasets and analytical outputs that will form the foundation of the statistical product. Teams identify which sources — such as surveys, registers, or model results — are relevant to the product being created. Next, the selected datasets are integrated and aligned. Integration involves merging data from different domains or time periods, ensuring compatibility in structure, concepts, and classifications. Cross-domain consistency checks confirm that values match across datasets and no contradictions exist. This step ensures that the product is built from reliable and compatible sources.

Value created: Integrated data foundation for consistent and comprehensive statistical products.

VS-05.2 Construct Composite Indicators or Indices

Type	Value Stream
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After integration, the organization develops composite indicators or derived indices that summarize complex information. Examples include economic indices, environmental sustainability measures, or composite well-being scores. This step involves designing formulas, applying weighting schemes, and ensuring methodological soundness. Each indicator is tested for robustness, interpretability, and comparability over time. Composite indicators add significant analytical value because they condense multiple measures into a single, easy-to-understand metric.

Value created: Clear, aggregated indicators that simplify complex statistical information.

VS-05.3 Perform Consistency and Coherence Checks

Type	Value Stream
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Next, the composed products are tested for internal and cross-domain consistency. This means checking that the values across datasets, tables, or indicators align logically and methodologically.

For example, totals should match their components, derived indices should follow expected trends, and time series should show plausible continuity. Any inconsistencies are reviewed and corrected. This ensures that all data and indicators in the statistical product tell a consistent and coherent story.

Value created: Reliable and internally consistent statistical products that maintain user confidence.

VS-05.4 Generate Analytical and Interpretive Outputs

Type	Value Stream
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Once the statistical products are verified, the organization develops analytical outputs and interpretations. This includes summaries, explanatory text, charts, and tables that help users understand the meaning and significance of the results. The outputs are designed to highlight trends, comparisons, and insights that arise from the data. Analysts work with communication experts to ensure clarity, neutrality, and accessibility. These interpretive outputs make the data more valuable by turning numbers into insightful narratives.

Value created: Analytical narratives that make complex statistical data understandable and relevant.

VS-05.5 Approve and Register Statistical Products

Type	Value Stream
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In step 5, each statistical product undergoes a formal review and approval process. Quality reviewers, domain experts, and governance bodies check that all data, methods, and interpretations meet internal standards and external obligations.

Once approved, the product is registered as an official statistical artefact. It is documented with detailed metadata — including data sources, methodology, quality indicators, and publication details.

This versioned and registered product becomes the official output of the organization, ready for dissemination through appropriate channels.

Value created: Verified, traceable, and approved statistical products ready for publication and reuse.

VS-06.0 Dissemination and Service Delivery Value Stream

Type	Value Stream
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The Dissemination & Service Delivery Value Stream is the final stage in the statistical value chain. At this point, the statistical products created in the previous stages are published, distributed, and made accessible to users through trusted and secure channels. The main goal of this value stream is to ensure that information reaches users in a way that is accurate, timely, and easy to understand. This includes policymakers, researchers, journalists, businesses, and the general public. Dissemination is not only about releasing data — it is about providing services that help users access, interpret, and apply statistical information responsibly. This step transforms statistical outputs into societal value by enabling informed decision-making, evidence-based policy, and transparency.

This value chain identifies the following value steps:

- Plan Dissemination and Access Strategy: Define target users, access channels, and release formats (open data, API, research access, publications).
- Prepare Data for Dissemination: Apply disclosure control, anonymization, and formatting for publication or access.
- Publish and Provide Access Services: Release data and metadata via portals, APIs,

dashboards, or secure research environments.

- Support and Engage Users: Provide user documentation, helpdesk support, and training for correct interpretation and use.
- Monitor Usage and Feedback: Track user interactions, collect feedback, and feed insights into continuous improvement loops.

The key value created here is impact and accessibility — ensuring that statistics and data services are usable, trusted, and beneficial to all.

The Dissemination & Service Delivery Value Stream relies heavily on several cross-cutting value streams:

- Governance and Compliance ensures that all dissemination complies with data protection and confidentiality laws.
- Infrastructure and Platform Enablement provides the technical platforms for portals, APIs, and research environments.
- Customer and Stakeholder Engagement ensures that dissemination services meet real user needs and expectations.

Together, these supporting streams ensure that dissemination is not just a technical process — but a strategic, user-centered activity that delivers societal value through trustworthy data.

VS-06.1 Plan Dissemination and Access Strategy

Type	Value Stream
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The process begins with planning how statistical products will be shared. Teams define who the users are, how they will access the data, and which formats or channels best meet their needs. This includes decisions about whether data will be published openly, shared under restricted access, or provided through secure research environments. It also involves setting release schedules, coordinating with communication teams, and ensuring compliance with release policies and embargo rules. The plan ensures that dissemination is purposeful, consistent, and aligned with user expectations.

Value created: A clear, controlled dissemination plan that balances accessibility with compliance and timing.

VS-06.2 Prepare Data for Dissemination

Type	Value Stream
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Once the strategy is set, the data must be prepared for publication. This means ensuring that all information to be released is safe, accurate, and properly formatted. Sensitive or confidential data is anonymized or aggregated. Disclosure control techniques — such as data masking or noise addition — are applied where necessary to protect privacy. All datasets, tables, and visualizations are formatted according to publication standards, ensuring clarity and consistency. Metadata and documentation are attached, describing the content, source, and methodology behind each release.

Value created: Secure and well-prepared data that is safe for public or controlled release.

VS-06.3 Publish and Provide Access Services

Type	Value Stream
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Next, the prepared data and products are released through publication and service platforms. This may include open data portals, interactive dashboards, downloadable datasets, APIs, and secure research access environments. Each access point is designed to support different types of users — from general audiences who want summaries and charts, to researchers who need detailed microdata. Systems are monitored to ensure reliability, accessibility, and data integrity. By providing multiple channels, the organization ensures that everyone can find and access the data they need in a way that suits them.

Value created: Trusted access to official statistics and data services through diverse, user-friendly channels.

VS-06.4 Support and Engage Users

Type	Value Stream
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Once the data is available, user support becomes essential. In this step, the organization provides guidance, documentation, and helpdesk services to assist users in understanding and using the data correctly. Users may have questions about definitions, methods, or technical details — these are addressed through FAQs, tutorials, and direct communication channels. For researchers and professional users, training or workshops may be offered. This step ensures that users not only access the data, but also interpret and apply it correctly, strengthening trust in official statistics.

Value created: Informed, confident users who can correctly understand and apply statistical data.

VS-06.5 Monitor Usage and Feedback

Type	Value Stream
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Finally, the organization continuously monitors how users interact with its data and services. Usage analytics, surveys, and feedback forms are used to learn which products are most useful, which channels are most effective, and where improvements are needed. This feedback is analyzed and shared with product owners, data producers, and management teams. Insights from this step help shape future dissemination strategies, product design, and data access services. By closing this loop, the organization ensures that dissemination evolves in step with user needs and technological change.

Value created: A feedback-driven improvement cycle that keeps data services relevant, accessible, and impactful.

VS-07.0 Governance and Compliance Value Stream

Type	Value Stream
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The Governance and Compliance Value Stream runs horizontally across all stages of the statistical value chain. Its purpose is to make sure that every process, dataset, and product within the organization operates under clear rules, standards, and accountability mechanisms. This value stream builds trust — both inside and

outside the organization — by ensuring that all data activities are ethical, transparent, and compliant with legal, methodological, and quality requirements. It provides the framework of control and assurance that keeps the statistical system reliable, consistent, and aligned with national and international standards such as the European Statistics Code of Practice, GDPR, GSBPM, GAMSO and ISO data quality principles.

This value chain identifies the following value steps:

- Define Governance Frameworks and Policies: Establish overarching governance structures, charters, and principles covering data, metadata, ethics, confidentiality, and statistical quality.
- Establish Legal and Regulatory Compliance Mechanisms: Identify applicable regulations (GDPR, official statistics legislation, data-sharing agreements) and define procedures to ensure compliance across all processes.
- Implement Data and Metadata Governance: Define roles (data owner, steward, custodian) and processes for metadata creation, maintenance, and lineage tracking.
- Monitor Quality and Compliance Indicators: Measure process and output quality (accuracy, timeliness, comparability) and track compliance status using formal scorecards or dashboards.
- Audit, Certify, and Report Compliance: Conduct internal or external audits, perform quality reviews, and document adherence to governance frameworks.
- Update and Improve Governance Mechanisms: Incorporate feedback, regulatory changes, and lessons learned into policies and control frameworks.

The value created here is integrity and accountability — ensuring that the statistical organization acts responsibly, protects privacy, and delivers products users can trust.

The Governance and Compliance Value Stream supports all other value streams by acting as the foundation of trust and assurance. It ensures that Data Acquisition & Ingestion follows lawful and ethical standards. It guarantees that Editing, Estimation, and Dissemination meet quality benchmarks. It provides the governance structure for Infrastructure, Innovation, and Stakeholder Engagement. In short, it underpins the entire SAF value chain — ensuring that every activity is compliant, transparent, and aligned with the mission of producing trustworthy statistics.

VS-07.1 Define Governance Frameworks and Policies

Type	Value Stream
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The process begins with defining the governance structure that guides how data and statistics are managed. This includes creating policies for data governance, metadata management, information security, and quality assurance.

The organization establishes roles and responsibilities — such as data owner, data steward, and quality coordinator — and defines how they interact. Governance committees or boards may be created to oversee compliance and resolve policy issues.

These frameworks form the foundation for consistent decision-making and control

across the statistical value chain.

Value created: A clear governance framework that defines ownership, accountability, and rules for data and statistical operations.

VS-07.2 Establish Legal and Regulatory Compliance Mechanisms

Type	Value Stream
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Next, the organization ensures that all its activities comply with relevant laws and regulations.

This includes privacy protection (e.g., GDPR), data-sharing agreements, intellectual property rights, and official statistics legislation.

Compliance mechanisms are formalized in procedures and checklists that apply to all projects and systems.

Legal experts and data protection officers review each data flow, ensuring that personal data and confidential information are handled safely and lawfully.

This step ensures that compliance is not optional or reactive, but embedded in every workflow.

Value created: Full alignment with legal and ethical standards, reducing risks and maintaining public trust.

VS-07.3 Implement Data and Metadata Governance

Type	Value Stream
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Once frameworks are in place, the organization puts governance into action by managing data and metadata consistently.

This includes establishing data catalogues, defining naming conventions, maintaining reference data, and tracking data lineage.

Each dataset is registered with metadata describing its source, transformations, and usage restrictions.

Stewards ensure that metadata remains complete, accurate, and up to date.

By doing so, data becomes discoverable, understandable, and traceable, supporting both operational efficiency and auditability.

Value created: Transparent and well-managed data and metadata that support reusability and accountability.

VS-07.4 Monitor Quality and Compliance Indicators

Type	Value Stream
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Next, the organization measures and monitors how well it adheres to governance and quality standards.

This involves defining key quality dimensions — such as accuracy, timeliness, consistency, comparability, and accessibility — and measuring performance against them.

Automated dashboards and audits are used to detect deviations or weaknesses in

processes.

When issues are identified, they are escalated for corrective action and recorded for reporting.

Continuous monitoring ensures that the organization stays within its quality and compliance targets at all times.

Value created: Measurable assurance that data and processes meet defined quality and compliance standards.

VS-07.5 Audit, Certify, and Report Compliance

Type	Value Stream
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Once quality and compliance are monitored, results are formally reviewed and documented.

Internal or external auditors verify adherence to policies, standards, and regulations.

Findings are summarized in audit reports and quality certifications that demonstrate accountability and transparency.

These results are shared with governance boards, management, and — when relevant — external stakeholders such as Eurostat or national authorities.

This step ensures that the organization's commitment to compliance is visible, verifiable, and continuously improved.

Value created: Documented and certified proof of compliance and statistical quality.

VS-07.5 Update and Improve Governance Mechanisms

Type	Value Stream
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Finally, governance is not static — it evolves. Based on audit results, user feedback, or changes in law and technology, the organization updates its governance frameworks. Policies are refined, new controls are introduced, and outdated ones are retired. This continuous improvement loop ensures that governance remains current, adaptive, and effective as the organization modernizes.

Value created: Adaptive governance that keeps the organization compliant and future-proof.

VS-08.0 Innovation and Methodological Development Value Stream

Type	Value Stream
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The Innovation and Methodological Development Value Stream makes sure the statistical organization does not stand still. Its purpose is to discover, test, and introduce new ways of working — new data sources, new methods, new tools, and new technologies. This value stream helps the organization move from “we have always done it this way” to “we can do this better, faster, and smarter.”

It connects research, methodology, IT, and business users. It also makes sure that innovation is controlled — so that only methods that are sound, documented, and compliant end up in production.

This value chain identifies the following value steps:

- Identify Innovation Needs and Opportunities: Scan technological, methodological, and policy developments (AI, ML, new data sources, cloud computing) for potential improvements in statistical production.
- Conduct Research and Proof-of-Concepts: Perform controlled experiments and pilot studies to test new methodologies or technologies in limited scope.
- Evaluate Methodological Soundness and Feasibility: Assess research outcomes for accuracy, scalability, and alignment with quality frameworks and ethical principles.
- Develop and Document New Methodologies or Tools: Formalize validated innovations into standardized procedures, algorithms, or reference implementations.
- Integrate Innovation into Production Environments: Transition successful methods or technologies from pilot to production, ensuring interoperability and governance compliance.
- Promote Knowledge Sharing and Capability Building: Disseminate results, best practices, and guidelines across teams; build staff competencies to adopt new methods.

The value created here is continuous improvement. Thanks to this stream, the SAF remains future-proof, able to work with new data (IoT, admin, data spaces), new techniques (ML, small-area estimation), and new operating models (secure processing environments, data sharing).

How This Supports the Other Value Streams.

- It gives Data Acquisition new ways to connect to data spaces and new sources.
- It gives Standardization & Integration new mapping or classification techniques.
- It gives Editing & Enrichment smarter imputation and outlier detection.
- It gives Estimation & Analysis new models and ML-based methods.
- It gives Dissemination new channels (APIs, linked data, interactive views).

So, this value stream keeps the entire SAF alive and evolving.

VS-08.1 Identify Innovation Needs and Opportunities

Type	Value Stream
Innovation starts with looking outward and forward. In this step, the organization scans for gaps and opportunities: - Are there new data sources we could use? - Are there methods that are outdated? - Are users asking for things we cannot yet deliver? - Are there EU / UNECE developments we must align with? Sources can be internal (analysts, methodologists, IT architects) or external (Eurostat pilots, UNECE HLG projects, EHDS / dataspace initiatives). This step turns vague ideas into concrete innovation topics. Value created: A prioritized list of innovation and method-development opportunities.	

VS-08.2 Conduct Research and Proof-of-Concepts

Type	Value Stream
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Next, ideas must be tested safely. In this step, small teams run experiments, PoCs, or pilot studies. They try out a new imputation method, a new AI-based classification, a new data ingestion pattern, or a new statistical package. The work is exploratory, but not chaotic. Scope, success criteria, and risks are defined upfront. Security and privacy rules still apply.

The goal is simple: Does this work for us?

If yes — continue.

If no — document and stop.

Value created: Evidence about what works and what does not, based on real tests — not assumptions.

VS-08.3 Evaluate Methodological Soundness and Feasibility

Type	Value Stream
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Not every idea that works technically is statistically acceptable. So, after the pilot, the result is reviewed methodologically.

Experts check:

- Is the method statistically valid?
- Does it meet quality dimensions (accuracy, coherence, reproducibility)?
- Can we explain it to users and auditors?
- Does it fit our legal and ethical frameworks?

If the answer is yes, the method can move forward. If not, it goes back for refinement.

Value created: Assurance that innovation does not weaken statistical quality or trust.

VS-08.4 Develop / Document New Methodologies or Tools

Type	Value Stream
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er a method is proven, it must be made reusable. In this step, the organization turns a tested idea into a documented, standardized asset. This can be:

- a methodological guideline,
- a processing pattern,
- a code template,
- a reference implementation,
- or an architectural building block.

Documentation includes: purpose, inputs, outputs, assumptions, quality effects, and links to governance rules. This makes the innovation usable by other teams, other domains, or even other national institutes.

Value created: Reusable, documented innovation assets — not one-off experiments.

VS-08.5 Integrate Innovation into Production Environments

Type	Value Stream
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Innovation only has impact if it reaches production. So, in this step, the new method or tool is embedded into the real statistical process.

That can mean:

- adding it to the standard GSBPM-based workflow,
- making it available in the data platform / data fabric,
- exposing it as a business or application service,
- or adding it to the “official toolstack” of the organization.

Security, performance, and support requirements are added here. This step is often done together with IT, architecture, and process owners.

Value created: Innovation that is operational, supported, and available to real statistical production.

VS-08.6 Promote Knowledge Sharing and Capability Building

Type	Value Stream
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Finally, innovation must be socialized. New methods are shared with statisticians, methodologists, and product teams. Short trainings, internal webinars, or method notes are published.

Feedback is collected from first users. This closes the loop: improvements can be made, and the method matures over time. Innovation becomes part of the organization, not just part of one project.

Value created: Organization-wide uptake of proven innovations and a culture of continuous improvement.

VS-09.0 Infrastructure and Platform Enablement Value Stream

Type	Value Stream
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The Infrastructure and Platform Enablement Value Stream provides the technical and digital foundation that supports every part of the statistical organization. Where the other value streams focus on data, methods, or governance, this one ensures that all those activities can actually run safely, efficiently, and at scale. It delivers the IT environments, data platforms, tools, and integration services that make statistical production possible — from data collection to dissemination. It ensures that the systems are secure, resilient, and compliant, while also flexible enough to handle new technologies and data sources as the organization modernizes. This value stream enables the organization to work faster, smarter, and with confidence. It transforms infrastructure from a background function into an active enabler of the statistical mission, ensuring that every user, process, and dataset has the tools and environment it needs to perform reliably.

This value chain identifies the following value steps:

- Define Digital and Technical Requirements: Identify infrastructure needs arising from business capabilities, data volumes, and analytical complexity.
- Design Platform and Infrastructure Architecture: Develop architectural blueprints for data storage, processing, integration, and secure access (data fabric, data market, compute platforms).
- Provision and Configure Platforms: Deploy and configure environments (on-premise, hybrid, or cloud) with necessary governance, monitoring, and security controls.
- Enable Integration and Orchestration Services: Implement APIs, workflow engines, and automation frameworks to connect business services and processes.
- Ensure Performance, Security, and Continuity: Monitor platform performance,

implement data protection, manage incidents, and guarantee availability and disaster recovery.

- Evolve and Modernize the Technical Ecosystem: Introduce new technologies, upgrade components, and decommission obsolete infrastructure in alignment with modernization strategy.

The value created here is operational capability and continuity — a stable, modern, and adaptive foundation for the entire SAF ecosystem.

Connecting to Other Value Streams

The Infrastructure and Platform Enablement Value Stream supports every other part of the SAF. It underpins the business and data processes by providing the digital environment in which they operate.

- It gives Data Acquisition the secure pipelines and storage needed for data collection.
- It gives Standardization, Editing, and Analysis the computing and orchestration needed to process large datasets efficiently.
- It gives Dissemination the platforms, portals, and APIs through which users access results.
- It ensures Governance and Compliance have the tools to monitor security, quality, and data lineage, and
- It enables Innovation by offering scalable, modern platforms for experimentation and research.

In short, it is the engine room of the statistical organization — invisible to most users, but vital for everything that happens above it.

VS-09.1 Define Digital and Technical Requirements

Type	Value Stream
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The first step begins with understanding what the business and data processes need from the technical environment.

Architects and system owners work closely with business and data experts to identify requirements for computing power, data storage, networking, security, scalability, and interoperability.

This step translates business goals into technical specifications.

For example, it defines how much storage is needed for large datasets, what security level is required for sensitive data, and what tools are needed for analytical work.

Requirements are collected for both current systems and future modernization projects, ensuring that the infrastructure can evolve with organizational needs.

Value created: Clear, shared understanding of what the infrastructure must deliver to support all business and data processes effectively.

VS-09.2 Design Platform and Infrastructure Architecture

Type	Value Stream
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Once requirements are known, the organization designs the technical architecture that will meet them.

This architecture defines how platforms, services, and technologies work together

— both conceptually and physically.

It includes the data platform (data lake, data fabric, data market), processing environments (secure analysis zones, compute clusters), and access layers (APIs, portals, dashboards).

The architecture defines integration patterns, security boundaries, identity management, and compliance zones.

The design is not static — it evolves over time, guided by architectural principles such as modularity, interoperability, reusability, and automation.

At this point, diagrams, blueprints, and documentation are produced to guide implementation teams.

Value created: A coherent, future-proof technical blueprint that connects business needs with technological capabilities.

VS-09.3 Provision and Configure Platforms

Type	Value Stream
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After the design is approved, the focus shifts to building and configuring the actual environments.

IT and platform teams provision servers, containers, databases, and analytical tools according to the architecture plan.

They set up data storage systems, computing clusters, integration layers, and APIs. At the same time, they configure security controls — encryption, firewalls, role-based access, and monitoring tools.

Automation scripts are used to deploy resources quickly and consistently, following infrastructure-as-code principles.

The goal is to create platforms that are reliable, reproducible, and compliant, while remaining adaptable to evolving needs.

Value created: Operational platforms that are configured, secure, and ready to support statistical production and analysis.

VS-09.4 Enable Integration and Orchestration Services

Type	Value Stream
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Once the core platforms are in place, they must be connected.

This step focuses on integration and orchestration — ensuring that data, applications, and services can work together smoothly.

Integration services enable the flow of data between systems using APIs, message queues, or data pipelines.

Orchestration tools automate sequences of actions — for example, automatically triggering data validation after ingestion, or sending completed datasets into the next processing stage.

These services form the “nervous system” of the organization’s technical ecosystem — coordinating activities, maintaining consistency, and ensuring efficiency.

Without them, systems would remain isolated, and processes would require manual intervention.

Value created: Seamless integration and automation across systems, enabling smooth and consistent data and process flows.

VS-09.5 Ensure Performance, Security, and Continuity

Type	Value Stream
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Once platforms are operational, maintaining them becomes critical. This step ensures that the infrastructure remains secure, reliable, and high-performing.

Monitoring tools continuously track performance metrics such as response time, resource usage, and availability.
Security teams watch for intrusions, vulnerabilities, or misconfigurations, applying updates and patches regularly.
Disaster recovery procedures are tested to ensure data and operations can be restored quickly in case of failure.
Service-level agreements (SLAs) are defined to maintain expected uptime and performance levels.

These activities protect the organization's data assets and guarantee operational resilience, even in the face of incidents or cyber threats.

Value created: Secure, stable, and resilient platforms that ensure continuous statistical operations.

VS-09.6 Evolve and Modernize the Technical Ecosystem

Type	Value Stream
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Technology never stands still — and neither should the organization's infrastructure.
In this step, the organization reviews, upgrades, and modernizes its technology landscape regularly.

Outdated systems are phased out, and new technologies — such as cloud computing, containerization, AI services, and confidential computing — are evaluated and introduced where appropriate.
Lessons learned from operations, user feedback, and innovation pilots are used to guide upgrades.

This ongoing evolution ensures that the infrastructure remains aligned with strategic goals, supports innovation, and complies with new security and interoperability standards.

Value created: A continuously improving infrastructure that adapts to technological progress and future organizational needs.

VS-10.0 Customer and Stakeholder Engagement Value Stream

Type	Value Stream
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The Customer and Stakeholder Engagement Value Stream ensures that the statistical organization remains connected, relevant, and responsive to the needs of those it serves. It focuses on building and maintaining mutual understanding and trust between the organization and its users, data providers, partners, and policy stakeholders. In the SAF, this value stream is essential because official statistics exist to create public value. To achieve that, the organization must understand what stakeholders need, communicate transparently, and ensure that products, data services, and policies remain aligned with real-world demands.

This value stream is not a one-way process. It is a continuous dialogue — listening to users, understanding their challenges, and translating those insights into better products, improved services, and stronger relationships.

This value chain identifies the following value steps:

- Identify and Segment Stakeholders: Map the organization's ecosystem of users, data providers, partners, and policymakers; understand their roles and expectations.
- Collect and Analyze User Needs: Gather feedback, conduct consultations, and translate stakeholder needs into data or service requirements.
- Define and Manage Service Portfolios: Develop catalogues of business services and data products that address stakeholder needs and align with strategic goals.
- Deliver Services and Manage Relationships: Operate customer support, service desks, and account management; monitor satisfaction and service performance.
- Engage and Communicate with Stakeholders: Conduct outreach, workshops, and knowledge-sharing events to promote awareness and understanding of statistical outputs.
- Monitor Impact and Satisfaction: Measure how data and services are used, assess societal impact, and feed insights back into planning and innovation.

The value created here is relevance and trust — ensuring that the organization's work is understood, used, and appreciated by its partners, users, and society at large.

Connecting to Other Value Streams

The Customer and Stakeholder Engagement Value Stream is the voice and mirror of the statistical organization.

- It connects external expectations with internal capabilities, closing the feedback loop across the SAF.
 - It informs Data Acquisition by revealing what data sources or topics matter most to users.
 - It influences Product Composition and Dissemination by identifying which outputs and formats users prefer.
 - It strengthens Governance and Compliance through transparent communication about data ethics, confidentiality, and quality. It supports Innovation and Methodology by gathering ideas and feedback from real users and partners.
- Through this value stream, the organization ensures that it doesn't just produce data — it produces public value. It turns statistics into impact by aligning what the organization creates with what stakeholders need.

VS-10.1 Identify and Segment Stakeholders

Type	Value Stream
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The first step is to map out who the stakeholders are. Statistical organizations serve many different audiences — from government ministries and researchers to journalists, businesses, international bodies, and citizens.

Each group has different expectations, data literacy levels, and use cases. Therefore, the organization classifies its stakeholders into segments such as data providers, policymakers, academic users, internal staff, or the general public.

For each segment, their needs, communication preferences, and influence are identified.

This mapping forms the foundation for a targeted engagement strategy.

Value created: A clear understanding of who the organization's stakeholders are and what they need from its data and services.

VS-10.2 Collect and Analyze User Needs

Type	Value Stream
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Once stakeholders are identified, the next step is to understand their needs and expectations. This is done through surveys, workshops, interviews, feedback forms, and digital analytics that track how users interact with published data. The organization studies how users search for, interpret, and apply statistical information. It identifies pain points — such as difficult data formats, missing indicators, or limited accessibility — and areas where new services could add value. Insights from this analysis are shared internally so that products and services can be adapted accordingly. In this way, user feedback becomes a strategic input into planning and product development.

Value created: Actionable insights that guide product design, data publication, and service improvement.

VS-10.3 Define and Manage Service Portfolios

Type	Value Stream
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After understanding user needs, the organization translates them into structured services and offerings. It defines and manages a service portfolio that includes statistical publications, open data portals, APIs, customized data delivery, and research access environments. Each service is described in terms of its target audience, scope, service levels, and access conditions.

Roles and responsibilities are assigned for ownership, maintenance, and user support. Clear documentation and service catalogues help users navigate and request the services they need. This step turns abstract user requirements into concrete, well-managed service offerings.

Value created: A transparent and organized set of services that clearly show what users can access and how.

VS-10.4 Deliver Services and Manage Relationships

Type	Value Stream
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Once services are defined, they must be delivered effectively. This step focuses on operational excellence and relationship management. Users interact with the organization through helpdesks, contact points, and online platforms. Requests are logged, processed, and tracked; performance against service level agreements (SLAs) is monitored. Regular communication ensures that stakeholders know who to contact, how to request data, and when updates will be available. In parallel, key account management is used for major stakeholders — such as ministries, research institutes, and data providers — to maintain direct relationships and ensure satisfaction.

Value created: Consistent, high-quality user experience and strong, trust-based relationships with key stakeholders.

VS-10.5 Engage and Communicate with Stakeholders

Type	Value Stream
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Beyond service delivery, the organization must actively engage and communicate. This means going beyond data provision to create real understanding and collaboration. Regular newsletters, webinars, conferences, and social media are used to share updates, explain new methods, and promote transparency about ongoing projects. Workshops and user forums are organized to discuss needs and co-create solutions. Feedback mechanisms are kept open so that stakeholders can continuously contribute to improving services. This two-way engagement strengthens the organization's role as a trusted, responsive, and visible public institution.

Value created: Strong, transparent communication that builds understanding, collaboration, and trust.

VS-10.6 Monitor Impact and Satisfaction

Type	Value Stream
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Engagement does not end when data is delivered — it continues through impact assessment and evaluation. This step measures how statistical data and services are being used, what decisions they inform, and how satisfied users are with their experience. Metrics include website usage, downloads, citations, customer satisfaction surveys, and social media reach. Qualitative evaluations assess how well data supports evidence-based policy and societal understanding. Results are shared internally to guide continuous improvement across all value streams. By closing the loop, the organization ensures that stakeholder engagement remains evidence-driven and outcome-focused.

Value created: Clear insight into how statistics create societal value and how stakeholder relationships can be further strengthened.

Business Layer

Aggregate Data

Type	Business Process
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Anonymize Data

Type	Business Process
------	------------------

Classify Data

Type	Business Process
------	------------------

Collect Survey

Type	Business Process
------	------------------

Compose Data

Type	Business Process
------	------------------

Data Aquisition

Type	Business Process
------	------------------

Data Delivery

Type	Business Process
------	------------------

Data Editing

Type	Business Process
------	------------------

Data Processing

Type	Business Process
------	------------------

Data Publication

Type	Business Process
------	------------------

Dissemination

Type	Business Process
------	------------------

Estimate Data

Type	Business Process
------	------------------

Extract Data

Type	Business Process
------	------------------

Imputate Data

Type	Business Process
------	------------------

Integrate Data

Type	Business Process
-------------	------------------

Output Data (statistics)

Type	Business Object
-------------	-----------------

Processed Data

Type	Business Object
-------------	-----------------

Published Output Data (Released Data)

Type	Business Object
-------------	-----------------

Raw Data

Type	Business Object
-------------	-----------------

Receive Data

Type	Business Process
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Source Data

Type	Business Object
-------------	-----------------

Standardization

Type	Business Process
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Standardized Data

Type	Business Object
-------------	-----------------

Transform Data

Type	Business Process
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Motivation

A clear dissemination plan that balances accessibility, compliance and timing

Type	Value
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A clear governance framework that defines ownership, accountability, and rules for data and statistical operations

Type	Value
------	-------

A clear understanding of who the organization's stakeholders are and what they need from its data and services

Type	Value
------	-------

A coherent, future-proof technical blueprint that connects business needs with technological capabilities

Type	Value
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A continuously improving infrastructure that adapts to technological progress and future organizational needs

Type	Value
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A feedback-driven improvement cycle that keeps data services relevant, accessible, and impactful

Type	Value
------	-------

A prioritized list of innovation and method-development opportunities

Type	Value
------	-------

A transparent and organized set of services that clearly show what users can access and how

Type	Value
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Actionable insights that guide product design, data publication, and service improvement

Type	Value
------	-------

Adaptive governance that keeps the organization compliant and future-proof

Type	Value
------	-------

Analytical narratives that make complex statistical data understandable

Type	Value
------	-------

Assurance that innovation does not weaken statistical quality or trust

Type	Value
------	-------

Balanced datasets that accurately represent real-world populations

Type	Value
------	-------

Clean, coherent data that accurately represents real-world information

Type	Value
------	-------

Clear understanding of data quality issues and areas requiring correction

Type	Value
------	-------

Clear understanding of what data is needed and where it can be obtained.

Type	Value
------	-------

Clear, aggregated indicators that simplify complex statistical information

Type	Value
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Clear, shared understanding of what the infrastructure must deliver to support the business

Type	Value
------	-------

clear, well-documented plan for estimation and analysis

Type	Value
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Collected dataset with basic metadata and source quality indicators

Type	Outcome
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Communication and outreach materials

Type	Outcome
------	---------

Compliance reports and audit trails

Type	Outcome
------	---------

Confidence that received data is complete, structured, and fully traceable

Type	Value
------	-------

Configured data and analytics environments

Type	Outcome
------	---------

Consistent, high-quality user experience and strong, trust-based relationships with key stakeholders

Type	Value
------	-------

Consistent, interoperable data aligned with recognized standards

Type	Value
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Data governance policy set

Type	Outcome
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Data lineage and metadata records

Type	Outcome
-------------	---------

Datasets that preserve statistical representativeness and minimize bias

Type	Value
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Deeper analytical insights and predictive understanding of data relationships

Type	Value
-------------	-------

Documented and certified proof of compliance and statistical quality

Type	Value
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Edited and enriched dataset with quality and lineage documentation.

Type	Outcome
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Enriched datasets that support deeper analysis and insight generation

Type	Value
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Estimated or analyzed dataset with methodological and quality documentation.

Type	Outcome
-------------	---------

Evidence about what works and what does not, based on real tests — not assumptions

Type	Value
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Finalized statistical product or analytical dataset, ready for dissemination.

Type	Outcome
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Full alignment with legal and ethical standards, reducing risks and maintaining public trust

Type	Value
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Full understanding of the dataset structure and format

Type	Value
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Governance documentation and certification logs

Type	Outcome
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Harmonized meaning and context across datasets

Type	Value
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Informed, confident users who can correctly understand and apply statistical data

Type	Value
------	-------

Innovation pipeline documentation

Type	Outcome
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Innovation that is operational, supported, and available to real statistical production

Type	Value
------	-------

Insight into how statistics create societal value and how stakeholder relationships can be strengthened

Type	Value
------	-------

Integrated data foundation for consistent and comprehensive statistical products

Type	Value
------	-------

Integration and orchestration services

Type	Outcome
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Knowledge base, guidance, and training materials

Type	Outcome
------	---------

Legal, ethical, and transparent access to data under controlled conditions

Type	Value
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Measurable assurance that data and processes meet defined quality and compliance standards

Type	Value
------	-------

Methodological research reports and prototypes

Type	Outcome
------	---------

Monitoring dashboards and performance metrics

Type	Outcome
------	---------

Operational platforms that are configured, secure, and ready to support statistical production and analysis

Type	Value
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Organization-wide uptake of proven innovations and a culture of continuous improvement

Type	Value
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Pilot evaluation results and transition plans

Type	Outcome
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Platform and infrastructure blueprints

Type	Outcome
------	---------

Published dataset, statistical release, or data service accessible to users.

Type	Outcome
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Reliable and internally consistent statistical products

Type	Value
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Reliable and validated analytical results ready for composition, interpretation, and dissemination

Type	Value
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Reusable, documented innovation assets

Type	Value
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SAF-D1 Evolving Data Ecosystems and Partnerships

Type	Driver
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The environment for producing official statistics is shifting from one of isolated data collection toward a complex and interdependent data ecosystem. Traditional surveys, once the cornerstone of statistical production, now coexist with administrative, commercial, platform, and community-driven data sources. This evolution expands opportunities for collaboration but also introduces dependencies on actors, technologies, and infrastructures outside the direct control of statistical organizations.

Open-source development has become a key element of this ecosystem. It provides a shared foundation for innovation, transparency, and cost efficiency, while fostering cooperation across institutional and national boundaries. As part of a wider collaboration model, open source allows NSIs to co-develop tools and methods with peers, academia, and private partners, ensuring that advances in technology benefit the entire statistical community.

SAF addresses this driver by promoting architectures that support secure data sharing, standardized interfaces, and transparent governance models. It enables NSIs to participate confidently in broader ecosystems, from national data platforms to European data spaces, where production may take place across multiple environments and jurisdictions. By combining openness with strong governance and clear accountability, SAF ensures that collaboration and shared development

strengthen rather than compromise the integrity of official statistics.

SAF-D2 Artificial Intelligence and Technological Acceleration

Type	Driver
Artificial intelligence, automation, and rapid technological progress are transforming how data is created, processed, and consumed. For statistical organizations, these developments bring both opportunities and dependencies. Emerging tools can automate repetitive tasks, enhance analytical capabilities, and strengthen quality assurance. At the same time, they redefine what it means for official statistics to remain relevant and trusted in an environment increasingly shaped by intelligent systems. The accelerating pace of innovation requires statistical offices to be both adopters and enablers of AI. Internally, this means developing architectures and competencies that support automation, machine learning, and advanced analytics within secure and transparent frameworks. Externally, it requires ensuring that official statistics are AI-ready - that is, machine-readable, machine-understandable, and interoperable with the digital platforms and applications used by policymakers, researchers, and citizens. SAF addresses this driver by enabling National Statistical Institutes to adopt a risk-based approach to modernization. It supports architectures where critical processes that demand methodological rigor, transparency, and control can be tightly governed, while areas that are less sensitive can be automated to a higher degree. This balance ensures that innovation strengthens quality and efficiency without compromising security, privacy, or trust. Through modular and standards-based design, SAF provides the flexibility needed to integrate emerging technologies at an appropriate pace. It allows NSIs to benefit from automation and AI-driven tools while maintaining full accountability for statistical quality, data protection, and institutional independence.	

SAF-D3 Trust, Scientific Rigor, and Data Protection

Type	Driver
Trust is the foundation of official statistics. In an environment shaped by misinformation, privacy concerns, and competition from unregulated data providers, maintaining public confidence requires more than compliance. It depends on scientific rigor, transparency, and clear accountability for how data is handled and communicated. SAF addresses this driver by embedding trust and protection directly into architectural design. It promotes governance and technical controls that safeguard confidentiality, ensure methodological integrity, and make processes traceable and transparent. Through a risk-based approach, SAF allows NSIs to apply strict oversight where sensitivity and quality demands are high, while enabling greater automation and shared solutions where risks are lower. This balance ensures that modernization strengthens trust rather than compromising it.	

SAF-D4 Uncertain and Polarized Global Environment

Type	Driver
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Statistical organizations operate in an increasingly complex and unpredictable world. Global tensions, political polarization, and shifting power dynamics influence both the demand for and the perception of official statistics. At the same time, digital interdependence and cross-border data flows require statistical systems to remain open and interoperable while safeguarding national and institutional integrity.

SAF addresses this driver by providing a stable and transparent foundation for cooperation and resilience. It promotes interoperable architectures based on shared standards, enabling NSIs to exchange data and methods securely even in times of uncertainty. By embedding governance and design principles that protect independence and transparency, SAF helps organizations maintain credibility, comparability, and trust in environments where facts themselves are often contested.

SAF-D5 Resource Constraints and Organizational Agility

Type	Driver
Many statistical organizations face increasing expectations with limited budgets, aging systems, and challenges in recruiting and retaining specialized skills. At the same time, the demand for more timely, detailed, and multidomain outputs continues to grow. Meeting these needs requires not only efficiency, but the ability to adapt rapidly to changing priorities and technologies. SAF addresses this driver by promoting architectures that make flexibility and reuse a core principle. It supports modular designs, shared services, and scalable solutions that reduce duplication and enable more effective use of existing resources. By fostering an agile organizational mindset and clear governance around priorities, SAF allows NSIs to reallocate capacity, modernize incrementally, and continue to deliver high-quality outputs even under financial and staffing constraints.	

SAF-G1 Cost Efficiency and Resource Sharing

Type	Goal
With limited budgets and growing demands, statistical organizations must operate as efficiently as possible. SAF promotes cost reduction and improved effectiveness through shared services, reuse of solutions, and stronger interoperability. By pooling resources and adopting common components, NSIs can access capabilities that would be difficult to develop individually while avoiding duplication of effort. SAF also encourages the use of open-source components and community-driven platforms where this strengthens transparency, collaboration, and cost efficiency. Through modular architectures and clear governance, organizations can adapt to changing needs, reallocate capacity, and modernize incrementally without large-scale reinvestments. This approach ensures that efficiency is achieved in ways that reinforce both quality and long-term sustainability.	

SAF-G2 Harmonized Reference Framework

Type	Goal
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Without a shared architectural reference, statistical organizations risk developing isolated solutions that reduce comparability and increase long-term costs. SAF provides a harmonized foundation of standards, patterns, and best practices that can be adapted to local contexts while supporting interoperability across borders and domains.

As collaboration increasingly takes place across shared infrastructures such as national data platforms and European data spaces, harmonization also becomes a prerequisite for participation. By promoting transparency and consistent design principles, SAF helps NSIs protect the integrity of official statistics and ensure that shared and open solutions can be reused safely and effectively. A common reference framework enables collaboration, strengthens trust, and supports coherence even as the data landscape evolves.

SAF-G3 Modernization of Statistical Systems

Type	Goal
Many statistical organizations continue to rely on fragmented and legacy IT environments that limit innovation and responsiveness. SAF provides guidance for transitioning toward modern, modular, and service-based architectures that are secure, scalable, and easier to maintain. Modernization reduces technical debt, improves reliability, and allows faster delivery of new and higher-quality statistical outputs. Modernization increasingly takes place in environments where multiple actors operate, from national infrastructures to cross-border data spaces. SAF ensures that such settings can be used safely by defining principles for access control, data protection, and accountability. It also supports a risk-based approach to automation and integration of emerging technologies. Critical processes can remain tightly governed to preserve methodological rigor, while less sensitive stages can be automated or shared. By ensuring that systems are machine-readable, interoperable, and adaptable to both open and controlled contexts, SAF helps NSIs remain relevant and resilient.	

SAF-G4 Support for Data Innovation

Type	Goal
Innovation in data and methods is central to the future of official statistics. SAF promotes an environment where experimentation, collaboration, and reuse are encouraged within secure and well-governed boundaries. By providing a clear architectural foundation, SAF enables NSIs to incorporate new data sources, tools, and analytical methods without undermining consistency or quality. Innovation is not only about creating new solutions but also about making change possible within production. Many statistical organizations find it difficult to introduce new data sources or develop new statistical products because their production systems are optimized for predictability and stability. While these are necessary principles, they can limit agility. SAF addresses this by supporting architectures that balance robustness with flexibility, allowing new components, data, or workflows to be integrated more rapidly while maintaining full control over quality and compliance.	

To make this possible, the framework emphasizes mechanisms and guardrails that enable controlled integration of innovative methods into production chains. This includes versioning, validation, and automated testing approaches that preserve reliability even as systems evolve. SAF also seeks to narrow the gap between innovation and production by promoting shared platforms and processes that allow prototypes to transition smoothly into operational environments.

Open source plays a key role in this innovation process. It allows organizations to build on each other's work, increase transparency, and strengthen the collective capacity of the statistical community. SAF supports this through architectural patterns that enable safe reuse of open components and integration with shared platforms. In doing so, it helps turn innovation into a structured, sustainable part of statistical production.

SAF-G5 Sustainable Compliance

Type	Goal
	Compliance with regulations, standards, and ethical principles is fundamental to maintaining the trust and legitimacy of official statistics. SAF ensures that modernization aligns with legal, security, and quality frameworks while promoting transparency and accountability.
	Within the statistical domain, well-established standards such as GSBPM, GSIM, and SDMX continue to provide a foundation for consistent processes, metadata management, and data exchange. These remain essential for ensuring methodological rigor and comparability across the statistical system.
	However, as the production of statistics becomes increasingly integrated into the broader data and technology ecosystem, compliance must also extend beyond traditional statistical standards. SAF encourages active engagement with and adoption of relevant standards from adjacent fields such as data management, artificial intelligence, and cloud computing. This outward-looking approach ensures that NSIs remain interoperable with modern data platforms and can benefit from technological progress developed outside the statistical community.
	Through consistent governance and the use of open, verifiable technologies, SAF supports compliance as an enabler rather than a constraint. It provides the structures necessary to ensure that innovation, data sharing, and open collaboration occur within controlled and auditable environments. This approach reinforces public trust by embedding transparency, scientific rigor, and privacy-by-design across all processes.

SAF-M1 Bridge Between Strategy and Technology

Type	Meaning
	SAF connects high-level strategic objectives, such as improving statistical relevance and efficiency, with concrete technology choices. It ensures that IT investments are always justified by their contribution to the organization's mission.

SAF-M2 Catalyst for Collaboration

Type	Meaning
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SAF is a unifying framework for the statistical community. It enables organizations to work together on shared goals, whether through joint platforms, interoperable systems, or harmonized standards, thereby amplifying the collective capacity of UNECE members.

SAF-M3 Foundation for Digital Transformation

Type	Meaning
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SAF is not just a set of technical models; it is the foundation for digital transformation in official statistics. It enables NSIs to embrace innovation without sacrificing methodological rigor or institutional trust.

SAF-M4 Unifying Language and Notation

Type	Meaning
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By adopting TOGAF and ArchiMate, SAF ensures that diverse stakeholders can communicate using a consistent language and notation. This reduces misunderstandings, enables shared modelling practices, and promotes architectural maturity across the community.

SAF-O1 Accelerated Modernization

Type	Outcome
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SAF provides a roadmap for NSIs to modernize systematically, enabling them to prioritize and sequence modernization efforts. With clear guidance, organizations can replace outdated technologies incrementally while maintaining operational continuity.

SAF-O2 Enhanced Data Quality

Type	Outcome
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Embedding governance, metadata, and lineage into architecture ensures that statistical outputs maintain the highest standards of quality and trustworthiness. SAF makes quality an architectural outcome rather than an afterthought.

SAF-O3 Improved Interoperability

Type	Outcome
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By adopting SAF, NSIs will achieve higher levels of interoperability across systems, domains, and borders. This facilitates cross-country collaborations, enhances comparability of statistics, and simplifies the integration of external data sources into official workflows.

SAF-O4 Improved Trust and Transparency

Type	Outcome
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By embedding principles such as openness, security, and transparency, SAF helps to build and maintain public trust in statistics. This outcome strengthens the reputation of NSIs and the legitimacy of their outputs.

SAF-O5 Increased Efficiency

Type	Outcome
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SAF reduces duplication of effort by promoting reuse, harmonization, and standardization. This outcome means lower IT costs, faster system development cycles, and more efficient data processing pipelines across the statistical system.

SAF-O6 Shared Knowledge Base

Type	Outcome
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SAF fosters a community of practice across UNECE members. Through shared architectural frameworks, NSIs can exchange lessons learned, reusable components, and knowledge — leading to continuous improvement at lower cost.

SAF-P01 Reuse before open source, open source before buying, buying before making it yourself

Type	Principle
archimate_element_identifier	EAID_538E9594_00EA_4c9d_A1DB_7B81F75965B5
archimate_element_identifier	EAID_A2F6A7F7_719C_468f_A4DA_2226D49E9035

In case of equal suitability (Business Case), reuse of (parts of) applications takes precedence. The use of open source software takes precedence over purchasing. Purchasing is then preferred over making it yourself. Non-statistical processes only use standard applications.

Reuse of an application or parts thereof is sustainable and cost-efficient, and also leads to standardization in service provision and information provision. If reuse is not possible, investigate whether an Open Source solution is available before considering a Closed Source (Commercial) solution.

This is to comply with the government policy "Open, unless". Only if there are no existing solutions available, is it permitted to develop the application yourself.

SAF-P02 Active support

Type	Principle
archimate_element_identifier	EAID_9006BAE5_1965_46f5_BE83_1BE907EE5B09
archimate_element_identifier	EAID_5F7EF182_8D94_4697_8C59_FB0620114A3D

Systems (hardware and software) have an active user community and/or vendor support. The rule of thumb is that we use the latest or the second-to-last major version. Among other things, for business continuity it is necessary to be able to get

quick and good support in resolving a disruption in the event of a disruption. It is also important from a security perspective that identified security risks are resolved in a timely manner.

SAF-P03 Active Life Cycle management

Type	Principle
archimate_element_identifier	EAID_33319E45_E03C_48ac_AA8B_4840ACC083E3
archimate_element_identifier	EAID_A6909E75_5DE2_4814_A308_2517C4FF8E1C

With active Life Cycle management we prevent overdue maintenance in the IV landscape so that continuity & security risks are mitigated, among other things. Without Life Cycle management, systems or parts thereof can become outdated and thus become vulnerable in terms of security. Without active Life Cycle management, a technological debt is also built up that, when it eventually has to be repaid, will require disproportionate effort or introduce continuity risks. Keeping systems and components up-to-date in accordance with established LCM policy mitigates these risks.

SAF-P04 Maximize the benefits for the organization

Type	Principle
archimate_element_identifier	EAID_E294275B_F597_47b0_9420_9E8BBBBB2C1AE
archimate_element_identifier	EAID_1FD8C967_0E20_4c8a_8271_104EC0F66693

IV decisions are made to maximize the benefit of the organization as a whole. This principle embodies 'service above self'. Decisions made from an organization-wide perspective have greater long-term value than decisions made from a departmental perspective. To maximize the return on the investment, IV decisions must align with the organization-wide mission and priorities.

SAF-P05 IT and technology is everyone's business

Type	Principle
archimate_element_identifier	EAID_886E0FB7_2D03_4d78_A7C6_54A2BF53CB6F
archimate_element_identifier	EAID_8825255C_B63A_42b7_9A2F_1D04E97A7C84

All departments in the organization participate in IV decisions that are necessary to achieve organizational goals.
IT and technology users are the key stakeholders in the application of technology to meet an organization's needs.
To ensure that IT and technology are aligned with business operations, all departments in the organization must be involved in all aspects of the IT and technology landscape. Business experts from across the organization and the technical staff responsible for developing and maintaining the IT and technology capabilities must work together as a team to jointly define the goals and objectives.

SAF-P06 Business Continuity

Type	Principle
archimate_element_identifier	EAID_081EA8EE_0CD6_4db4_AF8B_F850C3507526
archimate_element_identifier	EAID_EA934F19_067D_45e4_8308_2A3A40C70AE2

Organizational activities are maintained despite system or employee outages. As systems and processes become increasingly important to operations, we become more dependent on them; therefore, we must consider the reliability of such systems and processes in their design and operation. Departments throughout the organization must have the ability to continue their activities, regardless of external events, within agreed lead times. Hardware failures, natural disasters, and data corruption must not disrupt or stop operations. The enterprise must be able to operate on the basis of alternative IV systems/processes.

SAF-P07 Implement loosely coupled processes and systems

Type	Principle
archimate_element_identifier	EAID_D9EA1AFD_0D07_4d4d_8713_AC2A651D6B31
archimate_element_identifier	EAID_D6298C0C_DDC5_4e50_80AA_761580ED3F89

Processes and systems should be built as independently as possible. This means that processes and systems are designed in such a way that they are not intertwined and dependent on each other.

Functionality and logic should be shared with the rest of the organization via standard interfaces. Loosely coupling processes and systems offers more flexibility to adapt or replace individual processes and systems in the future. In addition, there are advantages in terms of scalability and robustness. Differences in security and privacy requirements can also be better enforced.

SAF-P08 Data is shared property

Type	Principle
archimate_element_identifier	EAID_6D643CD8_3F83_407e_AE22_906A56969096
archimate_element_identifier	EAID_481DB596_A5B5_420e_8D65_0CCD3F210402

Statistical data that has been determined to add value for use by internal or external users is shared. This implies re-use of data, both in observation to reduce the burden and in further processing to prevent duplication of work in improving and analyzing data.

SAF-P09 No data without metadata and classification

Type	Principle
archimate_element_identifier	EAID_91798D0F_4A40_48d7_9924_4615C19DBEBE
archimate_element_identifier	EAID_84B866E4_F45D_4622_9DE4_EEF87CCB71B0

All data that needs to be shared, needs to be provided with metadata information

that describes the content and meaning of the data as well as possible. In addition, the data also needs to be classified. Metadata makes it possible to make data FAIR and the BIV classification (Availability, Integrity and Confidentiality) provides direction for the necessary technical and organizational measures. This is to comply with various laws, requirements and security standards.

SAF-P10 Security By Design

Type	Principle
archimate_element_identifier	EAID_8BB3F74C_B40B_4c8e_916F_2C9919E7CFFB
archimate_element_identifier	EAID_4911417C_5655_4db0_81CF_254FCC1E9130

Security should not be an afterthought in IV solutions, but should be part of those solutions. Coherent security mechanisms should span all layers of the architecture and be scalable  from small objects to large objects. Security should be designed as an integrated part of the system architecture.

SAF-P11 Privacy By Design

Type	Principle
archimate_element_identifier	EAID_0BB196F4_7E3C_4be4_8CEB_111E21E6D965
archimate_element_identifier	EAID_3F8A4C54_4F7A_432f_8403_DDDC31E051B5

Privacy by design should be designed as an integrated part of the system architecture and business processes. Privacy should not be an afterthought in IV solutions, but should be part of those solutions. The result is that privacy becomes an essential part of the core functionality that is delivered. Privacy is an integral part of the system without compromising functionality.

SAF-ST01 CIOs & CTOs of NSIs

Type	Stakeholder
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Technology leaders within NSIs need to make strategic IT investment decisions. SAF supports them by providing a clear framework to evaluate options, align IT with business goals, and ensure compliance with international standards.

SAF-ST02 Citizens & Society

Type	Stakeholder
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Ultimately, society is the beneficiary of SAF, as it ensures that statistical organizations produce trustworthy, transparent, and modern statistics that support democratic accountability and informed public discourse.

SAF-ST03 Government Policy Makers

Type	Stakeholder
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Policy makers rely on official statistics to design, implement, and evaluate policies. SAF indirectly benefits them by ensuring that NSIs deliver timely, accurate, and comparable data in formats that are accessible and actionable.

SAF-ST04 IT & Data Professionals

Type	Stakeholder
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Architects, engineers, and data scientists will use SAF to guide their daily work in designing, implementing, and managing systems. SAF provides them with common patterns, tools, and practices that increase efficiency and reduce the learning curve when collaborating across institutions.

SAF-ST05 National Statistical Institutes (NSIs)

Type	Stakeholder
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NSIs are the primary beneficiaries of SAF, as it provides them with structured guidance for their enterprise architecture. By using SAF, NSIs can align with international practices, improve efficiency, and strengthen their ability to deliver high-quality statistics to stakeholders.

SAF-ST06 Researchers & Academia

Type	Stakeholder
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Research communities increasingly rely on official data for scientific inquiry and innovation. SAF supports them by ensuring that statistical systems are designed with openness, data sharing, and high-quality metadata in mind, facilitating advanced research use cases.

SAF-ST07 UNECE & Other International Bodies

Type	Stakeholder
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UNECE acts as a custodian of SAF, ensuring that it remains relevant, comprehensive, and widely adopted. Other bodies, such as Eurostat or the UN Statistics Division, are also stakeholders, as SAF complements their work in statistical modernization.

SAF-V1 Cross-Border Comparability

Type	Value
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Statistics gain value when they are comparable across borders. SAF enables this by promoting common standards, models, and processes, ensuring that statistics from different countries can be meaningfully compared.

SAF-V2 Innovation Enablement

Type	Value
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SAF is designed to support the integration of advanced technologies such as AI/ML, real-time analytics, and big data. By lowering the barriers to innovation, it enables NSIs to experiment safely while maintaining methodological soundness.

SAF-V3 Risk Reduction

Type	Value
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By embedding compliance, security, and governance into the architecture, SAF reduces risks associated with system failures, data breaches, or regulatory non-compliance. This value is particularly critical in maintaining the trust of citizens and stakeholders.

SAF-V4 Strategic Alignment

Type	Value
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SAF ensures that all IT initiatives within NSIs directly contribute to their mission of producing official statistics. This alignment improves governance, prioritization, and the impact of IT investments.

SAF-V5 Sustainability

Type	Value
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SAF promotes efficiency and modernization while reducing technical debt. By providing clear pathways for evolution, it ensures that NSIs can sustain their IT landscapes without repeated costly overhauls.

SAF-V6 Trust & Legitimacy

Type	Value
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Ultimately, the value of SAF lies in enhancing trust in official statistics. By ensuring transparency, ethical practices, and robust security, SAF reinforces the legitimacy of NSIs as authoritative providers of public data.

Seamless integration and automation across systems, enabling smooth and consistent data and process flows

Type	Value
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Secure and verifiable transfer of authentic data into the organization

Type	Value
------	-------

Secure and well-prepared data that is safe for public or controlled release

Type	Value
------	-------

Secure, organized, and traceable storage of raw data

Type	Value
------	-------

Secure, stable, and resilient platforms that ensure continuous statistical operations

Type	Value
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Security and continuity management documentation

Type	Outcome
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Service catalogues and SLAs

Type	Outcome
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Stakeholder maps and engagement strategies

Type	Outcome
------	---------

Standardized dataset with aligned metadata and classification mappings.

Type	Outcome
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Statistical quality framework and scorecards

Type	Outcome
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Statistically valid and transparent estimates

Type	Value
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Strong, transparent communication that builds understanding, collaboration, and trust

Type	Value
------	-------

Traceable, reusable, and standardized dataset

Type	Value
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Transparent and well-managed data and metadata that support reusability and accountability

Type	Value
------	-------

Trusted access to official statistics and data services through various channels

Type	Value
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User requirements documentation

Type	Outcome
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User satisfaction and impact assessments

Type	Outcome
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Validated new statistical or analytical methods

Type	Outcome
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Verified and quality-assured data that meets all Standards

Type	Value
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Verified, high-quality dataset with clear lineage and confidence in its reliability

Type	Value
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Verified, traceable, and approved statistical products ready for publication and reuse

Type	Value
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Relations

Access relation

Type	Access relation
Source	Data Processing
Target	Output Data (statistics)

Access relation

Type	Access relation
Source	Data Editing
Target	Processed Data

Access relation

Type	Access relation
Source	Dissemination
Target	Output Data (statistics)

Access relation

Type	Access relation
Source	Data Aquisition
Target	Raw Data

Access relation

Type	Access relation
Source	Data Editing
Target	Standardized Data

Access relation

Type	Access relation
Source	Data Aquisition
Target	Source Data

Access relation

Type	Access relation
Source	Dissemination
Target	Published Output Data (Released Data)

Access relation

Type	Access relation
Source	Standardization
Target	Standardized Data

Access relation

Type	Access relation
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Source	Data Processing
Target	Processed Data

Access relation

Type	Access relation
Source	Standardization
Target	Raw Data

Aggregation relation

Type	Aggregation relation
Source	1.5. Dissemination
Target	1.5.1. Publication management
archimate_connector_id ntifier	EAID_97F19B05_93DB_43f9_8002_70423F986ACF
archimate_connector_id ntifier	EAID_39324632_308B_4f8c_99FF_4907C6F036C9

Aggregation relation

Type	Aggregation relation
Source	3.4. Customer and chain interaction
Target	3.4.2. Contact management
archimate_connector_id ntifier	EAID_A4DC5F65_E7AC_4f80_A94D_4D3E49982B21
archimate_connector_id ntifier	EAID_95156BF4_C3E9_49be_9DA3_4EA9F238EF23

Aggregation relation

Type	Aggregation relation
Source	3. Supporting business capacities
Target	3.1. Business operations
archimate_connector_id ntifier	EAID_8319D0F7_B7C7_4abb_89B9_D9A706BF0E17
archimate_connector_id ntifier	EAID_74EAA754_03F8_43db_ADFA_27F2CDA6B515

Aggregation relation

Type	Aggregation relation
Source	3.4. Customer and chain interaction
Target	3.4.4. Self-service
archimate_connector_id ntifier	EAID_26EA7A7A_2A30_4c44_AAA0_33EDD17A781D
archimate_connector_id ntifier	EAID_0F2DE06D_DBA0_49ae_BE2A_8C61570FDD4C

Aggregation relation

Type	Aggregation relation
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Source	2.2. Research
Target	2.2.2. Product Innovation
archimate_connector_id ntifier	EAID_7E48486E_6DA0_43fc_8F71_84E85759407B
archimate_connector_id ntifier	EAID_14F94289_9C8D_4e59_85CF_33508F5C41AF

Aggregation relation

Type	Aggregation relation
Source	2.2. Research
Target	2.2.5. User research
archimate_connector_id ntifier	EAID_6D4C7176_5DD1_4647_8478_08ADCA447B6F
archimate_connector_id ntifier	EAID_879B8254_8957_425c_93E1_8509BBC90EBA

Aggregation relation

Type	Aggregation relation
Source	1. Primary statistical capabilities
Target	1.1. Statistical design and build
archimate_connector_id ntifier	EAID_3CD75267_C5A1_4b33_A7BE_BC811D6DB0DA
archimate_connector_id ntifier	EAID_25CC4413_8314_4609_ABDF_BADE15E537D8

Aggregation relation

Type	Aggregation relation
Source	1. Primary statistical capabilities
Target	1.4. Analysis
archimate_connector_id ntifier	EAID_A86C2C0C_F47C_4b4f_B158_A1363589BE03
archimate_connector_id ntifier	EAID_1CF0768F_8061_4a5b_A74B_4B0A95AE6737

Aggregation relation

Type	Aggregation relation
Source	3.1. Business operations
Target	3.1.1. Personnel management (HR)
archimate_connector_id ntifier	EAID_7CE2D5EE_A04F_4989_8BDA_62575FF51E83
archimate_connector_id ntifier	EAID_D040EE69_B385_47ee_83E4_3B94188DDEB0

Aggregation relation

Type	Aggregation relation
Source	1.1.1 Needs inventory
Target	1.1.1 Needs inventory

archimate_connector_id ntifier	EAID_3E3DFB62_130B_4bb1_8D98_D3F4E3CEB5DB
archimate_connector_id ntifier	EAID_A0AA9B1D_12A0_4ddf_9A96_00094B4438F3

Aggregation relation

Type	Aggregation relation
Source	3.4. Customer and chain interaction
Target	3.4.1. Services management
archimate_connector_id ntifier	EAID_E4614A1F_1786_4b3c_BECE_565CB6028D52
archimate_connector_id ntifier	EAID_BF057CEF_3071_4e4c_9621_504D4A146EF7

Aggregation relation

Type	Aggregation relation
Source	3.1. Business operations
Target	3.1.4. Communication management
archimate_connector_id ntifier	EAID_3C8183FD_7FF4_47b3_96D0_CAE371BA78C1
archimate_connector_id ntifier	EAID_3C735FF8_3F3A_419b_B657_D065BAF8183A

Aggregation relation

Type	Aggregation relation
Source	2.1. Data management
Target	2.1.2. Metadata management
archimate_connector_id ntifier	EAID_60641A1C_4B60_4c1b_9996_5EB86D0AB27E
archimate_connector_id ntifier	EAID_945C47A2_FF28_4e67_8F50_19AD37473430

Aggregation relation

Type	Aggregation relation
Source	3.2. Computerization and automation
Target	3.2.2. Functional management
archimate_connector_id ntifier	EAID_38809BCA_EE0D_4c75_9A70_E5ADC879322E
archimate_connector_id ntifier	EAID_A80BA036_3CD3_4d53_B7F5_EAF1FB4C0F1C

Aggregation relation

Type	Aggregation relation
Source	3.4. Customer and chain interaction
Target	3.4.3. Service Agreement

archimate_connector_id ntifier	EAID_8D369CF9_343C_4512_8314_FAC1C6DC363D
archimate_connector_id ntifier	EAID_DC8D941A_E2F0_4f88_ACD4_1CD1EF67EE9B

Aggregation relation

Type	Aggregation relation
Source	2.2. Research
Target	2.2.2. Product Innovation
archimate_connector_id ntifier	EAID_29B72830_E631_41b6_8B48_875736BFDF02
archimate_connector_id ntifier	EAID_5695A7A6_75B9_43ac_BA16_FDDECA8D3F36

Aggregation relation

Type	Aggregation relation
Source	4. Governance capabilities
Target	4.1.2. Policy and planning
archimate_connector_id ntifier	EAID_B3C5109E_7851_4ac5_A6AA_0BF88DE4C3A6
archimate_connector_id ntifier	EAID_944568D8_DE25_4551_B012_1C77245BA737

Aggregation relation

Type	Aggregation relation
Source	4. Governance capabilities
Target	4.1.3. Change management
archimate_connector_id ntifier	EAID_41186B00_9C59_4801_90D2_0FFD52DFCACC
archimate_connector_id ntifier	EAID_39EAD962_6FA0_47c1_9F1F_192FD5AA0B65

Aggregation relation

Type	Aggregation relation
Source	1.2. Data collection
Target	1.2.2. Secondary data collection
archimate_connector_id ntifier	EAID_E7A8AFAB_83C2_4cd6_A227_FE2B9AD6623F
archimate_connector_id ntifier	EAID_F0C02C6F_6BD7_4066_8AF3_21BDC81CD02B

Aggregation relation

Type	Aggregation relation
Source	1.1. Statistical design and build
Target	1.1.1 Needs inventory

archimate_connector_id ntifier	EAID_11681EE8_AD3E_4f80_B14E_D1BCAF00CEBE
archimate_connector_id ntifier	EAID_0A85C9A8_38FC_4e44_9497_143461E5EAFE

Aggregation relation

Type	Aggregation relation
Source	1. Primary statistical capabilities
Target	1.3. Processing
archimate_connector_id ntifier	EAID_48346A7B_9D23_4757_B646_DFC09263A640
archimate_connector_id ntifier	EAID_BE196EAB_B59C_41c4_A352_2C20CEE2B752

Aggregation relation

Type	Aggregation relation
Source	3.3. Security and Privacy
Target	3.3.4. Logical identity and access control
archimate_connector_id ntifier	EAID_2DF7ECB2_85E5_4276_88FF_5E18982B62FE
archimate_connector_id ntifier	EAID_73503181_C1F9_4426_B576_3A0A7C34F637

Aggregation relation

Type	Aggregation relation
Source	1.4. Analysis
Target	1.4.1. Statistical (output) analysis
archimate_connector_id ntifier	EAID_AFB24493_71FA_4274_85AF_0F12066DFB96
archimate_connector_id ntifier	EAID_098D56BD_CE97_4fe4_BCEF_DD41AFB2023C

Aggregation relation

Type	Aggregation relation
Source	2.2. Research
Target	2.2.1. Data source exploration
archimate_connector_id ntifier	EAID_5E2182CD_F10E_43b6_87B6_3643502EE188
archimate_connector_id ntifier	EAID_A97D433A_83DB_4748_A85A_AB6725B9CCFD

Aggregation relation

Type	Aggregation relation
Source	3. Supporting business capacities
Target	3.4. Customer and chain interaction

archimate_connector_id ntifier	EAID_A9678DF5_92D7_4790_AA71_52506B260EB0
archimate_connector_id ntifier	EAID_EF30F84C_56B0_4203_A103_EA84B4983621

Aggregation relation

Type	Aggregation relation
Source	3.2. Computerization and automation
Target	3.2.3. Technical management
archimate_connector_id ntifier	EAID_DF4EC9D5_7898_402e_B751_4E09E173467A
archimate_connector_id ntifier	EAID_631CA0D3_BCA5_4401_9F7D_ADB6338095BE

Aggregation relation

Type	Aggregation relation
Source	1.2. Data collection
Target	1.2.2. Secondary data collection
archimate_connector_id ntifier	EAID_8B3D80D3_404B_4941_B471_C224E29B5F9D
archimate_connector_id ntifier	EAID_78895893_9A4D_47ed_98D0_F352BBF6494F

Aggregation relation

Type	Aggregation relation
Source	3.1. Business operations
Target	3.1.7. Legal support
archimate_connector_id ntifier	EAID_6E7DEA62_83D4_4bcd_87D7_6810033DD48C
archimate_connector_id ntifier	EAID_38B5D059_30C1_4cb4_B51D_EAF7E702AC42

Aggregation relation

Type	Aggregation relation
Source	3.1. Business operations
Target	3.1.3. Financial management
archimate_connector_id ntifier	EAID_F87CDEE7_067E_49e2_AE3E_759DFF229CB1
archimate_connector_id ntifier	EAID_6420196D_EB54_4e57_B8CA_AEA5922F913F

Aggregation relation

Type	Aggregation relation
Source	4. Governance capabilities
Target	4.1.4. Improve management

archimate_connector_id ntifier	EAID_2788BD61_ABA6_425a_8915_BC6A862B502A
archimate_connector_id ntifier	EAID_A34E10F0_44E8_4569_865F_157B97159511

Aggregation relation

Type	Aggregation relation
Source	3.1. Business operations
Target	3.1.3. Financial management
archimate_connector_id ntifier	EAID_9611F27A_78B5_4860_8D3C_493463685AE4
archimate_connector_id ntifier	EAID_6E981534_0153_40c3_A1CA_55EB33B55F9A

Aggregation relation

Type	Aggregation relation
Source	4. Governance capabilities
Target	4.1.6. Collaboration
archimate_connector_id ntifier	EAID_B1FB7742_01E3_4201_BA23_1CB4F18FE33D
archimate_connector_id ntifier	EAID_CE34FBC1_3647_4dad_A06D_D78A3237EA9C

Aggregation relation

Type	Aggregation relation
Source	3.1. Business operations
Target	3.1.6. Facility management
archimate_connector_id ntifier	EAID_7358C245_15EB_4ddc_9DD5_109B59D456CA
archimate_connector_id ntifier	EAID_56A4C1EB_C5B2_41ab_889A_D0AB04DC320C

Aggregation relation

Type	Aggregation relation
Source	1.3. Processing
Target	1.3.1. Data standardization
archimate_connector_id ntifier	EAID_70FF7E47_9063_4507_AD3E_55C13CB930EB
archimate_connector_id ntifier	EAID_61FD7F4F_72BD_4195_B515_83D50B9983C8

Aggregation relation

Type	Aggregation relation
Source	3.3. Security and Privacy
Target	3.3.1. Privacy management

archimate_connector_id ntifier	EAID_91DD14AA_818E_40d5_A164_D24444082A43
archimate_connector_id ntifier	EAID_8E40513F_26CC_4f00_9F44_411E508782B8

Aggregation relation

Type	Aggregation relation
Source	2.2. Research
Target	2.2.3. Methodology
archimate_connector_id ntifier	EAID_880E8F8F_974B_4826_A0DB_9817F19F64D8
archimate_connector_id ntifier	EAID_5579F4AE_1D07_49f0_818E_DDE717A7FCFE

Aggregation relation

Type	Aggregation relation
Source	1.3. Processing
Target	1.3.7. Statistical security
archimate_connector_id ntifier	EAID_F6A235DF_5513_45a5_84A3_1C32B7DE3028
archimate_connector_id ntifier	EAID_AF76CC6B_CA5C_42de_BD86_15E67B0E049F

Aggregation relation

Type	Aggregation relation
Source	3. Supporting business capacities
Target	3.3. Security and Privacy
archimate_connector_id ntifier	EAID_E9778F18_8054_41dd_8E55_34FB0DB8E704
archimate_connector_id ntifier	EAID_FF2326A9_D5FF_4949_A32E_2D4BD15DAC41

Aggregation relation

Type	Aggregation relation
Source	1.2. Data collection
Target	1.2.1. Primary data collection
archimate_connector_id ntifier	EAID_7F406563_EE0D_4c9a_81A5_8D86A29A5373
archimate_connector_id ntifier	EAID_3B38D3D1_1BA7_4b8d_ABCD_87AC27A536E1

Aggregation relation

Type	Aggregation relation
Source	3.3. Security and Privacy
Target	3.3.2. Security planning

archimate_connector_id ntifier	EAID_4348AB51_3720_4528_98B7_6F641CA49CB7
archimate_connector_id ntifier	EAID_01936B7A_1437_4849_896E_923F0600E935

Aggregation relation

Type	Aggregation relation
Source	1.2. Data collection
Target	1.2.1. Primary data collection
archimate_connector_id ntifier	EAID_673033C5_35CB_4851_AB48_ACF752933D2D
archimate_connector_id ntifier	EAID_9A605532_5526_4248_8065_A4FBAAC5AEEE

Aggregation relation

Type	Aggregation relation
Source	1.5. Dissemination
Target	1.5.1. Publication management
archimate_connector_id ntifier	EAID_25B0B81B_AF46_4a7d_9FD7_A4AEAD1E51CF
archimate_connector_id ntifier	EAID_45A74D7B_B985_4ac0_A03B_00313242E6F8

Aggregation relation

Type	Aggregation relation
Source	2.1. Data management
Target	2.1.1. Data lifecycle management
archimate_connector_id ntifier	EAID_39F4F9AF_ACC6_4422_843E_FD174C846589
archimate_connector_id ntifier	EAID_60070876_E805_4d24_B26B_AEFC0E573E9C

Aggregation relation

Type	Aggregation relation
Source	1.5. Dissemination
Target	1.5.2. Promotion to use
archimate_connector_id ntifier	EAID_E075BE1F_9C24_4fd4_841B_5E4114B027CB
archimate_connector_id ntifier	EAID_9499D8A9_BB68_4285_BD7F_1A4848D84879

Aggregation relation

Type	Aggregation relation
Source	2.2. Research
Target	2.2.1. Data source exploration

archimate_connector_id ntifier	EAID_6C579B2F_63D1_4422_8480_65DFF8CB83C8
archimate_connector_id ntifier	EAID_E94E93E5_8BFE_45cc_A975_86D84FB1646E

Aggregation relation

Type	Aggregation relation
Source	2. Supporting Data capabilities
Target	2.2. Research
archimate_connector_id ntifier	EAID_25372267_48B2_4ffe_BB63_65DFBD3CAEE4
archimate_connector_id ntifier	EAID_AC08DF79_C7FC_41ee_922D_AC9521779FF1

Aggregation relation

Type	Aggregation relation
Source	1.5. Dissemination
Target	1.5.3. Facilitate data access
archimate_connector_id ntifier	EAID_B9D5BF4B_2FBD_4646_8743_E32AF9751C88
archimate_connector_id ntifier	EAID_51FE68C7_F0F3_4575_965F_6BEEF886A25F

Aggregation relation

Type	Aggregation relation
Source	1.1. Statistical design and build
Target	1.1.2. Design statistics product
archimate_connector_id ntifier	EAID_EC1CEE4F_9C7D_402e_9EA8_C52ECED1003C
archimate_connector_id ntifier	EAID_1EBB99C3_64B9_421d_A099_EA136EF8F8DF

Aggregation relation

Type	Aggregation relation
Source	1.4. Analysis
Target	1.4.1. Statistical (output) analysis
archimate_connector_id ntifier	EAID_CB19AC1D_866A_4943_88F7_EB697E350F48
archimate_connector_id ntifier	EAID_9A13C0CC_1AFA_4ab1_8260_23020E87B137

Aggregation relation

Type	Aggregation relation
Source	3. Supporting business capacities
Target	3.1. Business operations

archimate_connector_id ntifier	EAID_F473C75B_D9CD_4d62_8F7A_D326762B71C9
archimate_connector_id ntifier	EAID_C292566E_B825_4654_AD41_CB69B362E967

Aggregation relation

Type	Aggregation relation
Source	1.3. Processing
Target	1.3.1. Data standardization
archimate_connector_id ntifier	EAID_7582A9AC_919E_4ac9_B298_014319EE02A7
archimate_connector_id ntifier	EAID_AFCFBB55_7A5C_4852_98BE_05B51A1EA6E8

Aggregation relation

Type	Aggregation relation
Source	2. Supporting Data capabilities
Target	2.1. Data management
archimate_connector_id ntifier	EAID_798D8FB4_0A63_4cca_86CF_655A63ADFB2F
archimate_connector_id ntifier	EAID_50BCA611_5192_4c37_A5DF_141CB72A2DF9

Aggregation relation

Type	Aggregation relation
Source	3.2. Computerization and automation
Target	3.2.1. Application development
archimate_connector_id ntifier	EAID_B23E577B_EE0D_43fe_9C00_34640AEDAA7

Aggregation relation

Type	Aggregation relation
Source	4. Governance capabilities
Target	4.1.1. Strategy and governance
archimate_connector_id ntifier	EAID_A3D311C3_26FC_4e11_9B84_2B9DBA7DF918
archimate_connector_id ntifier	EAID_7EB2C2FE_24F7_413a_A5C7_00C0DA1B36B9

Aggregation relation

Type	Aggregation relation
Source	1. Primary statistical capabilities
Target	1.3. Processing
archimate_connector_id ntifier	EAID_1C232491_EEE0_46ff_8516_76AF7713AB45
archimate_connector_id ntifier	EAID_3825A6D4_A01F_4208_ABAA_400F7559F062

Aggregation relation

Type	Aggregation relation
Source	1.1. Statistical design and build
Target	1.1.3. Build statistics product
archimate_connector_id ntifier	EAID_00CB5ED7_FB82_4fb7_BF0D_24AEEF1618AE
archimate_connector_id ntifier	EAID_71573B32_3C3A_47a3_AB06_5C3A0ED9C34C

Aggregation relation

Type	Aggregation relation
Source	2.1. Data management
Target	2.1.2. Metadata management
archimate_connector_id ntifier	EAID_B734674F_BCA4_4b6d_9712_50932B66ACC4
archimate_connector_id ntifier	EAID_6E9BC931_3B9E_4966_B299_4253134E28CE

Aggregation relation

Type	Aggregation relation
Source	1. Primary statistical capabilities
Target	1.2. Data collection
archimate_connector_id ntifier	EAID_995F4521_B383_477c_AEC8_5CEA4F8758E2
archimate_connector_id ntifier	EAID_3C2A1DF7_DD7A_45b8_B2F8_E1476B34FA59

Aggregation relation

Type	Aggregation relation
Source	1. Primary statistical capabilities
Target	1.2. Data collection
archimate_connector_id ntifier	EAID_60C0A0F0_219C_46ac_8D58_3311BFC5A397
archimate_connector_id ntifier	EAID_E148ACA7_6901_45c7_8945_868C7E41090D

Aggregation relation

Type	Aggregation relation
Source	2. Supporting Data capabilities
Target	2.2. Research
archimate_connector_id ntifier	EAID_14547305_095C_43a5_8BA3_D5DD0A263CA2
archimate_connector_id ntifier	EAID_02B82B2A_7A15_44f0_9AF0_80169647F5F3

Aggregation relation

Type	Aggregation relation
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Source	2.1. Data management
Target	2.1.3. Statistical quality management
archimate_connector_id ntifier	EAID_FF3F05AB_1ADE_4fe6_AA75_8350E22BA546
archimate_connector_id ntifier	EAID_BBC9CF06_AE25_4d1e_97F9_6FE3DAFD97D5

Aggregation relation

Type	Aggregation relation
Source	2.2. Research
Target	2.2.3. Methodology
archimate_connector_id ntifier	EAID_A9362C22_9E8A_4566_BC6D_76D63B0AD87B
archimate_connector_id ntifier	EAID_ED32F2D6_05B8_4b03_97EE_DFFDDD5D2BC0

Aggregation relation

Type	Aggregation relation
Source	SAF-P09 No data without metadata and classification
Target	SAF-P08 Data is shared property
archimate_connector_id ntifier	EAID_02EA1CBB_6B13_4c74_9547_E0B0D2B97E48

Aggregation relation

Type	Aggregation relation
Source	2.1. Data management
Target	2.1.1. Data lifecycle management
archimate_connector_id ntifier	EAID_8A1D51F5_6970_4717_9E51_1083C6446FCB
archimate_connector_id ntifier	EAID_11DFBF14_F66A_410d_8CF6_F0E239DB1208

Aggregation relation

Type	Aggregation relation
Source	3.1. Business operations
Target	3.1.2. Purchasing & contract management
archimate_connector_id ntifier	EAID_8E2C87AC_614C_40b8_9DAA_742ED1CE652C
archimate_connector_id ntifier	EAID_81A23F1C_65BF_4724_895F_416BA91CF981

Aggregation relation

Type	Aggregation relation
Source	3.1. Business operations
Target	3.1.5. Administrative support

archimate_connector_id ntifier	EAID_E00A5858_F230_426d_AD05_CABD3186E67F
archimate_connector_id ntifier	EAID_2EBB285C_79B8_4f33_AF11_1045FA077F01

Aggregation relation

Type	Aggregation relation
Source	3.1. Business operations
Target	3.1.7. Legal support
archimate_connector_id ntifier	EAID_68E2136C_AD33_4c20_8393_C10CF082E148
archimate_connector_id ntifier	EAID_7B95A23E_F66E_4984_A225_8BF49ACB66B0

Aggregation relation

Type	Aggregation relation
Source	3.2. Computerization and automation
Target	3.2.5. IT infrastructure
archimate_connector_id ntifier	EAID_A63F9166_CB93_4b47_89DB_195D1423E04C
archimate_connector_id ntifier	EAID_3FE7F439_BB23_451d_8501_29001ED2110E

Aggregation relation

Type	Aggregation relation
Source	3.1. Business operations
Target	3.1.8. Information management and archive management
archimate_connector_id ntifier	EAID_BD250864_4A19_4cbf_8FA4_0B1797F79E2E
archimate_connector_id ntifier	EAID_911DB7C5_5F9C_4de6_8530_15BCBF04B544

Aggregation relation

Type	Aggregation relation
Source	1.5. Dissemination
Target	1.5.4. Statistical information management
archimate_connector_id ntifier	EAID_FED1FF1D_CD24_4271_9849_3BEA1FB442E9
archimate_connector_id ntifier	EAID_22AD27F2_6F4C_4f6e_8384_ED8DAB06D88C

Aggregation relation

Type	Aggregation relation
Source	2.2. Research
Target	2.2.4. Data services

archimate_connector_id ntifier	EAID_BFC20E0E_9262_40d8_925C_394568C43B26
archimate_connector_id ntifier	EAID_8FC9AD54_4654_4759_8B3A_5D16043CCC7D

Aggregation relation

Type	Aggregation relation
Source	1. Primary statistical capabilities
Target	1.5. Dissemination
archimate_connector_id ntifier	EAID_791BA2C9_C933_4ca4_B256_E59A75DEF841
archimate_connector_id ntifier	EAID_D2B9722D_D444_4a62_B8F7_919E3140C0FD

Aggregation relation

Type	Aggregation relation
Source	1.3. Processing
Target	1.3.7. Statistical security
archimate_connector_id ntifier	EAID_48ADC171_63DD_4af9_A144_A0E9D4F334E7
archimate_connector_id ntifier	EAID_7E79C844_5B48_4ec7_9D1D_1FA3FE86A8B2

Aggregation relation

Type	Aggregation relation
Source	3.3. Security and Privacy
Target	3.3.5. Physical security & access control
archimate_connector_id ntifier	EAID_A7B9BCD1_B627_4d9e_B0C8_157A1F65E2B4
archimate_connector_id ntifier	EAID_DA5FA71A_316C_4968_A3A6_BB1D75D6F6F5

Aggregation relation

Type	Aggregation relation
Source	3. Supporting business capacities
Target	3.2. Computerization and automation
archimate_connector_id ntifier	EAID_4AE77222_3A73_40d3_B862_1AE06A2B7D19
archimate_connector_id ntifier	EAID_80FFD3BD_30E2_4dc1_B2A6_326DD562B7AF

Aggregation relation

Type	Aggregation relation
Source	1. Primary statistical capabilities
Target	1.4. Analysis

archimate_connector_id ntifier	EAID_2106D401_2FD2_4457_B3C1_054638EAB57B
archimate_connector_id ntifier	EAID_D9DFD963_DE34_4334_9B8E_3D0D31D112CE

Aggregation relation

Type	Aggregation relation
Source	3.3. Security and Privacy
Target	3.3.3. Security incident management
archimate_connector_id ntifier	EAID_1FD0B843_10C4_4ec5_94B5_47A98C3F7F22
archimate_connector_id ntifier	EAID_BDE5ADE1_3C2E_4645_8ECC_FC5C84AD9DA7

Aggregation relation

Type	Aggregation relation
Source	2. Supporting Data capabilities
Target	2.1. Data management
archimate_connector_id ntifier	EAID_DE3EB3C0_4075_4dfa_A8A0_F9AFC382F9CC
archimate_connector_id ntifier	EAID_396411C5_6351_4110_AA45_4086ED1053EB

Association relation

Type	Association relation
Source	VS-08.6 Promote Knowledge Sharing and Capability Building Organization-wide uptake of proven innovations and a culture of continuous improvement

Association relation

Type	Association relation
Source	VS-01.2 Establish Data Agreement and Access Right Legal, ethical, and transparent access to data under controlled conditions

Association relation

Type	Association relation
Source	VS-09.1 Define Digital and Technical Requirements Clear, shared understanding of what the infrastructure must deliver to support the business

Association relation

Type	Association relation
Source	VS-04.4 Perform Model-Based Analysis Deeper analytical insights and predictive understanding of data relationships

Association relation

Type	Association relation
Source	VS-02.3 Align Metadata and Semantics
Target	Harmonized meaning and context across datasets

Association relation

Type	Association relation
Source	VS-05.3 Perform Consistency and Coherence Checks
Target	Reliable and internally consistent statistical products

Association relation

Type	Association relation
Source	VS-09.5 Ensure Performance, Security, and Continuity
	Secure, stable, and resilient platforms that ensure continuous statistical operations

Association relation

Type	Association relation
Source	VS-02.4 Validate Structural and Semantic Conformance
Target	Verified and quality-assured data that meets all Standards

Association relation

Type	Association relation
Source	VS-07.5 Update and Improve Governance Mechanisms
	Adaptive governance that keeps the organization compliant and future-proof

Association relation

Type	Association relation
Source	VS-08.4 Develop / Document New Methodologies or Tools
Target	Reusable, documented innovation assets

Association relation

Type	Association relation
Source	VS-04.2 Apply Weighting and Calibration
Target	Balanced datasets that accurately represent real-world populations

Association relation

Type	Association relation
Source	VS-05.4 Generate Analytical and Interpretive Outputs
	Analytical narratives that make complex statistical data understandable

Association relation

Type	Association relation
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Source	VS-07.5 Audit, Certify, and Report Compliance
Target	Documented and certified proof of compliance and statistical quality

Association relation

Type	Association relation
Source	VS-06.3 Publish and Provide Access Services
	Trusted access to official statistics and data services through various channels

Association relation

Type	Association relation
Source	VS-06.5 Monitor Usage and Feedback
	A feedback-driven improvement cycle that keeps data services relevant, accessible, and impactful

Association relation

Type	Association relation
Source	VS-10.2 Collect and Analyze User Needs
	Actionable insights that guide product design, data publication, and service improvement

Association relation

Type	Association relation
Source	VS-10.4 Deliver Services and Manage Relationships
	Consistent, high-quality user experience and strong, trust-based relationships with key stakeholders

Association relation

Type	Association relation
Source	VS-09.6 Evolve and Modernize the Technical Ecosystem
	A continuously improving infrastructure that adapts to technological progress and future organizational needs

Association relation

Type	Association relation
Source	VS-06.2 Prepare Data for Dissemination
	Secure and well-prepared data that is safe for public or controlled release

Association relation

Type	Association relation
Source	VS-03.2 Perform Data Editing and Correction
Target	Clean, coherent data that accurately represents real-world information

Association relation

Type	Association relation
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Source	VS-07.3 Implement Data and Metadata Governance
	Transparent and well-managed data and metadata that support reusability and accountability

Association relation

Type	Association relation
Source	VS-02.1 Profile and Assess Data Structure
Target	Full understanding of the dataset structure and format

Association relation

Type	Association relation
Source	VS-03.3 Impute Missing or Non-Response Data
Target	Datasets that preserve statistical representativeness and minimize bias

Association relation

Type	Association relation
Source	VS-01.5 Store and Register Raw Data Artefacts
Target	Secure, organized, and traceable storage of raw data

Association relation

Type	Association relation
Source	VS-04.5 Validate and Document Results
	Reliable and validated analytical results ready for composition, interpretation, and dissemination

Association relation

Type	Association relation
Source	VS-01.4 Validate and Log Data Intake
	Confidence that received data is complete, structured, and fully traceable

Association relation

Type	Association relation
Source	VS-02.5 Register Standardized Artefacts
Target	Traceable, reusable, and standardized dataset

Association relation

Type	Association relation
Source	VS-03.4 Derive Additional Variables and Indicators
Target	Enriched datasets that support deeper analysis and insight generation

Association relation

Type	Association relation
Source	VS-06.4 Support and Engage Users
	Informed, confident users who can correctly understand and apply

statistical data

Association relation

Type	Association relation
Source	VS-08.2 Conduct Research and Proof-of-Concepts Evidence about what works and what does not, based on real tests — not assumptions

Association relation

Type	Association relation
Source	VS-05.1 Select and Integrate Source Datasets Integrated data foundation for consistent and comprehensive statistical products

Association relation

Type	Association relation
Source	VS-07.1 Define Governance Frameworks and Policies A clear governance framework that defines ownership, accountability, and rules for data and statistical operations

Association relation

Type	Association relation
Source	VS-01.3 Collect or Receive Data
Target	Secure and verifiable transfer of authentic data into the organization

Association relation

Type	Association relation
Source	VS-04.1 Design Estimation and Analytical Approach
Target	clear, well-documented plan for estimation and analysis

Association relation

Type	Association relation
Source	VS-10.3 Define and Manage Service Portfolios A transparent and organized set of services that clearly show what users can access and how

Association relation

Type	Association relation
Source	VS-07.4 Monitor Quality and Compliance Indicators Measurable assurance that data and processes meet defined quality and compliance standards

Association relation

Type	Association relation
Source	VS-08.1 Identify Innovation Needs and Opportunities
Target	A prioritized list of innovation and method-development opportunities

Association relation

Type	Association relation
Source	VS-05.5 Approve and Register Statistical Products
	Verified, traceable, and approved statistical products ready for publication and reuse

Association relation

Type	Association relation
Source	VS-10.5 Engage and Communicate with Stakeholders
	Strong, transparent communication that builds understanding, collaboration, and trust

Association relation

Type	Association relation
Source	VS-06.1 Plan Dissemination and Access Strategy
	A clear dissemination plan that balances accessibility, compliance and timing

Association relation

Type	Association relation
Source	VS-09.2 Design Platform and Infrastructure Architecture
	A coherent, future-proof technical blueprint that connects business needs with technological capabilities

Association relation

Type	Association relation
Source	VS-02.2 Apply Standard Classifications and Formats
Target	Consistent, interoperable data aligned with recognized standards

Association relation

Type	Association relation
Source	VS-03.1 Conduct Error Detection/ Quality Diagnostics
	Clear understanding of data quality issues and areas requiring correction

Association relation

Type	Association relation
Source	VS-10.1 Identify and Segment Stakeholders
	A clear understanding of who the organization's stakeholders are and what they need from its data and services

Association relation

Type	Association relation
Source	VS-09.3 Provision and Configure Platforms
	Operational platforms that are configured, secure, and ready to

support statistical production and analysis

Association relation

Type	Association relation
Source	VS-05.2 Construct Composite Indicators or Indices
	Clear, aggregated indicators that simplify complex statistical information

Association relation

Type	Association relation
Source	VS-04.3 Estimate and Aggregate Data
Target	Statistically valid and transparent estimates

Association relation

Type	Association relation
Source	VS-01.1 Identify Data Needs and Sources
	Clear understanding of what data is needed and where it can be obtained.

Association relation

Type	Association relation
Source	VS-08.3 Evaluate Methodological Soundness and Feasibility
Target	Assurance that innovation does not weaken statistical quality or trust

Association relation

Type	Association relation
Source	VS-09.4 Enable Integration and Orchestration Services
	Seamless integration and automation across systems, enabling smooth and consistent data and process flows

Association relation

Type	Association relation
Source	VS-07.2 Establish Legal and Regulatory Compliance Mechanisms
	Full alignment with legal and ethical standards, reducing risks and maintaining public trust

Association relation

Type	Association relation
Source	VS-03.5 Perform Quality Evaluation and Versioning
	Verified, high-quality dataset with clear lineage and confidence in its reliability

Association relation

Type	Association relation
Source	VS-08.5 Integrate Innovation into Production Environments
	Innovation that is operational, supported, and available to real

statistical production

Association relation

Type	Association relation
Source	VS-10.6 Monitor Impact and Satisfaction
	Insight into how statistics create societal value and how stakeholder relationships can be strengthened

Composition relation

Type	Composition relation
Source	VS-10.0 Customer and Stakeholder Engagement Value Stream
Target	VS-10.1 Identify and Segment Stakeholders

Composition relation

Type	Composition relation
Source	Standardization
Target	Classify Data

Composition relation

Type	Composition relation
Source	VS-07.0 Governance and Compliance Value Stream
Target	VS-07.4 Monitor Quality and Compliance Indicators

Composition relation

Type	Composition relation
Source	VS-05.0 Statistical Product Composition Value Stream
Target	VS-05.5 Approve and Register Statistical Products

Composition relation

Type	Composition relation
Source	VS-10.0 Customer and Stakeholder Engagement Value Stream
Target	VS-10.3 Define and Manage Service Portfolios

Composition relation

Type	Composition relation
Source	VS-01.0 Data Acquisition and Ingestion Value Stream
Target	VS-01.5 Store and Register Raw Data Artefacts

Composition relation

Type	Composition relation
Source	VS-08.0 Innovation and Methodological Development Value Stream
Target	VS-08.5 Integrate Innovation into Production Environments

Composition relation

Type	Composition relation
Source	VS-05.0 Statistical Product Composition Value Stream
Target	VS-05.2 Construct Composite Indicators or Indices

Composition relation

Type	Composition relation
Source	VS-09.0 Infrastructure and Platform Enablement Value Stream
Target	VS-09.4 Enable Integration and Orchestration Services

Composition relation

Type	Composition relation
Source	VS-09.0 Infrastructure and Platform Enablement Value Stream
Target	VS-09.2 Design Platform and Infrastructure Architecture

Composition relation

Type	Composition relation
Source	VS-08.0 Innovation and Methodological Development Value Stream
Target	VS-08.3 Evaluate Methodological Soundness and Feasibility

Composition relation

Type	Composition relation
Source	VS-04.0 Estimation, Analysis, and Statistical Modeling Value Stream
Target	VS-04.1 Design Estimation and Analytical Approach

Composition relation

Type	Composition relation
Source	VS-02.0 Data Standardization and Integration Value Stream
Target	VS-02.2 Apply Standard Classifications and Formats

Composition relation

Type	Composition relation
Source	Data Editing
Target	Integrate Data

Composition relation

Type	Composition relation
Source	VS-10.0 Customer and Stakeholder Engagement Value Stream
Target	VS-10.6 Monitor Impact and Satisfaction

Composition relation

Type	Composition relation
Source	VS-02.0 Data Standardization and Integration Value Stream
Target	VS-02.1 Profile and Assess Data Structure

Composition relation

Type	Composition relation
Source	VS-02.0 Data Standardization and Integration Value Stream
Target	VS-02.5 Register Standardized Artefacts

Composition relation

Type	Composition relation
Source	VS-01.0 Data Acquisition and Ingestion Value Stream
Target	VS-01.1 Identify Data Needs and Sources

Composition relation

Type	Composition relation
Source	VS-03.0 Data Editing and Enrichment Value Stream
Target	VS-03.2 Perform Data Editing and Correction

Composition relation

Type	Composition relation
Source	Data Editing
Target	Impute Data

Composition relation

Type	Composition relation
Source	VS-10.0 Customer and Stakeholder Engagement Value Stream
Target	VS-10.2 Collect and Analyze User Needs

Composition relation

Type	Composition relation
Source	VS-07.0 Governance and Compliance Value Stream
Target	VS-07.3 Implement Data and Metadata Governance

Composition relation

Type	Composition relation
Source	VS-08.0 Innovation and Methodological Development Value Stream
Target	VS-08.4 Develop / Document New Methodologies or Tools

Composition relation

Type	Composition relation
Source	VS-03.0 Data Editing and Enrichment Value Stream
Target	VS-03.1 Conduct Error Detection/ Quality Diagnostics

Composition relation

Type	Composition relation
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Source	VS-01.0 Data Acquisition and Ingestion Value Stream
Target	VS-01.4 Validate and Log Data Intake

Composition relation

Type	Composition relation
Source	VS-06.0 Dissemination and Service Delivery Value Stream
Target	VS-06.3 Publish and Provide Access Services

Composition relation

Type	Composition relation
Source	VS-07.0 Governance and Compliance Value Stream
Target	VS-07.5 Audit, Certify, and Report Compliance

Composition relation

Type	Composition relation
Source	Data Acquisition
Target	Extract Data

Composition relation

Type	Composition relation
Source	VS-04.0 Estimation, Analysis, and Statistical Modeling Value Stream
Target	VS-04.2 Apply Weighting and Calibration

Composition relation

Type	Composition relation
Source	VS-08.0 Innovation and Methodological Development Value Stream
Target	VS-08.2 Conduct Research and Proof-of-Concepts

Composition relation

Type	Composition relation
Source	VS-10.0 Customer and Stakeholder Engagement Value Stream
Target	VS-10.5 Engage and Communicate with Stakeholders

Composition relation

Type	Composition relation
Source	Standardization
Target	Transform Data

Composition relation

Type	Composition relation
Source	VS-07.0 Governance and Compliance Value Stream
Target	VS-07.1 Define Governance Frameworks and Policies

Composition relation

Type	Composition relation
Source	VS-09.0 Infrastructure and Platform Enablement Value Stream
Target	VS-09.6 Evolve and Modernize the Technical Ecosystem

Composition relation

Type	Composition relation
Source	VS-05.0 Statistical Product Composition Value Stream
Target	VS-05.3 Perform Consistency and Coherence Checks

Composition relation

Type	Composition relation
Source	VS-08.0 Innovation and Methodological Development Value Stream
Target	VS-08.6 Promote Knowledge Sharing and Capability Building

Composition relation

Type	Composition relation
Source	VS-09.0 Infrastructure and Platform Enablement Value Stream
Target	VS-09.5 Ensure Performance, Security, and Continuity

Composition relation

Type	Composition relation
Source	VS-05.0 Statistical Product Composition Value Stream
Target	VS-05.1 Select and Integrate Source Datasets

Composition relation

Type	Composition relation
Source	Data Aquisition
Target	Receive Data

Composition relation

Type	Composition relation
Source	VS-08.0 Innovation and Methodological Development Value Stream
Target	VS-08.1 Identify Innovation Needs and Opportunities

Composition relation

Type	Composition relation
Source	VS-04.0 Estimation, Analysis, and Statistical Modeling Value Stream
Target	VS-04.4 Perform Model-Based Analysis

Composition relation

Type	Composition relation
Source	VS-06.0 Dissemination and Service Delivery Value Stream
Target	VS-06.2 Prepare Data for Dissemination

Composition relation

Type	Composition relation
Source	VS-10.0 Customer and Stakeholder Engagement Value Stream
Target	VS-10.4 Deliver Services and Manage Relationships

Composition relation

Type	Composition relation
Source	VS-06.0 Dissemination and Service Delivery Value Stream
Target	VS-06.1 Plan Dissemination and Access Strategy

Composition relation

Type	Composition relation
Source	VS-01.0 Data Acquisition and Ingestion Value Stream
Target	VS-01.3 Collect or Receive Data

Composition relation

Type	Composition relation
Source	VS-04.0 Estimation, Analysis, and Statistical Modeling Value Stream
Target	VS-04.5 Validate and Document Results

Composition relation

Type	Composition relation
Source	VS-06.0 Dissemination and Service Delivery Value Stream
Target	VS-06.5 Monitor Usage and Feedback

Composition relation

Type	Composition relation
Source	VS-03.0 Data Editing and Enrichment Value Stream
Target	VS-03.3 Impute Missing or Non-Response Data

Composition relation

Type	Composition relation
Source	Data Processing
Target	Estimate Data

Composition relation

Type	Composition relation
Source	VS-09.0 Infrastructure and Platform Enablement Value Stream
Target	VS-09.3 Provision and Configure Platforms

Composition relation

Type	Composition relation
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Source	Data Processing
Target	Aggregate Data

Composition relation

Type	Composition relation
Source	Dissemination
Target	Data Publication

Composition relation

Type	Composition relation
Source	Data Aquisition
Target	Collect Survey

Composition relation

Type	Composition relation
Source	VS-07.0 Governance and Compliance Value Stream
Target	VS-07.2 Establish Legal and Regulatory Compliance Mechanisms

Composition relation

Type	Composition relation
Source	VS-06.0 Dissemination and Service Delivery Value Stream
Target	VS-06.4 Support and Engage Users

Composition relation

Type	Composition relation
Source	Dissemination
Target	Data Delivery

Composition relation

Type	Composition relation
Source	VS-04.0 Estimation, Analysis, and Statistical Modeling Value Stream
Target	VS-04.3 Estimate and Aggregate Data

Composition relation

Type	Composition relation
Source	VS-02.0 Data Standardization and Integration Value Stream
Target	VS-02.4 Validate Structural and Semantic Conformance

Composition relation

Type	Composition relation
Source	Data Processing
Target	Compose Data

Composition relation

Type	Composition relation
Source	VS-05.0 Statistical Product Composition Value Stream
Target	VS-05.4 Generate Analytical and Interpretive Outputs

Composition relation

Type	Composition relation
Source	VS-02.0 Data Standardization and Integration Value Stream
Target	VS-02.3 Align Metadata and Semantics

Composition relation

Type	Composition relation
Source	VS-01.0 Data Acquisition and Ingestion Value Stream
Target	VS-01.2 Establish Data Agreement and Access Right

Composition relation

Type	Composition relation
Source	Standardization
Target	Anonymize Data

Composition relation

Type	Composition relation
Source	VS-03.0 Data Editing and Enrichment Value Stream
Target	VS-03.5 Perform Quality Evaluation and Versioning

Composition relation

Type	Composition relation
Source	VS-09.0 Infrastructure and Platform Enablement Value Stream
Target	VS-09.1 Define Digital and Technical Requirements

Composition relation

Type	Composition relation
Source	VS-03.0 Data Editing and Enrichment Value Stream
Target	VS-03.4 Derive Additional Variables and Indicators

Composition relation

Type	Composition relation
Source	VS-07.0 Governance and Compliance Value Stream
Target	VS-07.5 Update and Improve Governance Mechanisms

Flow relation

Type	Flow relation
Source	VS-04.4 Perform Model-Based Analysis
Target	VS-04.5 Validate and Document Results

Flow relation

Type	Flow relation
Source	VS-10.1 Identify and Segment Stakeholders
Target	VS-10.2 Collect and Analyze User Needs

Flow relation

Type	Flow relation
Source	VS-01.2 Establish Data Agreement and Access Right
Target	VS-01.3 Collect or Receive Data

Flow relation

Type	Flow relation
Source	VS-05.4 Generate Analytical and Interpretive Outputs
Target	VS-05.5 Approve and Register Statistical Products

Flow relation

Type	Flow relation
Source	VS-03.3 Impute Missing or Non-Response Data
Target	VS-03.4 Derive Additional Variables and Indicators

Flow relation

Type	Flow relation
Source	VS-08.1 Identify Innovation Needs and Opportunities
Target	VS-08.2 Conduct Research and Proof-of-Concepts

Flow relation

Type	Flow relation
Source	VS-01.1 Identify Data Needs and Sources
Target	VS-01.2 Establish Data Agreement and Access Right

Flow relation

Type	Flow relation
Source	VS-08.4 Develop / Document New Methodologies or Tools
Target	VS-08.5 Integrate Innovation into Production Environments

Flow relation

Type	Flow relation
Source	VS-02.1 Profile and Assess Data Structure
Target	VS-02.2 Apply Standard Classifications and Formats

Flow relation

Type	Flow relation
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Source	VS-09.4 Enable Integration and Orchestration Services
Target	VS-09.5 Ensure Performance, Security, and Continuity

Flow relation

Type	Flow relation
Source	VS-10.4 Deliver Services and Manage Relationships
Target	VS-10.5 Engage and Communicate with Stakeholders

Flow relation

Type	Flow relation
Source	VS-07.3 Implement Data and Metadata Governance
Target	VS-07.4 Monitor Quality and Compliance Indicators

Flow relation

Type	Flow relation
Source	VS-07.4 Monitor Quality and Compliance Indicators
Target	VS-07.5 Audit, Certify, and Report Compliance

Flow relation

Type	Flow relation
Source	VS-04.3 Estimate and Aggregate Data
Target	VS-04.4 Perform Model-Based Analysis

Flow relation

Type	Flow relation
Source	VS-10.2 Collect and Analyze User Needs
Target	VS-10.3 Define and Manage Service Portfolios

Flow relation

Type	Flow relation
Source	VS-02.2 Apply Standard Classifications and Formats
Target	VS-02.3 Align Metadata and Semantics

Flow relation

Type	Flow relation
Source	VS-01.4 Validate and Log Data Intake
Target	VS-01.5 Store and Register Raw Data Artefacts

Flow relation

Type	Flow relation
Source	VS-02.3 Align Metadata and Semantics
Target	VS-02.4 Validate Structural and Semantic Conformance

Flow relation

Type	Flow relation
Source	VS-06.1 Plan Dissemination and Access Strategy
Target	VS-06.2 Prepare Data for Dissemination

Flow relation

Type	Flow relation
Source	VS-07.5 Audit, Certify, and Report Compliance
Target	VS-07.5 Update and Improve Governance Mechanisms

Flow relation

Type	Flow relation
Source	VS-06.3 Publish and Provide Access Services
Target	VS-06.4 Support and Engage Users

Flow relation

Type	Flow relation
Source	VS-10.5 Engage and Communicate with Stakeholders
Target	VS-10.6 Monitor Impact and Satisfaction

Flow relation

Type	Flow relation
Source	VS-07.1 Define Governance Frameworks and Policies
Target	VS-07.2 Establish Legal and Regulatory Compliance Mechanisms

Flow relation

Type	Flow relation
Source	VS-03.1 Conduct Error Detection/ Quality Diagnostics
Target	VS-03.2 Perform Data Editing and Correction

Flow relation

Type	Flow relation
Source	VS-03.2 Perform Data Editing and Correction
Target	VS-03.3 Impute Missing or Non-Response Data

Flow relation

Type	Flow relation
Source	VS-09.2 Design Platform and Infrastructure Architecture
Target	VS-09.3 Provision and Configure Platforms

Flow relation

Type	Flow relation
Source	VS-08.3 Evaluate Methodological Soundness and Feasibility
Target	VS-08.4 Develop / Document New Methodologies or Tools

Flow relation

Type	Flow relation
Source	VS-05.1 Select and Integrate Source Datasets
Target	VS-05.2 Construct Composite Indicators or Indices

Flow relation

Type	Flow relation
Source	VS-02.4 Validate Structural and Semantic Conformance
Target	VS-02.5 Register Standardized Artefacts

Flow relation

Type	Flow relation
Source	VS-10.3 Define and Manage Service Portfolios
Target	VS-10.4 Deliver Services and Manage Relationships

Flow relation

Type	Flow relation
Source	VS-03.4 Derive Additional Variables and Indicators
Target	VS-03.5 Perform Quality Evaluation and Versioning

Flow relation

Type	Flow relation
Source	VS-08.2 Conduct Research and Proof-of-Concepts
Target	VS-08.3 Evaluate Methodological Soundness and Feasibility

Flow relation

Type	Flow relation
Source	VS-05.3 Perform Consistency and Coherence Checks
Target	VS-05.4 Generate Analytical and Interpretive Outputs

Flow relation

Type	Flow relation
Source	VS-04.2 Apply Weighting and Calibration
Target	VS-04.3 Estimate and Aggregate Data

Flow relation

Type	Flow relation
Source	VS-09.3 Provision and Configure Platforms
Target	VS-09.4 Enable Integration and Orchestration Services

Flow relation

Type	Flow relation
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Source	VS-09.1 Define Digital and Technical Requirements
Target	VS-09.2 Design Platform and Infrastructure Architecture

Flow relation

Type	Flow relation
Source	VS-06.2 Prepare Data for Dissemination
Target	VS-06.3 Publish and Provide Access Services

Flow relation

Type	Flow relation
Source	VS-05.2 Construct Composite Indicators or Indices
Target	VS-05.3 Perform Consistency and Coherence Checks

Flow relation

Type	Flow relation
Source	VS-08.5 Integrate Innovation into Production Environments
Target	VS-08.6 Promote Knowledge Sharing and Capability Building

Flow relation

Type	Flow relation
Source	VS-09.5 Ensure Performance, Security, and Continuity
Target	VS-09.6 Evolve and Modernize the Technical Ecosystem

Flow relation

Type	Flow relation
Source	VS-06.4 Support and Engage Users
Target	VS-06.5 Monitor Usage and Feedback

Flow relation

Type	Flow relation
Source	VS-01.3 Collect or Receive Data
Target	VS-01.4 Validate and Log Data Intake

Flow relation

Type	Flow relation
Source	VS-04.1 Design Estimation and Analytical Approach
Target	VS-04.2 Apply Weighting and Calibration

Flow relation

Type	Flow relation
Source	VS-07.2 Establish Legal and Regulatory Compliance Mechanisms
Target	VS-07.3 Implement Data and Metadata Governance

Realization relation

Type	Realization relation
Source	VS-10.0 Customer and Stakeholder Engagement Value Stream
Target	Communication and outreach materials

Realization relation

Type	Realization relation
Source	VS-07.0 Governance and Compliance Value Stream
Target	Governance documentation and certification logs

Realization relation

Type	Realization relation
Source	VS-09.0 Infrastructure and Platform Enablement Value Stream
Target	Configured data and analytics environments

Realization relation

Type	Realization relation
Source	VS-07.0 Governance and Compliance Value Stream
Target	Statistical quality framework and scorecards

Realization relation

Type	Realization relation
Source	VS-09.0 Infrastructure and Platform Enablement Value Stream
Target	Security and continuity management documentation

Realization relation

Type	Realization relation
Source	VS-01.0 Data Acquisition and Ingestion Value Stream
Target	Collected dataset with basic metadata and source quality indicators

Realization relation

Type	Realization relation
Source	VS-08.0 Innovation and Methodological Development Value Stream
Target	Knowledge base, guidance, and training materials

Realization relation

Type	Realization relation
Source	VS-03.0 Data Editing and Enrichment Value Stream
Target	Edited and enriched dataset with quality and lineage documentation.

Realization relation

Type	Realization relation
Source	VS-07.0 Governance and Compliance Value Stream
Target	Data governance policy set

Realization relation

Type	Realization relation
Source	VS-10.0 Customer and Stakeholder Engagement Value Stream
Target	User requirements documentation

Realization relation

Type	Realization relation
Source	VS-09.0 Infrastructure and Platform Enablement Value Stream
Target	Platform and infrastructure blueprints

Realization relation

Type	Realization relation
Source	VS-02.0 Data Standardization and Integration Value Stream
	Standardized dataset with aligned metadata and classification mappings.

Realization relation

Type	Realization relation
Source	VS-08.0 Innovation and Methodological Development Value Stream
Target	Methodological research reports and prototypes

Realization relation

Type	Realization relation
Source	VS-08.0 Innovation and Methodological Development Value Stream
Target	Pilot evaluation results and transition plans

Realization relation

Type	Realization relation
Source	VS-06.0 Dissemination and Service Delivery Value Stream
	Published dataset, statistical release, or data service accessible to users.

Realization relation

Type	Realization relation
Source	VS-07.0 Governance and Compliance Value Stream
Target	Compliance reports and audit trails

Realization relation

Type	Realization relation
Source	VS-04.0 Estimation, Analysis, and Statistical Modeling Value Stream
	Estimated or analyzed dataset with methodological and quality documentation.

Realization relation

Type	Realization relation
Source	VS-05.0 Statistical Product Composition Value Stream
	Finalized statistical product or analytical dataset, ready for dissemination.

Realization relation

Type	Realization relation
Source	VS-10.0 Customer and Stakeholder Engagement Value Stream
Target	Stakeholder maps and engagement strategies

Realization relation

Type	Realization relation
Source	VS-08.0 Innovation and Methodological Development Value Stream
Target	Innovation pipeline documentation

Realization relation

Type	Realization relation
Source	VS-09.0 Infrastructure and Platform Enablement Value Stream
Target	Integration and orchestration services

Realization relation

Type	Realization relation
Source	VS-09.0 Infrastructure and Platform Enablement Value Stream
Target	Monitoring dashboards and performance metrics

Realization relation

Type	Realization relation
Source	VS-10.0 Customer and Stakeholder Engagement Value Stream
Target	User satisfaction and impact assessments

Realization relation

Type	Realization relation
Source	VS-07.0 Governance and Compliance Value Stream
Target	Data lineage and metadata records

Realization relation

Type	Realization relation
Source	VS-10.0 Customer and Stakeholder Engagement Value Stream
Target	Service catalogues and SLAs

Realization relation

Type	Realization relation
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Source	VS-08.0 Innovation and Methodological Development Value Stream
Target	Validated new statistical or analytical methods